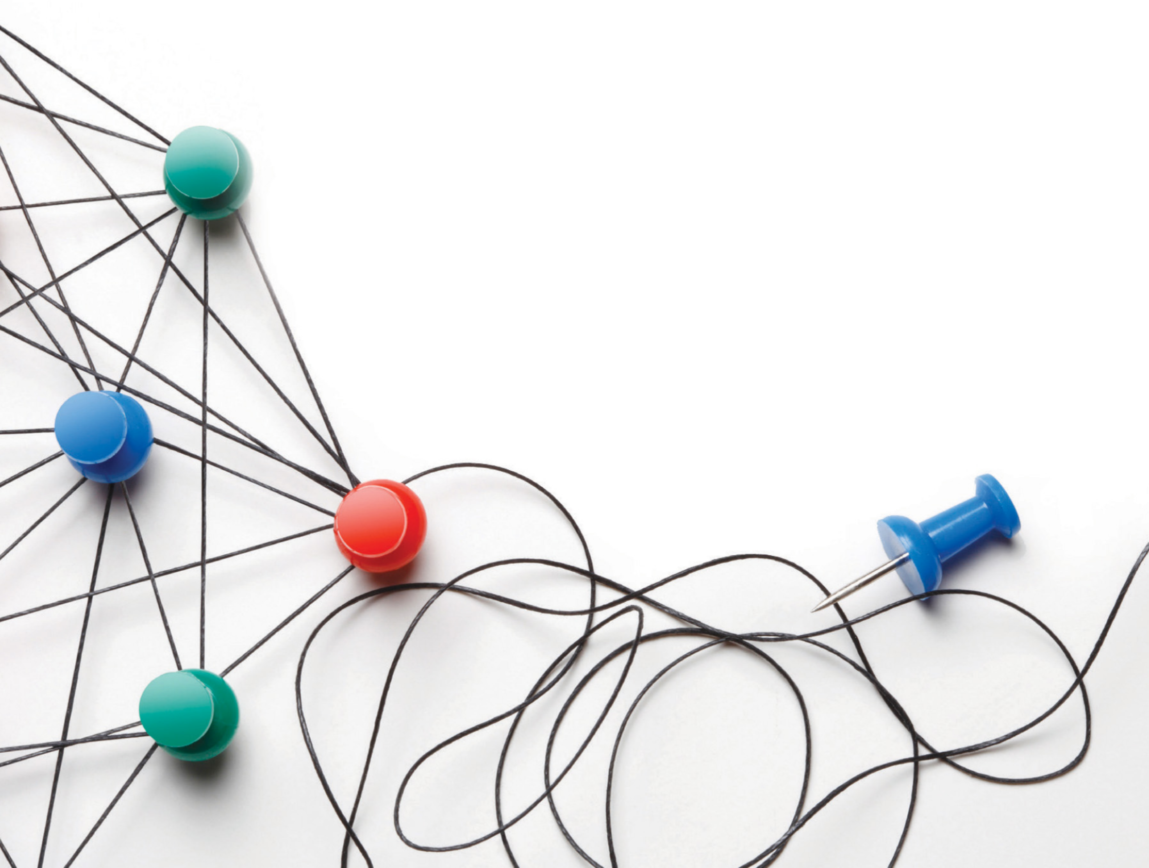
AMY C. EDMONDSON JEAN-FRANCOIS HARVEY

Foreword by HENRY W. CHESBROUGH,
Author of *Open Innovation*

EXTREME TEAMING

LESSONS IN COMPLEX, CROSS-SECTOR LEADERSHIP



EXTREME TEAMING

Lessons in Complex, Cross-Sector Leadership

To all leaders of extreme teaming, especially those who led Projects Anna, Bianca, Fiona, Sofia, and Willa, and inspired us to write this book.

EXTREME TEAMING

Lessons in Complex, Cross-Sector Leadership

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emerald
PUBLISHING

United Kingdom — North America — Japan — India — Malaysia — China

Emerald Publishing Limited
Howard House, Wagon Lane, Bingley BD16 1WA, UK

First edition 2017

British Library Cataloguing in Publication Data
A catalogue record for this book is available from the British Library

ISBN: 978-1-78635-450-1 (Print)
ISBN: 978-1-78635-449-5 (Online)
ISBN: 978-1-78714-898-7 (Epub)

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ABOUT THE AUTHORS

Amy C. Edmondson is the Novartis Professor of Leadership and Management at the Harvard Business School. Her work explores teaming – the dynamic forms of collaboration needed in environments characterized by uncertainty and ambiguity. She has also studied the role of psychological safety in teamwork and innovation. Before her academic career, she was Director of Research at Pecos River Learning Centers, where she worked with founder and CEO Larry Wilson to design change programs in large companies. In the early 1980s, she worked as Chief Engineer for architect/inventor Buckminster Fuller.

Jean-François Harvey is Assistant Professor in the Department of Entrepreneurship and Innovation at HEC Montréal. He joined the faculty after completing a two-year postdoc at Harvard Business School. His work on innovation management has been published in several academic and management journals, and he has consulted on this topic for public and private organizations.

FOREWORD

I had the privilege to teach at Harvard Business School with Amy Edmondson in the Technology and Operations Management (TOM) group 20 years ago. Amy was polishing off her research on psychological safety, examining a critical, but heretofore overlooked, factor that influenced the performance of work teams. That work would later establish her as a global management thought leader. Meanwhile I was down the hall performing the field research on companies like Xerox, IBM, Intel, Lucent, Procter & Gamble, and Genzyme that would culminate later in Open Innovation. At that time, we didn't seem to have much to say to each other, and so were friendly but distant colleagues. After all, it seemed that we were working on quite different problems, and investigating quite disparate phenomena.

I moved on to Berkeley from Harvard in 2003, and had the pleasure of watching my book *Open Innovation* (Chesbrough, 2003) develop into an important contribution to the study of innovation. But apart from saying hello to Amy at occasional academic conferences, things remained distant between us. Something happened fairly recently to change this state of affairs. Jean-François Harvey came to Berkeley a few short years ago as a visiting scholar. He was bright, well-trained, and full of ideas. While I learned a lot from him, he got infected with the Open Innovation virus during his time with me, and he is now pushing the open innovation concept forward in new and important ways. It is his insights, his energy, and his passion that have brought the work of two previously distant colleagues much closer together. One example of this comes from a very recent case study he conducted with Amy (Edmondson & Harvey, 2016b), and the other comes from this new book.

Now that I see Amy and JF's latest work in this new book, it is now obvious to me that Open Innovation and the organization of teams have a lot of common interests. Amy's work on teaming as a process (not just as an entity), combined with her exploration of the ways in which people from different organizations come together to pursue a common project, has positioned her to make new contributions to my own field of innovation studies. For most of the innovating in Open Innovation is done not by single individuals, nor by entire organizations, but by groups of people working across organizational boundaries. If you want to move knowledge across boundaries, you need to organize, motivate, and coordinate people in groups. With this new book, innovation scholars will find a wealth of insights about the core innovation work activity that takes place in collaborative innovation initiatives across those boundaries.

With my colleague Marcel Bogers, I have recently modified my definition of Open Innovation as follows: "…*a distributed innovation process that involves purposively managed knowledge flows across organizational boundaries, using pecuniary and non-pecuniary mechanisms in line with the organization's business model*" (Chesbrough & Bogers, 2014). This definition goes beyond the well-known phenomenon of knowledge spillovers long studied in the economics of R&D to purposive, intentional flows of knowledge of interest to management scholars. These intentional flows involve both flows of knowledge from outside the organization, or outside-in knowledge flows, and also flows from inside the organization to the outside, or inside-out knowledge flows. What Amy and JF's book reminds us is that we must look at the work of groups of people, if we are to understand these flows of knowledge.

One type of outside-in open innovation – so-called *crowdsourcing* – seeks to engage the problem-solving abilities of individuals located around the world. Crowdsourcing can take the form of contests that post a perplexing problem on a website in the hopes of eliciting novel solutions from remote sources. NASA and Samsung come to mind: the first has established a partnership with a number of crowdsourcing platforms to reach outside-of-the-box ideas for some of its most pressing problems, while the second seeks innovative solutions for existing electronic products and technologies. Crowdsourcing can also be more playful and everyday, such as imagining a new burger for McDonald's or creating a new flavor for Lays potato chips.

Similarly, open-source software development allows individuals to write code remotely, offering modular elements that can be compiled and combined to create robust software programs. Contributors, motivated by everything from fame to fortune to altruism to learning, work autonomously to push the technical envelope. And this open approach is not limited to software. In the data center hardware industry, Facebook's Open Compute Project (OCP) has achieved some important breakthroughs and has mobilized knowledge from numerous external contributors. Indeed, by developing an initial proposal, contributing initial reference designs, and offering test deployments of OCP designs – all examples of inside-out open innovation – Facebook initiated and then subsequently orchestrated a lot of outside-in open innovation.

But what if a problem is inherently multidisciplinary and complex, a so-called wicked problem? If solutions require people to work interdependently across disciplines or locations, crowdsourcing is unlikely to work. And so, a new kind of open innovation is needed to bring together people from several organizations in projects targeting such challenges. This book looks at the behavior of humans in groups and teams at the core of these kinds of strategic, complex innovation projects.

Edmondson and Harvey bring a new perspective to the field of open innovation with this book. Using their research into a handful of open innovation projects, they begin to identify what project leaders can do to overcome the major hurdles that lie ahead.

In many ways, open innovation has a lot to do with team development and leadership. Connecting these streams of research appears important to develop knowledge that will help support the increasingly important cross-boundary collaborations organizations must engage in today. Team-based configurations have become so fluid that some have argued for the vanishing of corporations, with software that can now combine the gig economy model with artificial intelligence to assemble "flash teams."[1] Yet, these flash teams have their limits, and I doubt they can be successful without good leadership. We need to understand what leaders can do to make the most of the new forms of collaboration that attempt to create value and innovate rapidly in our increasingly complex world.

This book is timely. It is also incomplete: it opens up a new field of inquiry. It should resonate with both researchers and practitioners. Researchers can find a rigorous approach to qualitative research enabling both theoretically robust constructs and convincing empirical findings. Practitioners from diverse fields can find actionable insights that promise to improve the management of complex innovation projects across the permeable organizational boundaries that arise in this increasingly Open Innovation landscape.

Henry Chesbrough
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NOTE

1. Readers of a certain age might recognize an echo of an earlier trend, of the so-called virtual organization of the 1990s. David Teece and I sought to understand but also qualify the limits of what virtual organizations can do with our 1996 Harvard Business Review article, "When is Virtual Virtuous?" There we reminded readers that real organizations had an ability to orchestrate complex, systemic technologies in ways that virtual organizations could not. I suspect that these "flash teams" will follow a similar pattern to the virtual organization. They will perform important work, but will not supplant all the other types of teams, due to their likely inability to orchestrate the actions of disparate actors in complex, interdependent situations.

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INTRODUCTION

Research on team effectiveness in the social sciences – notably, in psychology, sociology, and economics – is extensive and enduring. Teams fascinate scholars and practitioners alike because of their potential to achieve far more than the sum of what individual team members can do alone. Potential is not inevitability, however, and what it takes to achieve the desired synergy in teams remains a topic of considerable research. Achieving synergy requires integrating and leveraging diverse expertise and perspectives. Yet, the presence of diverse expertise and perspectives poses a barrier to doing so; people may not adequately understand each other's thoughts and ideas, and they lack the shared norms, values, or timeframes that facilitate interaction. Herein lies both the promise and the challenge of extreme teaming – project teams that cross disciplinary, organizational, and industry boundaries to innovate.

The late Harvard psychologist Richard Hackman, a preeminent scholar of team effectiveness, conducted numerous quantitative studies to pinpoint features that influence team performance, and this work provided foundational insights (e.g., Hackman, 1983, 1990; Hackman & Morris, 1975). The basic theoretical framework employed is an input-process-output model, in which a set of inputs such as task design or organizational support give rise to certain behavioral and interactional processes, which lead to various performance outputs. Decision-making, conflict-resolution, and information-management are some of the important processes Hackman and his colleagues studied. Factors that shape these processes (inputs) included variables at the individual, group, and organizational levels. For example, individual factors include members' skills, group factors include the size of the team, the team task, and the clarity of the team's goal, and organizational factors are such variables as access to resources and a supportive environment. Inputs and team processes both give rise to team outputs. The most obvious of these is team task performance, but Hackman (1990) conceptualized team effectiveness as more than task performance, identifying three vital dimensions of effectiveness. The first, the extent to which a team's work satisfies the needs of its customers, is what most managers would expect to matter, and is fundamentally about performance. The other two are a team's ability to work well together in the future (a kind of team-level learning outcome) and individual team members' satisfaction with the team experience (an individual level job satisfaction variable). The essence of the theory is that well-designed teams will be more likely to have desired processes and outcomes. In short, well-designed teams perform well.

Hackman identified three essential defining features of a team. First, teams have clear boundaries that distinguish members from non-members. Second, members are interdependent in working toward a common goal, such that they are collectively responsible for what they produce together. Third, teams are relatively stable entities, giving members the opportunity to learn how to work well together (Wageman, Hackman, & Lehman, 2005). Although teamwork and collaboration are needed in settings other than within formal teams, it is helpful to distinguish between these phenomena and actual team structures, as well as between groups and teams. Organizations encompass many kinds of groups, but Hackman proposed reserving the term "team" for groups that meet these three criteria. Others have proposed that teams be defined by the sense of identity that derives from being part of a group: a self-conception that is shared by members and gives rise to a self-inclusive category that causes them to identify with the group (Tajfel, 1978; Tajfel & Turner, 1986). Unclear social identity, or multiple social identities, can be detrimental to team performance (Brewer, 1996). We concur with this perspective, but see shared identity as an emergent state, rather than a defining criterion and structural input.

Most research on teams and teamwork focuses on the design of teams, including studies of structural inputs like team size, composition, and task. Although some team research is conducted in laboratories, much of it examines real teams in real organizations. Some perspectives emphasize processes, and process interventions related to coaching and facilitation, such as training individuals to operate in complex team environments (Cannon-Bowers & Salas, 1998), or focus on team norms and climate, including psychological safety (Edmondson, 1999).

One important area of study is team leadership, which has gained considerable attention over the past two decades as a vital force in helping teams achieve their potential. Most of the this literature examines how leaders influence the performance of groups that are reasonably stable performance units with clearly defined boundaries, that is, teams that meet Hackman's team-defining criteria. The kinds of teams studied in the field range from home improvement store teams (Chen et al., 2007), to financial services teams (Schaubroeck et al., 2007), customer services teams (Wageman, 2001), and more. The advantages of stable teams with clear boundaries and consistent membership have been well documented; members of such teams can leverage long-lasting relationships and contextual knowledge to communicate and execute effectively (Griffith & Neale, 2001; Lewis, 2004). Nonetheless, fewer teams in today's dynamic workplaces are stable or clearly bounded (Mortensen, 2014; O'Leary, Mortensen, & Woolley, 2011). Many teams change fast and members have little time to establish shared understanding about tasks, context, or each other (Wageman, Gardner, & Mortensen, 2012).

Recent work, therefore, has shifted to include a different perspective on teamwork in organizations, one that includes the interpersonal interactions taking place in shifting groups of people working collaboratively toward shared goals. This perspective calls attention to *teaming* as a process – rather than teams as entities (Edmondson, 2012) – and requires research to understand what leaders can do to support teamwork in shifting configurations and contexts, including teamwork that brings people from different organizations together on a novel project.

FROM TEAMS TO TEAMING

Recognizing that a great deal of collaborative work in organizations occurs outside of formal teams, recent work has employed a "teaming" perspective on managing interdependent work, which emphasizes the processes of teamwork rather than the structures (Edmondson, 2012). Teaming takes various forms. To begin with, stable intact teams are often tasked with carrying out interdependent work that requires back-and-forth ("reciprocal") coordination to do it well (Thompson, 1967), and one can reasonably call that highly interdependent coordination a form of teaming. This form is the one that has received most of the attention in prior literature, and our understanding of how such teamwork is reasonably well established.

Second, in addition to coordination that occurs within stable teams, teaming in today's workplaces occurs in fluid configurations as well. In some cases, people serve on multiple teams at once and thus confront the need to manage the various relationships they encounter in these different groups (Mortensen, 2014). In other cases, people work in hyper-fluid or extremely temporary team-like arrangements, such as in a hospital emergency department, where each patient is treated by a newly formed small team of professionals, involving various hand offs, where the teams convene and disband constantly (Valentine & Edmondson, 2015). A small but rapidly growing literature is examining such teams.

A third form of teaming, which we believe is also on the rise, involves people coming together from diverse backgrounds and organizations to address a complex and usually novel problem (Edmondson & Harvey, 2017). Such situations bring people together who are not only diverse in expertise but also are employed by different organizations. This book is focused on understanding both the challenges and the opportunities presented by such cross-sector teaming arrangements, which present an extreme form of teaming. For instance, in the economic development context, experts in agriculture, economic development, finance, marketing, supply chain management and project management from Coca-Cola, the United States Agency for International Development, the Inter-American Development Bank, and the nonprofit organization TechnoServe teamed up on an ambitious project to improve 25,000 Haitian mango farmers' business practices and double their income (Edmondson & Harvey, 2016). This third form of teaming often stems from necessity to tackle complex, multifaceted problems. The success of these projects depends on learning – that is, on the ability to adapt rapidly and efficiently to new knowledge. In the absence of past experience and knowledge, such projects must make recourse to learning to shape their responses to threats and opportunities. Each project participant lacks not only a body of project-specific knowledge but also contextual knowledge about viable paths to success. As a result, such projects shift rapidly in ways that are difficult, or even impossible, to predict.

In these types of projects, people work together temporarily, spanning boundaries that include expertise, function, organization, and sometimes industry. The newly formed teams must develop *in situ* – that is, in a specific context where leadership plays an important role. Many such teams face highly novel challenges and therefore must learn quickly and effectively to succeed. Through field research studying several such extreme cases of teaming, we hope to provide insights and new directions to further this stream of research to contribute both to theory and practice.

EXTREME TEAMING FOR INNOVATION

Edmondson (2012) emphasized the importance of understanding and enabling teaming processes to complement the extensive research on team structures. We build on this prior work to elaborate the phenomena and leadership functions associated with effective teaming on innovation projects that span occupations, organizations, and industries. We refer to this type of cross-boundary collaboration as *extreme teaming*. Considerable research has investigated cross-functional teams, especially in the context of new product development and other innovation work (e.g., see Edmondson & Nembhard, 2009 for an exploration of the challenges and opportunities such teams face). Other work has examined the challenge of teaming across boundaries between hierarchical levels (e.g., Nembhard & Edmondson, 2006). Both kinds of cross-boundary teamwork are challenging, but extreme teaming takes these challenges to a new level.

Innovation is essential for staying relevant in today's challenging, fast paced environment. Few industries remain untouched by dynamism, uncertainty, and turbulence. The list of once-successful organizations that are no longer in business today – outpaced by more innovative rivals – is long and growing. In almost every industry, the demand for innovation is thus intensifying (Teece, 2012). Increasingly, companies must tap into ideas that are generated outside their organizational boundaries and find themselves collaborating to develop new products or services (Chesbrough, 2003) and even in exploiting them through open business models (Chesbrough, 2006a). The goal of extreme teaming is usually related to this impetus. By assembling groups of people with various backgrounds, those driving extreme teaming hope to set the stage for complex problem solving and innovation that affects more than one organization. How to do this well, especially facing ambitious goals and timelines, is a question of importance for both research and practice. This book thus builds on prior research to improve understanding of a particular type of teaming and to suggest new ideas for future research and practice. Our primary goal is to shed light on the leadership practices that help a group of individuals with very different backgrounds (notably, occupation, organization, and industry) develop into a high performing, albeit temporary, team. In short, we hope to explain how extreme teaming can be nurtured, despite its inherent challenges.

As business ecosystems evolve in ways that force organizations to become more fluid and flexible (Tucci, Chesbrough, Piller, & West, 2016), people working on innovation often move across contracts, projects, departments, and organizations (e.g., Hargadon & Bechky, 2006; O'Mahony & Bechky, 2008). A rise in project-based organizing has been noted across the public and private sectors alike (Hobday, 2000; Wheelright & Clark, 1992). When projects are the primary unit for execution and exploration of innovation work, fluidity of roles and tasks often follows (Bechky, 2006). Some projects are long in duration, going on for years, as in certain new product development projects and most construction projects. Others are very short, as in most patient-care teams, some task forces, event planning groups, and more. As more and more people work in multiple teams, projects, departments, or organizations, often simultaneously, it is important to understand the processes and practices that enable teaming across not just departmental but also organizational and industry boundaries. A growing portion of the innovation landscape requires organizations to work beyond their usual disciplinary or organizational boundaries, including a growing number of projects that bring people together from multiple organizations to work together (Ferraro, Etzion, & Gehman, 2015; Senge et al., 2008).

Because leadership plays out in social or organizational settings (Pettigrew, 1992), and its effectiveness depends on fit with the situation or context (Fiedler, 1967), we need to know more about leadership in such contexts to better deal with its associated technological uncertainty (Fleming, 2001), coordination costs (Cummings & Kiesler, 2005), and logistical challenges (Lingo & O'Mahony, 2010).

The central message in this book is that extreme teaming is as challenging as it is necessary and, therefore, that leadership is vital to doing it well. We present qualitative research on a handful of cases of extreme teaming (complex cross-industry innovation projects) as a starting point for understanding the leadership functions that enable success in such challenging endeavors. This research does not allow us to draw firm conclusions about cause and effect, but rather to open a new area of study by observing some common practices shared by a diverse set of projects in extreme contexts. This research will also help in developing precise contextualized recommendations for leading extreme teaming. Most of the past research on team leadership has been in the context of leading stable, bounded teams rather than the more complex work arrangements created by extreme teaming. Leadership theory on how leaders tackle the challenges of teaming in newly formed, temporary work groups that span diverse skill sets and organizations is limited. Our work in this book takes a small step toward developing this theory; we hope that the leadership functions we describe will help those in the trenches to lead successful extreme teaming in innovation efforts around the world.

OVERVIEW OF THIS BOOK

Crossing the conceptual and physical boundaries between organizations heightens the already well-recognized challenges of cross-disciplinary work. Understanding the interpersonal and technical dynamics of extreme teaming is thus an important new area for research and practice. How do diverse groups of people come together to accomplish challenging innovation goals, requiring them to master new content, build new relationships, and integrate their ideas and expertise to produce high-value output? This book explores these questions in three parts to offer new directions for scholarly research and practical application.

Part I elaborates the need for extreme teaming, describing how business environments are evolving to make teaming across boundaries an important new activity for success. It also describes what makes this so difficult to do, and what insights can be drawn from prior work on leadership theory in the context of teams and teaming. Chapter 1 opens with a powerful illustration of extreme teaming to motivate our explanation of why organizations increasingly face the need for cross-boundary teaming. The chapter reflects the shift underway from a focus on business industries to a focus on innovation systems. We also explain why teams are the performance unit *par excellence* for innovation. Chapter 2 considers the main leadership theories that have been developed over the past few decades. One stream of research emphasizes leadership functions (rather than traits or other attributes), and we explain why this is the most appropriate approach to inform the activities of those leading cross-boundary teaming projects. We stress the need for a taxonomy centered on the extreme teaming context. Chapter 3 synthesizes team development and team diversity research to reveal the essential challenges to extreme teaming from both an interpersonal and a technical perspective. We explain why gaining additional insight into these dynamics is essential for managers who lead extreme teaming efforts. These three opening chapters thus set the stage for our study of extreme teaming in a variety of complex, cross-sector innovation projects.

Part II presents the findings from our multiyear study of extreme teaming in a set of remarkably varied industries. Our case-study approach used qualitative analyses of a series of unusual experiences of extreme teaming to develop a set of four interdependent leadership functions fostering extreme teaming and innovation results. These leadership functions are presented in Chapter 4: Build an Engaging Vision; Chapter 5: Cultivate Psychological Safety; Chapter 6: Develop Shared Mental Models, and Chapter 7: Empower Agile Execution. Each of these chapters opens with a brief story from one of our case studies to illustrate the leadership function elaborated in the chapter and to offer an in-depth account of the practices it entails.

In Part III, we discuss and extend the implications of our findings. Chapter 8 pulls the four leadership functions together to explain the overarching framework and rationale for their use as a system of leadership practices. We also explain how this framework contributes to leadership theory and practice. To foster extreme teaming, leadership can – and must – motivate people to extend themselves in these challenging tasks. Leaders also must enable this work by removing the natural, very real, barriers to collaborating across boundaries. At the same time, leaders must help teams overcome both the interpersonal and technical challenges of extreme teaming. These dual challenges give rise to a 2x2 matrix outlining the four leadership functions we described in Part II. Chapter 9 concludes with suggestions for future research stemming from our findings, and how team-diversity and team-leadership scholars may develop studies that inform the practice of managers involved in extreme teaming efforts.

CHAPTER TAKEAWAYS

- An input-process-output view of team performance has dominated the literature on teams and team effectiveness. The theory posits that teams will perform well if they are well designed.

- Well-designed means a clear boundary, a shared goal, an interdependent task, some stability, appropriate composition for the task, and adequate resources. Correspondingly, the research on team effectiveness has tended to study teams as reasonably stable performance units with clearly defined boundaries.
- More and more teams today do not qualify as "well-designed" according to traditional definitions – because of the shifting nature of the work, not because managers have failed to do their jobs.
- Business ecosystems change rapidly and people from different occupations and organizations increasingly move from project to project, and collaborate in temporary, team-based arrangements with fluid membership.
- This new reality calls for additional emphasis on teaming as a process rather than teams as entities.
- Teaming is a dynamic activity; teams are bounded entities. Teaming is largely determined by the mindset and practices of teamwork, not by the design and structures of effective teams.
- Extreme teaming refers to cross-sector collaboration. Increasingly, companies tap into ideas and skills outside their organizational boundaries. By assembling groups of people with various backgrounds, extreme teaming sets the stage for complex problem solving and innovation that affect more than one organization.
- The goal of this book is to shed light on the leadership practices that help a group of individuals with very different backgrounds (notably, occupation, organization, and industry) develop into a high performing, albeit temporary, team.

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PART I

TRENDS GIVING RISE TO EXTREME TEAMING

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1

WHY EXTREME TEAMING MATTERS

The accelerated pace of knowledge development, well documented by scholars in management, sociology, history of science and other fields, gives rise to complexity and uncertainty in today's business and scientific environments (Powell & Snellman, 2004). In most fields of expertise, the rate of new knowledge development requires people to invest considerable time just to stay current. In technical fields, the explosion of new knowledge leads inexorably to greater specialization. Specialized jargon proliferates and experts struggle to keep up with developments in even closely related fields.

On the one hand, the so-called *knowledge explosion* leads to narrower and deeper areas of specialization (Edmondson & Nembhard, 2009). Fields thus spawn subfields. For instance, as new medical discoveries proliferate, the number of subspecialties grew from about 40 in 1985 to 100 in 2000 (Donini-Lenhoff & Hedrick, 2000), with the promise of more to come. Internal medicine divides into subspecialties like cardiology, endocrinology, gastroenterology, hematology, infectious diseases, nephrology, oncology, pulmonary diseases, rheumatology, and so on. Radiology today encompasses angiography and intervention radiology, body imaging, diagnostic radiology, diagnostic roentgenology, diagnostic ultrasonography, neuroradiology, radiation oncology, radiation therapy, vascular-interventional radiology, nuclear radiology, and pediatric radiology. Each of these subfields faces continued change. Technologies that were barely used 10 years ago (e.g., virtual reality) are having a major impact in various medical subfields today (e.g., cybertherapy). The minimum number of years of study required to graduate in those fields and subfields has slowly but steadily increased, and beyond formal training, experts must invest ongoing effort in remaining current. The promise of specialization lies in the depth of understanding experts can offer.

On the other hand, concurrent with the rise of narrow and deep expertise, the problems facing organizations and society have not, of course, narrowed accordingly. Instead, they are increasingly complex and multifaceted. Addressing them requires multidisciplinary approaches. Consider what it takes to design, build, or maintain high-tech infrastructure projects, intelligent buildings, avionics systems, mobile telephone networks, or banking communication systems. Even products such as shoes that appear simple can become tremendously complex: for example, 50 biomechanical engineers, industrial designers, and electromechanical experts teamed up to make asymmetrical spikes for Olympic gold medalist Jeremy Wariner (Hochman, 2008). In healthcare, organizations such as Hacking Health show that varied professionals' perspectives must be considered, in addition to the patients' perspectives, to produce successful innovation (Dionne & Carlile, 2016). In short, dealing with today's complex products and systems requires solving dozens, sometimes-even hundreds, of interconnected problems (Davies & Hobday, 2005).

Thus, to solve complex problems and innovate in ways that reflect the increasing rate of change, today's organizations must take advantage of deep specialized knowledge and manage knowledge integration across these domains of expertise at the same time. These two opposing challenges create the need for organizations to master extreme teaming. This chapter thus builds a case for the importance of extreme teaming for a new and increasingly important type of initiative – one that brings together diverse areas of expertise to solve a particularly challenging societal or business problem.

In the pages that follow, we set the stage with a particularly compelling case study to illustrate the power of extreme teaming at its best. We then shift to discuss recent work on business ecosystems, to emphasize the utility of a new perspective on how organizations thrive and innovate as a result of their interactions with other organizations in highly interdependent systems. We finish the chapter by proposing essential features of innovation efforts employing an extreme teaming approach.

EXTREME TEAMING THAT SAVED 33 LIVES

The success of the now legendary mining rescue in Chile in 2010, while widely covered in the news and immortalized in a major Hollywood film, is not widely understood. We offer the rescue as an extraordinary example of extreme teaming, to illustrate the enormous potential of diverse experts coming together to innovate to overcome a nearly impossible challenge. The case illustrates the centrality of diverse perspectives in producing innovation, as well as the importance of leadership in making it happen.

An Unprecedented Challenge

Although mining accidents often present immense hurdles that make rescue unlikely, the situation at Chile's San Jose copper mine that began on August 5, 2010 was unprecedented on several dimensions. The most daunting of these was the extraordinary depth – 700 meters below ground – at which the miners were trapped in the aftermath of an explosion that left half a million tons of rock blocking the mine's entrance of the mine. The number of miners trapped (33), the hardness of the rock, the instability of the land, and the complete inadequacy of provisions for the trapped men (enough food for two men for 10 days) combined to make the possibility of rescue appear all but impossible to consulted experts. A mining rescue in the United States just a few years earlier, in which nine men were trapped 240 feet below ground, had been considered at the time a remarkable feat (Robbins, 2007). In Chile, early estimates of the possibility of finding anyone alive – put at 10% – diminished sharply two days later when rescue workers narrowly escaped the secondary collapse of a ventilation shaft, taking away the initial best option for extracting the miners. At that point, no expert considered rescue of the 33 men a reasonable possibility. Nonetheless, within 70 days all of them would be alive and reunited with their families.

This outcome was the result of an extraordinary cross-industry teaming effort by hundreds of individuals spanning physical, organizational, cultural, geographic, and professional boundaries. Engineers, geologists, drilling specialists, and more came together from different organizations, sectors, and nations to work on the immensely challenging technical problem of locating, reaching, and extracting the trapped miners. Senior leaders in the Chilean government provided resources to support the on-site efforts.

How Senior Leadership Triggered Extreme Teaming

In Santiago, Chile's capital city, President Piñera and Laurence Golborne, the Minister of Mining, met on the morning of August 6, 2010. Piñera then sent Golborne to the mine with a mandate to do whatever possible to bring the miners home, sparing no expense. Golborne and Piñera quickly reached out to their networks of colleagues around the world. As the president put it, "We were humble enough to ask for help" (Robbins, 2007). Michael Duncan, a deputy chief medical officer with the U.S. National Aeronautics and Space Administration (NASA), reported that the Chilean officials said, "Let's try to identify who the experts are in the field – let's get some consultants in here that can give us the best information possible." Duncan brought experience with long space flights to help solve concerns related to the miners' physical and psychological survival in their small quarters. NASA engineers played a crucial role in the design of the escape capsule that would be used in the final stage of the rescue to extract the miners from the refuge.

The Chilean *Carabineros* Special Operations Group – an elite Chilean police unit for rescue operations – had arrived a few hours after the first collapse. Yet their initial attempt at rescue had triggered that devastating secondary shaft collapse. As news of a mine cave-in spread, family members, emergency response teams, rescue workers, and reporters poured into the vicinity. Meanwhile, the Chilean mining community dispatched experts, drilling machines, and bulldozers. At the request of President Pinera, Codelco, the state-owned company, sent a senior mining engineer to lead the effort; Andre Sougarret, known for his engineering prowess, calm composure and ease with people, brought extraordinary technical and leadership competence to the project.

Parallel Teaming Efforts

Sougarret formed three teams to oversee different aspects of the operation. One searched for the men, poking drill holes deep into the earth in the hopes of hearing sounds to indicate that the men were alive. Another worked on how to keep them alive if found, and a third brainstormed solutions for how to extract them from the refuge. The first team came up with four possible rescue strategies. The most obvious through the ventilation shaft, was quickly rendered impossible. The second strategy, drilling a new mine ramp, also soon proved impossible as the instability of the rock was discovered. The third, tunneling from an adjacent mine a mile away, would have taken 8 months and was thus soon excluded.

The only hope left was the last strategy – drilling a series of holes at various angles to try to locate the refuge.

The extreme depth of the refuge, along with its small size, made the problem of location staggeringly difficult. With the drills' limited precision, the odds of hitting the refuge with each laborious drilling trial were about one in eighty. Even that was optimistic, because the location of the refuge was imprecisely known. Maps of the tunnels had not been updated in years. Additional technical challenges disallowed drilling straight down from the top of the mine, further exacerbating the drilling accuracy problem.

Rescuers soon divided into subteams to experiment with different strategies for drilling holes. More often than not, these teams failed to achieve their desired goals in any individual drilling attempt, but they soon learned to celebrate the valuable information each attempt provided, such as revealing features of the rock, to inform future action. For instance, the drillers and geologists discovered that fallen rock had trapped water and sedimentary rocks, increasing drill deviations and further exacerbating the odds of reaching the refuge in time. This was the kind of technical detail that engineers had to quickly incorporate into their plans, which shifted rapidly with each passing day. One dramatic change to procedure was the discovery and use of frequent, short action-assessment cycles. In normal drilling operations, precision was measured after a hole was completely drilled. Here, in contrast, drillers realized that to hit the refuge, they would have to make measurements every few hours and promptly abandon holes that deviated too much, discouraging as that might be. As they learned more about the search challenge, the odds of success diminished further, with one driller putting it at less than 1%.

In this remarkable story, different clusters of experts came up with with remarkably complementary pieces of an ultimately viable complex solution. Of course this didn't happen by accident, but rather was enabled by a particular type of leadership.

For example, a Chilean geologist named Felipe Matthews brought a unique technology for measuring drilling with high precision that he had recently developed. Matthews came to the site, and, working with several other strangers, discovered that his measurements were inconsistent with those of other groups; a rapidly improvised set of experiments showed that his equipment was most accurate. Matthews was then put in charge of measuring drilling efforts going forward. In this way, roles emerged and shifted as the teaming went on.

Leaders of different subgroupings met routinely every morning and called for additional quick meetings on an as-needed basis. They developed a protocol for transitioning between day and night drill shifts and for routine maintenance of machinery; "We structured, structured, structured all aspects of execution." As drill attempts continued to fail, one after another, Sougarret communicated gracefully with the families. Despite these failures, Sougarret and his new colleagues persevered.

A NASA engineer who went to Chile in late August teamed up with engineers in the Chilean navy to design the rescue capsule, after first going back to the United States to pull together a group of 20 NASA engineers. The engineers developed a twelve-page list of requirements, used by the Chilean navy in the final design for the capsule, called the Fenix. The Fenix interior, just large enough to hold a person, was equipped with a microphone, oxygen, and spring-loaded wheels to roll smoothly against the rock walls.

On October 13, 2010, the Fenix started its life-saving runs to bring miners one by one through the 15-minute journey to safety. Over the next two days, miners were hauled up one by one in the 28-inch-wide escape capsule painted with the red, white, and blue of the Chilean flag. After a few minutes to hug relatives, each was taken for medical evaluation. The resulting fervor of the national – and even global – celebration cannot be underestimated.

How Leadership Enables Extreme Teaming

The Chilean rescue presents a superb example of teaming at its best. Reflecting on the situation, one readily comprehends that a top-down, command-and-control approach would have failed to achieve this stunning outcome. No one person, or even one leadership team, or one organization or agency, could have successfully innovated to solve this problem. It's also clear that simply encouraging everyone to try anything they wanted could have led to chaos or harm. It required extreme teaming. Facing the unprecedented nature of the disaster, multiple temporary, constantly shifting groups of people working separately on different types of problems, and coordinating across groups, as needed, was the only viable approach. These separate efforts – managerial and technical – were intensely focused.

This approach necessitated progressive experimentation, a kind of rapid-cycle learning. Diverse technical experts worked collaboratively to design, test, modify, and abandon options, over and over again, until they got it right. They organized quickly to design and experiment with various solutions, and just as quickly admitted when these had failed. They willingly changed course based on feedback – some obvious (the collapse of the ventilation shaft), some subtle (being told that their measurements were inaccurate by Matthews intruding mid-process with a new technology). Perhaps most important, the engineers did not take repeated failure as evidence that a successful rescue was impossible. Unfortunately, extreme teaming involves risk. And risk necessarily brings both success and failure. Fortunately, there is nearly always much to be learned from the failures to inform next steps.

Finally, the support of senior leadership – not just the technical leadership on the front lines of innovation – was a critical input to the success of this extreme teaming process. Leadership's commitment to the initiative gave others motivation and the protection they needed to take technical and interpersonal risks that are integral to extreme teaming. This turns out to be important in many business organizations where extreme teaming is employed by diverse technical experts to innovate.

As this example demonstrates, extreme teaming can produce awe-inspiring results. The problem is that its success can be too easily thwarted by communication failures at the boundaries between professions, organizations, and industries. As individuals bring diverse expertise, skills, perspectives, and goals together in unique configurations to accomplish challenging goals, they must overcome subtle and not-so-subtle challenges of communicating across boundaries. Some boundaries are obvious – being in different countries with different time zones, for example. Others are subtle, such as when two engineers working for the same company in different facilities unknowingly bring different taken-for-granted assumptions about how to carry out this or that technical procedure to collaboration. To understand the challenge, it's helpful to review how industries operate in business ecosystems.

FROM INDUSTRIES TO ECOSYSTEMS

A new perspective on modern industries as interconnected networks of organizations, technologies, consumers, and products has led to a conceptualization of the activities and relationships taking place within industries as ecosystems (Iansiti & Levien, 2004). The computing industry is perhaps the most dramatic and well-documented example of this interconnectedness, exhibiting an astonishing degree of interaction between organizations bringing modular products and services together to produce value for consumers and business customers. Modularity allows interchangeability so long as relationships between parts are standardized and codified. The essential insight in this work is that the success of many companies is interdependent with (rather than at odds with) that of other firms in an ecosystem. In this way, organizations cannot thrive without a thriving ecosystem of suppliers, customers, and even competitors who spur learning and innovation. A company may thus experience a dramatic and unexpected loss due to failures occurring in suppliers or customers – rather than in their own operations. If the industrial economy was driven by economies of scale, the knowledge economy is increasingly driven by the economics of networks and platforms.

Historically, the way organizations structure activities to promote efficiency tended to encourage specialization. In a review of the historical development of the American legal profession, for example, Abbott (1988) showed that division of labor in large law firms came with a high degree of specialization and led to a dramatic increase in productivity. More generally, specialization occurs through an industrial organization approach that builds on the development of sophisticated contractual arrangements and provides organizations with external economies, that is a decrease in the average costs of doing business (Robertson & Langlois, 1995). Business and industrial activities become subject to more in-depth division and dispersion and organizations are thus prone to identify, cultivate, and exploit the core competencies that make their growth possible (Prahalad & Hamel, 1990). Organizations set out to be leaders in their field, focus on their strong points, and fine-tune them.

However, efficiency gains are not enough to undergird sustainable competitive advantage (Teece, 2014), and pressure is mounting for organizations to develop ways to deal with the increasingly complex problems that come with the constant demands for innovation. This is challenging. So much that executives from powerful multinational companies like GE claim that today's problems are too big for them to solve alone and that to do so they need to collaborate like they never have before (Nagji & Walters, 2012). Successful organizations have become fast, timely expertise integrators for fragmented, scattered, and thinly spread knowledge to be available to the right people in the right place at the right time (David & Foray, 2002).

Thanks to advances in evolutionary economics (e.g., Nelson & Rosenberg, 1993; Nelson & Wright, 1992), our understanding of the economic activity has evolved alongside the phenomena described above. It is gradually moving from a view that focuses on industries and markets to study enterprise performance to one that considers organizations as part of innovation systems, which support the development of knowledge between several actors (e.g., for-profit and nonprofit organizations, governments, universities, research centers, and consumers). These actors are seen as diverse and the relationships between them as well as the institutions influencing them form innovation systems.

From this perspective, research on learning and innovation emphasizes the dynamics of learning by interacting and its co-regulation (e.g., Freeman, 1987; Lundvall, 1988). For instance, Lundvall and his colleagues (e.g., Jensen, Johnson, Lorenz, & Lundvall, 2007; Lundvall, 1992; Lundvall & Johnson, 1994; Lundvall, Johnson, Andersen, & Dalum, 2002) posit that innovation should be regarded as an interactive process where organizations do not learn and innovate in isolation, but in interaction with other actors. Such viewpoint goes beyond the contributions of Arrow's analysis of learning by doing (1962) and of Rosenberg's learning by using (1982) by taking account of several parties interacting together as they seek solutions to complex problems. As reflected in abundant scholarly works that call for more cross-sector interactions (Googins & Rochlin, 2000; Selsky & Parker, 2005; Senge, Smith, Kruschwitz, Laur, & Schley, 2008), three-time Pulitzer-Prize Winner Thomas L. Friedman noted the enormous potential of today's innovation systems:

> *It is now possible for more people than ever to collaborate and compete in real time with more other people on more different kinds of work from more different corners of the planet and on more equal footing than at any previous time in the history of the world. (Friedman, 2006, p. 8)*

The premises of learning by interacting within innovation systems are threefold: (1) organizations are not self-sufficient; (2) they cannot generate all the necessary resources internally, and (3) they must mobilize resources from other entities in their environment if they are to survive (Meeus, Oerlemans, & Hage, 2001). Firms, governments, universities, consumers, etc. can all contribute in a different and interactive way to the innovation process through the cross-fertilization of knowledge around complex problems and approaches to solve them. More or higher interactive learning capabilities enable organizations to reach higher degree of novelty in their ways to solve complex problems and innovate. Organizations therefore need to master complex learning processes that allow them to proactively integrate dispersed knowledge for skills to be continually developed, refined, and updated (Amin & Cohendet, 2004).

This need to go outside organizational boundaries to innovate is now general knowledge among managers, thanks to the pioneering work on *open innovation* (Chesbrough, 2003). Organizations have been exploring and continue to explore the full potential of "purposive inflows and outflows of knowledge to accelerate internal innovation, and expand the markets for external use of innovation" (Chesbrough, 2006b: 1). However, while this literature has developed rapidly in recent years, it has remained mostly focused on contracts and other strategic mechanisms through which organizations may accelerate the generation and commercialization of innovation (e.g., licensing out and licensing in, joint ventures, spin-offs and spin-ins, and acquisitions). Organizational challenges associated with open innovation have been studied extensively but there is a dearth of research at the individual- and team-level of analysis (Vanhaverbeke, Chesbrough, & West, 2014; West, Vanhaverbeke, & Chesbrough, 2006). Understanding factors associated with effective extreme teaming, the topic of this book, expands the possibilities for open innovation, and leverages research on teams and leadership to do so.

By gaining a better understanding of how interorganizational collaboration works at the group-level, we hope to provide much needed support to managers. These activities, for the most part, are extremely challenging (Majchrzak, Jarvenpaa, & Bagherzadeh, 2014). Most managers remain ill equipped to effectively lead extreme teaming endeavors because these collaborations pose different challenges than those that managers typically face when leading teams inside their organization. We argue that gaining a better understanding of team-level activities and leadership is key for organizations to take advantage of the innovation systems in which they find themselves.

INNOVATION IN CROSS-BOUNDARY TEAMS

Cross-boundary teams allow organizations to leverage the potential of innovation systems. While a great variety of knowledge is accessible to organizations, it is the integration of knowledge that fuels innovative capabilities (Grant, 1996), just as happened in the Chilean mining rescue, and research shows that it is integration at the group-level that provides the most benefits (Van Den Bosch, Volberda, & De Boer, 1999). Teams have been described as "the basic building block of any intelligent organization" (Pinchot & Pinchot, 1993, p. 66), "the norm in a learning organization" (Senge, 1994, p. 355), and the unit of organizational learning (Edmondson, 2002). More recent work shows that teams are becoming increasingly common in the production of innovation (Jones, 2009; Wuchty, Jones, & Uzzi, 2007), and that teams are more likely than individuals to develop more innovative solutions (Singh & Fleming, 2010; Uzzi, Mukherjee, Stringer, & Jones, 2013). In a world where organizations are hard pressed to continuously produce innovation to maintain sustainable competitive advantage, the trend toward more dynamic and complex forms of teamwork will continue.

Recombining dispersed heterogeneous bits of knowledge is at the source of radical innovation. John Stuart Mill, two centuries ago, highlighted the creative friction that takes place at the interstices of knowledge domains when individuals interact together:

> *It is hardly possible to overrate the value... of placing human beings in contact with persons dissimilar to themselves, and with modes of thought and action unlike those with which they are familiar... Such communication has always been, and is peculiarly in the present age, one of the primary sources of progress. (Swedberg, 1990, p. 3)*

This quote highlights what takes place at the interstices of knowledge domains. Burt (1992, 2000, 2002) calls these spaces "structural holes" – gap between two discrete groups with nonredundant knowledge. Structural holes act like insulators in an electric circuit: knowledge within each group is buffered and develops within a distinct logic. Innovation emerges from selection and synthesis across the structural holes between groups (Burt, 2004). As a result, the knowledge recombination perspective has become a significant stream of research both in economics (e.g., Boschma, 2005; Cohendet & Llerena, 1997; Nooteboom, 2000) and management (e.g., Fleming, Mingo, & Chen, 2007; Gruber, Harhoff, & Hoisl, 2013; Nerkar, 2003; Nerkar & Paruchuri, 2005).

Newly formed, temporary teams that assemble members from various backgrounds often set the stage for knowledge recombination and innovation. While research has shown benefits of team member familiarity (Edmondson, Bohmer, & Pisano, 2001; Huckman, Staats, & Upton, 2009), studies that focus on creativity and innovation show that when people have worked together many times previously, their projects exhibit less innovation than those with unfamiliar team members (Skilton & Dooley, 2010). Social network theory suggests that highly diverse teams can obtain valuable knowledge from interpersonal relations outside the team. For example, among others (e.g., Guimera, Uzzi, Spiro, & Amara, 2005; Reagans & Zuckerman, 2001), a study of over 2000 artists who worked on 474 creative teams developing new material for Broadway from 1945 to 1989 by Uzzi and Spiro (2005) showed that teams with highly familiar members were less effective than teams with less familiar members, in both artistic and financial terms. Other research shows that when teams have spent a long time working together, their members become cut off from new sources of information, and their thinking starts to converge (Katz, 1982). This blinds them to alternative perspectives that could improve performance (Janis, 1971). Moreover, the longer team members work together, the more likely it is that they will get "stuck in a rut" of habit or routine, and end up exhausting the idea permutations available (Gersick & Hackman, 1990). While routines have value in terms of boosting efficiency and reducing uncertainty over team working approaches, they ultimately impede innovative performance because they limit teams' ability to generate new ideas to solve complex problems (Hirst, Van Knippenberg, Chen, & Sacramento, 2011).

From an interactionist perspective, it is the complex interaction between individuals and their work situations that drives creativity and innovation (Woodman, Sawyer, & Griffin, 1993). "By interacting and sharing tacit and explicit knowledge with others, the individual enhances the capacity to define a situation or problem, and apply his or her knowledge so as to act and specifically solve problems" (Nonaka, Von Krogh, & Voelpel, 2006, p. 1182). For instance, Hargadon and Sutton (1997) showed how designers managed to team across fields (mechanical, industrial, electrical, software engineering, as well as human factors and ergonomics experts) and made original connections between the old solutions of one industry and the new problems of another. As a group of people from eclectic backgrounds, such designers innovated by spanning boundaries and moving knowledge from one place to another – from energy to medical products, to financial services, to the public sector, and more.

In their study of open-ended problem solving, Hargadon and Bechky (2006) showed that management and design consultants relied on moments when people's perspectives and experiences were brought together in work groups to bear on problems rather than relying on the cognitive skills of participating individuals. Kurtzberg and Amabile (2001) also moved away from individuals' creative potential, emphasizing instead how creative minds interact in groups. Many other contributions in the creativity and innovation literature emphasize the benefits of social dynamics that occur during extreme teaming (e.g., Harvey, 2014; Paulus & Yang, 2000). Radical innovation thus occurs when cross-boundary teams make connections between heterogeneous knowledge.

Stories that portray the combinatorial nature of innovation in extreme teaming endeavors increasingly populate the news world. For example, to figure out how electronic devices can be adapted to the needs of people with mobility issues, Samsung brought together a team of people that was able to span a large variety of subfields of human-computer interaction and electrical engineering. That team has so far been able to develop a demo tablet that is controlled by a human brain (Young Rojahn, 2013). Spanning knowledge boundaries and successfully making connections between heterogeneous materials is also at the core of the work of teams in the healthcare and medical sectors: 3-D printing is now used to print teeth-straightening braces and expert knowledge in artificial intelligence and mobile technology is at the foundation of an application that serves as a diagnostic tool (Kotler, 2013).

Extreme teaming efforts are examples of what Nonaka and Konno (1998) termed "ba." Meaning "space" in Japanese, the concept of *ba* denotes a context in which people enter a space of shared emotions that encourages interaction to create new knowledge (Nonaka, Toyama, & Konno, 2000; Von Krogh, Nonaka, & Rechsteiner, 2012). Consistent with a recent stream of literature that has identified the importance of collective forms in the process of invention and innovation (e.g., Drazin, Glynn, & Kazanjian, 1999), a *ba* gives the opportunity to individuals to become active participants in knowledge creation activities, and engage in what Tsoukas has termed "productive dialogue" (2009). Extreme teaming leverages the opportunity for active boundary-crossing dialogue and inquiry that allow people to adjust and reframe their own knowledge, to examine their own perceptions in a different light and reflect on experience to generate ideas and produce innovation (Edmondson, Dillon, & Roloff, 2007). In accordance with our efforts with this book, Nonaka et al. (2006) also posit that research needs to help identify ways in which leaders can develop *Bas* to foster knowledge processes in teams.

CHAPTER TAKEAWAYS

- The so-called knowledge explosion has given rise to narrower and deeper areas of specialization. In this way, fields spawn subfields, and domain knowledge can quickly become obsolete, such that expertise is difficult to develop and to keep current.
- Problems, in turn, do not narrow along with expertise; instead many problems, especially so-called "wicked problems" are multifaceted and complex. Complex problems require tackling many interconnected issues.
- Integrating knowledge across domains is required for solving today's most pressing and complex problems.
- The 2010 Chilean mine rescue is an example of the power of collaboration across knowledge domains. Leadership played an important role in enabling the rescue against overwhelming odds.

- The business world has become more complex, and the need for innovation ever greater. Organizations can rarely focus on a single industry to capture value, and instead must continually collaborate beyond their boundaries. Ecosystems have replaced industries, and organizations must learn to develop knowledge "in the open," acquiring ideas and spinning others out in order to maintain and undergird their competitive advantage.
- Innovation takes place in new extreme teams where heterogeneous knowledge inputs are integrated to produce new and useful products, services, processes or business models. Unfamiliar connections between entities with different perspectives fuel creativity and innovation.

2

LEADING TEAMS AND TEAMING

Leadership has been the focus of scholarly research for many decades. Management and organizational behavior scholars have contributed many articles and books on leadership in organizations. This work has identified several of the individual characteristics and behaviors associated with effective leadership (see Avolio, Walumbwa, & Weber, 2009 for a review). In this book, we are more interested in what leaders do than in the personality traits that they display – traits such as those measured by the Big Five factor model of openness to experience, conscientiousness, extraversion, agreeableness, and neuroticism (Judge, Bono, Ilies, & Gerhardt, 2002) or Myers-Briggs (MBTI) preferences for extroversion, intuition, thinking, and judging (Atwater & Yammarinol, 1993). As we explain below, our work falls in the category of functional leadership.

TEAM LEADERSHIP THEORY

Kozlowski, Mak, and Chao (2016) argued that two mainstream generic leadership theories dominate the management literature: leader–member exchange (LMX) and transformational leadership. Further, two approaches have been explicitly applied to a team setting from the outset: shared leadership and functional leadership.

Leader–Member Exchange

First, LMX is built on dyadic, rather than group-based, relationships (Graen & Uhl-Bien, 1995). The theory is based on the assumption that leaders treat each of their followers differently (Dansereau, Graen, & Haga, 1975), building high-quality relationships with some group members but relying on contractual obligations with the rest (Liden & Graen, 1980). Both types of relationship are regarded as long-term and largely static (Kozlowski, Gully, McHugh, Salas, & Cannon-Bowers, 1996). They affect the current and future performance of the follower, as shown by a recent meta-analysis by Martin, Guillaume, Thomas, Lee, and Epitropaki (2016), and the differentiation of LMX relationships may be a more powerful predictor of team performance than mean level of LMX (e.g., Boies & Howell, 2006). With regard to cross-boundary teams, scholars have suggested that high LMX with all members minimize the drawbacks of diversity (Nishii & Mayer, 2009). Stewart and Johnson (2009), however, found that LMX differentiation had no effect on cross-boundary teams.

Process models have been introduced to account for the development of leader–member relationships (e.g., Graen & Scandura, 1987), but not at the team level. As suggested by various scholars (e.g., Burke, Stagl, Salas, Pierce, & Kendall, 2006; Zaccaro, Heinen, & Shuffler, 2009), while LMX has been used to study individual team members' attitudes, cognition, emotions, and behaviors (see Henderson, Liden, Glibkowski, & Chaudhry, 2009, for a review), "it is not designed theoretically to inform team-level processes and outcomes" (Kozlowski et al., 2016, p. 30). Hence, we cannot tap into this literature to inform our topic of extreme teaming.

Transformational Leadership

Transformational leadership theory, for its part, stems from political science, and considers that leaders can encourage people to surpass their own self-interest for the greater good. It is often presented in contrast to transactional leadership, which emphasizes extrinsic motivations to shape goal setting (Kuhnert & Lewis, 1987). The theory posits four leadership dimensions that positively influence followers through distinct psychological processes: individualized consideration (to attend to the individual needs of others), intellectual stimulation (to challenge assumptions and encourage others to take risks), idealized influence (to encourage others to identify with the leader themselves), and inspirational motivation (to present a vision meant to inspire others) (Bass, 1985; Bass & Avolio, 1995). Other scholars have offered similar operationalizations of the concept (e.g., Conger & Kanungo, 1994; Podsakoff, MacKenzie, Moorman, & Fetter, 1990; Rafferty & Griffin, 2004; Shamir, Zakay, Breinin, & Popper, 1998). In short, transformational leadership theory posits that good leaders seek to meet others' higher-order needs.

While the vast majority of empirical studies following this approach have examined individual-level perceptions of leadership and outcomes (i.e., attitudes and performance), it has been suggested that the two processes of "individualized consideration" and "intellectual stimulation" have relational effects, and that the processes of "idealized influence" and "inspirational motivation" are more likely to influence a collective as a whole (Kark & Shamir, 2013). Thus, some empirical studies conceptualize transformational leadership at the team level (e.g., Jung & Sosik, 2002; Schaubroeck, Lam, & Cha, 2007).

With regard to extreme teaming, scholars suggest that transformational leadership helps transcend differences and stimulate discussion that include divergent viewpoints and ideas, hence supporting the positive effects of knowledge diversity. Kearney and Gebert (2009) found that when it fosters identification with the group and facilitates team-learning processes from diverse team members, transformational leadership has a positive effect on performance-related variables. Similarly, Shin and Zhou (2007) found that a three-way interaction between transformational leadership, educational specialization heterogeneity, and team creativity: high transformational leadership supported teams with greater educational specialization heterogeneity to achieve superior team creativity. More recently, in a study of 68 teams in three Chinese companies, Shin, Kim, Lee, and Bian (2012) found team knowledge diversity to be positively related to individual creativity, but only when transformational leadership was high. Other scholars have looked at closely related categories, and posit that nondiverse and diverse teams might benefit from different leadership styles, the latter performing better under considerate leadership style – concern for team members' well-being and comfort (Homan & Greer, 2013). Similarly, a participative but not a directive leadership style was shown to positively influence team innovation (through more team reflection) in high but not low functionally diverse teams (Somech, 2006).

For the most part, however, these studies simply demonstrate that the processes and outcomes that have historically been examined at the individual level can be generalized to the team level (Kozlowski et al., 2016). Out of the 181 empirical studies on the topic Kozlowski and colleagues examined, 180 represent correlational field studies, and 128 assess variables collected at a single point in time. As lamented by various scholars with regard to team research in general (e.g., Cronin, Weingart, & Todorova, 2011; McGrath, Arrow, & Berdahl, 2000), this indicates a tendency to treat leadership as a static construct rather than a dynamic process, and it provides little insight into the teaming process that is our focus here. Some harsher critics of transformational leadership theory have gone as far as recommending that it be abandoned completely, and that the field forego this label in favor of studying more clearly defined and empirically distinct aspects of leadership. Van Knippenberg and Sitkin (2013), in particular, observe that the body of research on transformational leadership is riddled with serious problems, such as vague conceptual groundings; an unclear causal model for all its dimensions, mediating processes, and outcomes; profound confusion between leadership and its effects; and measurement tools and results that are not consistent with the theory.

Shared Leadership

Third, very much in line with the "power with" perspective of Mary Parker Follett (1940) and the later conceptualization of empowerment (Conger & Kanungo, 1988), the shared leadership approach suggests that leadership can be distributed between team members rather than being concentrated in one appointed person (Pearce & Conger, 2003). Thus, multiple individuals can exercise leadership at any given time. In the study of medical teams in an emergency trauma center by Klein, Ziegert, Knight, and Xiao (2006), for instance, it is shown that leadership roles can be assumed by different team members at various points during the team's lifecycle, depending on the situational demands of the task. The dynamic delegation of leadership proved to engender reliable performance and the development of junior leaders' skills, thus developing the team's capacity.

The view of shared leadership is not new (e.g., Bowers & Seashore, 1966; Gibb, 1954; Katz & Kahn, 1978), and continues to attract interest in recent literature (e.g., Avolio, Sivasubramaniam, Murry, Jung, & Garger, 2003; Carson, Tesluk, & Marrone, 2007; Pearce & Sims, 2002). The benefits of shared leadership are now well documented in studies conducted in a variety of settings, such as new-venture top-management teams (Ensley, Hmieleski, & Pearce, 2006), consulting teams (Carson et al., 2007), virtual teams (Hoch & Kozlowski, 2014), and others. Very recent studies, however, have found that shared leadership relates positively to certain teams' processes, such as team cohesion, but not directly to their performance (e.g., Mathieu, Kukenberger, D'Innocenzo, & Reilly, 2015). This is consistent with Wang, Waldman, & Zhang's meta-analysis of the extant literature on the subject (2014), which shows that shared leadership tends to be more strongly related to team attitudinal outcomes, behavioral processes, and emergent team states, compared with team performance.

Several scholars posit that the concept of shared leadership lacks clarity (Day, Gronn, & Salas, 2004). In particular, the inherent temporality of shared leadership has been highlighted by researchers, suggesting that it is an emergent phenomenon that develops over time (e.g., Carson et al., 2007; Contractor, DeChurch, Carson, Carter, & Keegan, 2012). For our purposes, we focus on the leadership behaviors that support extreme teaming – that is to say, those that can facilitate the development of shared leadership, its propagation across members, and its growth. This is in line with the behaviors presented by Kozlowski et al. (2016) who invite leadership scholars to engage in such investigation. We believe these leadership behaviors should fall into the functional leadership category.

Functional Leadership

Finally, we turn to the approach that we have selected for the book: functional leadership. Functional leadership is generally regarded as the oldest team-centric approach to leadership (Kozlowski et al., 2016), and remains the best approach to study team leadership in the eyes of many prominent scholars of teamwork (e.g., Burke et al., 2006; Morgeson, DeRue, & Karam, 2010). Rather than dwelling on the relationship between leader and follower, it focuses on leader-team interactions, and emphasizes the creation of necessary conditions for effective performance

(Hackman, 2010; Hackman, Walton, & Goodman, 1986; Zaccaro, Rittman, & Marks, 2001). As this view of leadership developed, it became agnostic as to who creates the conditions in question, as long as they are addressed, and has been described as event-driven (Morgeson, 2005) and goal-oriented (Hackman & Wageman, 2009). In other words, it is not necessarily the leader who accomplishes all functions; they may be distributed throughout the team. The leader's responsibility is to make sure the functions are accomplished.

Increasingly, scholars apply taxonomies identifying core team leadership functions to explore how leaders can promote team development. Several scholars have thus proposed general categories of team leadership functions. For example, McGrath (1962) developed a typology of leadership functions arranged in a two-by-two matrix, with the type of activity (monitoring/taking executive action) along one axis and the activity's orientation (internal/external to the group) along the other. McGrath described leadership as social problem solving with the purpose of ensuring that critical tasks are completed. Hackman and Walton (1986) suggested that the team leader's job is to keep an eye on conditions within the team and, when necessary, take action to foster the right conditions for team effectiveness. The five conditions they proposed were: a clear direction of engagement; a facilitating group structure; a supportive context; accessible expert coaching; and adequate material resources. Hackman (2002) later detailed three core conditions (clarity of boundaries, task, membership; compelling direction; and enabling structure with adequate norms and task-relevant resources) and two enabling conditions (supportive organizational context; and expert coaching). Similarly, Mumford and Connelly (1991) saw the leadership role as being about identifying goals, building viable paths to reach these goals, and guiding other people along those paths. Fleishman et al. (1991) put forward a taxonomy of functional leadership that organized activities according to four superordinate dimensions:

information search and structuring, information use in problem solving, managing personnel resources, and managing material resources. The authors also noted that leadership functions generally focused either on (a) the facilitation of group social interaction, and (b) objective task accomplishment.

Subsequent work built on these older taxonomies. Zaccaro and colleagues (2001), for instance, argued that the leader, through Fleishman et al. (1991)'s four leadership functions, facilitates the interaction processes of the team; such processes being essential for solving problems related to the task at hand. Burke and colleagues (2006) suggested a leadership model that updates Hackman's (Hackman, 2002; Hackman & Walton, 1986), wherein the leader's responsibilities to provide core and enabling conditions are integrated with the four leadership functions proposed by Fleishman et al. (1991). The authors also insist further on the two-category classification of task- and person-focused leadership functions.

More recently, Kozlowski, Watola, Jensen, Kim, and Botero (2009) took six classic functional leadership models and organized them into an integrative taxonomy reflecting leadership functions (plan/organize and monitor/act) as they are applied to particular team foci (internal/external and task/social). The authors argued for additional attention to the leadership process (rather than structure) in which each function may be directed internally or externally, and may be task- or social-focused. Finally, Morgeson et al. (2010) carried out an integrative review of functional leadership. They derived a wide-ranging set of 15 functions that aggregate to more than 80 activities, which are organized according to the phase of team lifecycle when they take place (i.e., transition or action; see Marks, Mathieu, & Zaccaro, 2001). They identified seven leadership functions falling within the transition phase: compose the team, define the mission, establish goals and expectations, structure and plan, train and develop, promote sense-making, and provide feedback. They also defined a further eight functions considered as facilitating team action phases: monitor the team, manage team boundaries, challenge members, perform team tasks, solve problems, provide resources, encourage team self-management, and support social climate.

Since functional leadership emphasizes "the cognitive capacities in the generation, selection, and implementation of influence attempts" (Fleishman et al., 1991, p. 259), as opposed to leaders' characteristics, traits, or styles, it is much more hands-on and less abstract than other leadership theories (Mumford, Zaccaro, Harding, Jacobs, & Fleishman, 2000; Zaccaro, Mumford, Connelly, Marks, & Gilbert, 2000). Yet, in part because most empirical studies that take this approach are based on survey-based, cross-sectional designs in stable, and relatively simple settings, we lack a rich account regarding the activities in which leaders engage to tackle cross-boundary teaming challenges. (Table 1 provides four examples of recent empirical research within this stream.)

While we appreciate that there are already plenty of functional leadership taxonomies out there, and that some have been tested in settings with similar characteristics to ours – such as teams entering novel environments (e.g., Marks, Zaccaro, & Mathieu, 2000; Morgeson, 2005) – we argue that we still lack a taxonomy to suit the tough challenges people face when engaging in extreme teaming. In fact, a central premise of the taxonomies developed so far is that leadership's purpose is to "ensure that the group fulfills all critical functions necessary to its own maintenance" (McGrath, 1962, p. 5). In other words, a team's continued ability to work effectively was long considered one dimension of effectiveness. Extreme teaming is about temporary projects, not permanent ones, and thereby falls outside the domain of traditional models of team effectiveness.

The leadership literature in general has neglected functions associated with leading projects that span multiple organizations, typically examining leadership within organizations instead. Most

Table 1. Four Recent Empirical Studies on Functional Leadership in Teams.

Authors	Study Details
Benoliel and Somech (2015)	*Team Context* 92 management teams from 92 public educational organizations in Israel *Leadership Functions* Internal Functions: ○ Relating: *How often the leader shows team members that he or she cares about them* ○ Scouting: *How often the leader gathers information about the needs and problems of the team* ○ Persuading: *How often the leader conveys information to the team about how their work helps the organization to achieve its goals* ○ Empowering: *How often the leader delegates responsibility and decision-making authority* External Functions: ○ Relating: *How well the leader understands school politics* ○ Scouting: *How often the leader seeks information outside the team* ○ Persuading: *How often the leader acts as a representative of the team* *Methods* Multisource Survey Design ○ Team members rated leaders' internal and external activities and interteam goal interdependence ○ Team leaders rated team effectiveness, team in-role performance, and team innovation

	Main Findings ○ Team functional heterogeneity and interteam goal interdependence explained 51% of the variance in internal activities ○ Interteam goal interdependence explained 34% of the variance in external activities ○ External activities explained 13% of the variance in team innovation ○ Internal activities explained 16% of the variance in team in-role performance ○ Leaders' external activities fully mediate relationship between interteam goal interdependence and team innovation
DeRue, Barnes, and Morgeson (2010)	*Team Context* 80 five-person teams of upper-level undergraduate students enrolled in an introductory management course at a large Midwestern university (four members; one leader) *Leadership Functions* ○ Coaching: Encourage team to manage itself, develop capacity to function without direct intervention from leader ○ Directive: Actively intervening in team, setting clear expectations and goals, monitoring team performance *Methods* Experimental and Survey-Based Design: ○ Dynamic, networked, military command-and-control simulation ○ 40 team leaders instructed to adopt coaching style; 40 others to adopt directive style ○ Team members rating their team leader's behavior, charisma, and their own feeling of self-efficacy *Main Findings* ○ Manipulation of team leadership style had no significant effect on team member effort or team performance ○ Coaching leadership was more effective than directive leadership when the leader had high charisma ○ Directive leadership was more effective when the leader has low charisma

Table 1. (*Continued*)

Authors	Study Details
Luciano, Mathieu, and Ruddy (2014)	*Team Context* 101 3-to-11-member customer service, self-managed teams of technicians reporting to 25 formally designated external leader (all employed by a major office equipment and technology company in four regions of the United States) *Leadership Functions* ○ Promote teamwork ○ Facilitate task work ○ Align efforts with broader organizational goals ○ Demonstrate consideration for employees *Methods* Survey-based design collected from individual team members assessing: ○ Member job satisfaction ○ Team performance ○ Team empowerment ○ Average external leadership (general behaviors that an external leader exhibits across teams) ○ Relative external leadership (variance in scores for one team in comparison to other teams reporting to the same leader)

	Main Findings ○ Both relative and average external leadership related positively (indirectly) to team empowerment, which in turn related positively to team performance and member job satisfaction ○ Relative external leadership was not directly related to performance ○ Average external leadership had negative direct influence on team performance ○ Relative external leadership was positively related to job satisfaction, average external leadership was not
Mohammed and Nadkarni (2011)	*Team Context* 71 teams in a business process outsourcing firm in India *Leadership Functions* ○ Temporal leadership: Scheduling deadlines, synchronizing team member behaviors, and allocating temporal resources *Methods* Time-lagged, survey-based design measuring: ○ Temporal individual differences ○ Diversity of temporal individual differences ○ Team temporal leadership ○ Team performance *Main Findings* ○ Influence of time urgency diversity and pacing style diversity on performance was more positive under conditions of stronger (as opposed to weaker) team temporal leadership ○ Team temporal leadership had a direct, positive influence on team performance

leadership theories, not surprisingly, were developed in contexts that did not involve multiple groups or organizations working together (Hogg, van Knippenberg, & Rast, 2012; Huxham & Vangen, 2000). Yet, such contexts will only gain importance as organizations continue to try to benefit from a strong innovation system by assembling diverse groups of people in new, temporary, and hyper-fluid teams. We believe it is now paramount to delineate the functions that are necessary to facilitate extreme teaming effectiveness in complex problem situations, and the specific ways in which leaders can fulfill each of the functions.

In their review of seminal leadership articles that appeared in *Journal of Applied Psychology* over the past century, Ford, Day, Zaccaro, Avolio, and Eagly (2017) also insist on the need to start comparing leadership functions in contemporary organizational forms with more traditional ones. With this in mind, we situate our work within the functional leadership tradition, and hope to characterize leadership functions for the extreme teaming challenges that we focus on in this book.

TOWARD A MODEL OF LEADERSHIP FOR EXTREME TEAMING

The broad array of leadership functions presented in the collection of taxonomies that have been produced throughout the years has been developed at the cost of precision regarding how leaders may improve a team's performance and effectiveness (Kozlowski et al., 1996; Zaccaro & Klimoski, 2002). To the best of our knowledge, none of the taxonomies is centered on the extreme teaming context, especially because little research has focused on how leaders help teams generate new solutions through complex knowledge integration (Mainemelis, Kark, & Epitropaki, 2015). In fact, the taxonomies of team leadership functions produced to date are usually "general" taxonomies that neglect the unique features of teams

(Kozlowski & Ilgen, 2006; Kozlowski et al., 2016). According to van Ginkel and van Knippenberg (2012), leadership research has largely neglected the role of leadership in cultivating teams' understanding of their work, the setting in which they work, and what it takes to become an effective team in that setting.

This is not particularly surprising. A context-free approach to research prevails in the field of organizational behavior more generally (Johns, 2006; Rousseau & Fried, 2001) and in team research in particular (Day, Gronn, & Salas, 2006; Edmondson et al., 2007; Hackman, 2012; Joshi & Roh, 2009; Mathieu, Maynard, Rapp, & Gilson, 2008; Stewart, 2010). Generally, scholars of teamwork provide rich and substantial details on internal, team-level features such as team composition, while treating external, higher-level features much more cursorily (Maloney, Bresman, Zellmer-Bruhn, & Beaver, 2016). In light of this, Edmondson et al. (2007) urged team researchers to venture out into the field to develop more contextually specific theories that take into account the particular challenges associated with a team's setting. Indeed, despite a handful of studies (e.g., Schippers, West, & Dawson, 2015), it is surprising to see such a central aspect of teamwork and innovation being left out of the equation so often. The setting in which a team works, such as that created by the various constraints it faces, can have an important influence on team processes and outcomes, as shown by recent research (e.g., Rosso, 2014; Young-Hyman, 2016). More precise taxonomies could therefore help address team leadership requirements for specific contexts by focusing on how leaders build and shape team development (Kozlowski et al., 2009).

Much of the work on leadership has not presented leadership as a practical activity in a complex setting (Denis, Langley, & Rouleau, 2010). Zaccaro et al. (2001) posit that despite all the research that has been done on leadership, we still know remarkably little about *how* leaders create and handle effective teams. For instance, our knowledge of how leaders make timely uses of objects, or how they enact certain leadership functions to deal with extreme teaming challenges, remains rather limited. Many of the taxonomies that have been developed are not accompanied by empirical data that would support the leadership functions put forward. As Kozlowski and his colleagues (2016, p. 40) argue: "It is ironic that functional leadership is the oldest approach to team leadership and has a substantial amount of theoretical diversity, but has the least amount of empirical research in comparison to transformational leadership, LMX, and shared leadership."

Several other scholars have also called for greater attention to leadership activities (Carroll, Levy, & Richmond, 2008; Denis, Langley, & Sergi, 2012; Fligstein, 2001). They posit that we need to shed light on the day-to-day activities that are at the foundation of leadership practice. Leadership may become manifest through mundane activities in which managers engage such as listening and chatting with colleagues and subordinates (Alvesson & Sveningsson, 2003). Additional micro-level focus must be put on situated activities and interactions and their consequences in the, here and now. Therefore, the goal of this book is to shed light on leadership activities that amount to leadership functions for extreme teaming.

CHAPTER TAKEAWAYS

- Leadership has been studied in multiple ways in the organizational literatures. Some research looks at leaders' traits, while other research focuses on leaders' behaviors.
- Most of the literature on team leadership falls into four theoretical streams: (a) leader–member exchange, which looks at the relationship leaders develop with each of their team members; (b) transformational leadership, which covers four processes that leaders enact to push team members to surpass themselves for the team's purpose; (c) shared leadership, which gives way to leadership distribution among team members depending on the demands of the tasks; (d) functional leadership, which

focuses on leader-team interactions and emphasizes the creation of necessary conditions for effective performance.

- This book takes a functional leadership approach, because of the need to build enabling conditions for extreme teaming, to overcome its unusual challenges.
- Numerous taxonomies of team leadership functions have been developed over many years. These are often classified into two categorical dimensions: leadership functions that are task- or person-focused, and leadership functions that are internal or external to the group.
- Most of the taxonomies that have been developed to date apply to stable, well-bounded teams. They are typically presented as context-free, meaning that they can be applied to any kind of work group or team.
- Building on the work of other scholars, we argue for situated taxonomies to specify leadership functions to support the diversity of teaming endeavors present in today's business world. Extreme teaming, in particular, is not well represented in prior taxonomies in the team leadership literature.
- Beyond leadership functions, it is also necessary to dig deeper to identify practical activities in which leaders engage to enact the functions that support effective performance.

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THE CHALLENGES OF EXTREME TEAMING

Chapter 1 described forces that have given rise to the need for cross-sector collaboration and explored the potential for radical innovation this way of working brings. Chapter 2 reviewed the literature on team leadership and insisted on the need for situated approaches to study leadership functions that match the unique features of extreme teaming. This chapter explores the challenges created by these features, that is, the challenges that lie ahead when diverse individuals and organizations engage in extreme teaming. Reviewing the literature on how teams, in general, develop over time, we set the stage for this way of working that extends beyond the constraints of ordinary teamwork.

HOW TEAMS DEVELOP OVER TIME

Much research has explored the way groups develop over time, and it seems there is general agreement on the dynamic, multifaceted nature of team development and effectiveness. Theoretical models generally speak of teams maturing through multiple stages that are either sequential (Bennis & Shepard, 1956; Tuckman, 1965) or nonsequential (Gersick, 1988; McGrath, 1991). Researchers in the sequential tradition propose unitary development paths that teams follow over their lifetimes, while researchers in the nonsequential stream focus on describing the factors behind pivotal changes in team development. However, the two streams are not incompatible, and they have been brought together in some prior work (e.g., Morgan, Salas, & Glickman, 1994). Both generally recognize the complexity and unpredictability of team development: Some teams make slow or faltering progress; not all evolve at the same speed; and some never mature at all. Many teams exhibit mid-point corrections (Gersick, 1989, 1991). All these events can transpire without contradicting other team-development models.

Most previous work on team effectiveness has owed something to the input-process-output (IPO) heuristic first put forward by McGrath (1964). Recent frameworks continue the trend, but have more to say about the inherent dynamics of IPO. For instance, Kozlowski and colleagues (Kozlowski et al., 1996; Kozlowski, Gully, Nason, & Smith, 1999), and Marks et al. (2001) pointed out the cyclical, episodic nature of IPO linkages. In particular, the temporally based model put forward by Marks et al. (2001) highlights the dynamics that underpin team effectiveness, including the interrelations between (a) emergent affective, cognitive, or motivational states such as team members' attitudes, values, cognitions, and motivation; and (b) team processes, which comprise team members' interactions with other members and their task environment as manifested in cognitive, verbal, and behavioral activities. Other frameworks emphasize the feedback loop that links team outputs and later inputs (Alexander & Van Knippenberg, 2014; Burke et al., 2006; Ilgen, Hollenbeck, Johnson, & Jundt, 2005; Mathieu et al., 2008).

There are several collective states that matter for teams to become effective. For instance, both team cohesion (Beal, Cohen, Burke, & McLendon, 2003; Webber & Donahue, 2001) and team mental models (Klimoski & Mohammed, 1994; Mathieu, Heffner, Goodwin, Salas, & Cannon-Bowers, 2000) play vital roles in supporting team performance. Team cohesion represents the commitment of team members to the team's goal and process to reach that goal, and team mental models comprise understandings shared by team members on task requirements, team procedures, and role responsibilities. Psychological safety, or the "shared belief held by members of a team that the team is safe for interpersonal risk taking" (Edmondson, 1999, p. 350), and team empowerment, or the team's "collective belief that they have the authority to control their proximal work environment and are responsible for their team's functioning" (Mathieu, Gilson, & Ruddy, 2006, p. 98), constitute additional examples of collective states generated by the shared team experience. Ultimately, these emergent states are the enablers for teams to develop and aim for specific goals (Hackman, 1990), as well as norms and routines that make them efficient at pursuing these goals (Gersick & Hackman, 1990). We do not know much, however, about how leaders can nurture the emergence of collective states, which further illustrates the great need for research and further practical understanding of the teaming process.

Overall, a newly formed team's states emerge as it develops through its members' interpersonal interactions. These states remain in flux throughout the entire teaming process, and can either support or impede team performance. Some benefits are available when those involved in the teaming effort originate from diverse backgrounds, yet working across diverse backgrounds also presents process challenges.

CONDITIONAL BENEFITS OF EXTREME TEAMING

Diversity, or "differences between individuals on any attribute that might lead to the perception that another person is different from self" (Van Knippenberg, De Dreu, & Homan, 2004, p. 1008), represents an important theme in research on teams. Two broad categories of attributes are at the roots of diversity in the literature: surface-level attributes, or readily detectable differences such as gender, age, and ethnicity; and deep-level attributes, or less visible underlying differences related to knowledge and work, such as occupational or organization affiliation (Harrison, Price, & Bell, 1998). In this book, we focus on the effects of deep-level attributes on teaming, which we term "knowledge diversity." It provides useful insights that inform the topic of extreme teaming. Numerous team-diversity studies emphasize the advantages of teams that encompass a range of distinct task-relevant resources. The premise is that teams expand their knowledge resources by bringing together diverse individuals, because each one offers a different set of ideas and perspectives; bringing these inventories of expertise together makes them all available to the team's work. However, meta-analyses and reviews of the team-diversity literature have shown weak or inconsistent support for the assumption that the presence of knowledge diversity necessarily leads to better performance outcomes (see Bell, Villado, Lukasik, Belau, & Briggs, 2011 for a meta-analysis; see Guillaume, Dawson, Otaye-Ebede, Woods, & West, 2015 for a review).

Early research assumed that knowledge diversity would be good for performance, but bad for interpersonal relationships (see Williams & O'Reilly, 1998 for a review). One influential study (Jehn, Northcraft, & Neale, 1999) posited that knowledge-based diversity (i.e., functional or educational background) could stimulate performance thanks to task-related conflict between team members, while demographic diversity (i.e., race, age, and gender) would lead to relational disruption within the team and negatively impact performance. In a sense, this work recognized the interpersonal discomfort that such diversity can bring, while assuming the diverse inputs could nonetheless be put to good use in accomplishing the team's work. Later evidence, however, showed that knowledge diversity sometimes had no effect on performance and other times even showed negative performance effects (Bowers, Pharmer, & Salas, 2000; Webber & Donahue, 2001).

The opportunity to integrate diverse knowledge offers the potential for complex problem solving, but achieving this potential is not a given. Diverse knowledge is often underutilized (Austin, 2003; Griffith & Sawyer, 2010) and its integration remains elusive (Dahlin, Weingart, & Hinds, 2005; Harvey, 2013; Srikanth, Harvey, & Peterson, 2016). For instance, diverse teams face the "common knowledge effect" – that is, the tendency of teams possessing diverse information to inadvertently spend their time discussing information that is common to all members, rather than identifying and using members' unique information (Gigone & Hastie, 1993). Indeed, laboratory experiments carried out by Gerald Stasser and his colleagues consistently showed that team members tend to discuss common rather than unique knowledge, even if that unique knowledge is essential to the team task (Stasser, Taylor, & Hanna, 1989; Stasser & Titus, 1985; Stewart & Stasser, 1995). Homan, Van Knippenberg, Van Kleef, and De Dreu (2007) coded the conversations held within such experimental groups to show that knowledge diversity was only associated with increased elaboration of information in groups where the members valued diversity, meaning that they consciously searched out new information and actively listened to others' views.

Scholars now insist that there is not "good" or "bad" diversity (Van Knippenberg et al., 2004). Its effect depends on how performance is measured (Van Dijk, Van Engen, & Van Knippenberg, 2012), but most importantly if the right team processes are in place to influence the relationship between any kind of diversity and performance (Mannix & Neale, 2005; Van Knippenberg & Schippers, 2007). For example, Drach-Zahavy and Somech's study of 48 administrative teams (2001) showed that knowledge diversity influenced team innovation by affecting three team processes: exchanging information, learning, and negotiating. Similarly, a study of top management teams from 57 manufacturing companies found that knowledge diversity was associated with *lower* performance, if the teams did not foster the internal debate that would leverage the positive aspects of diversity (Simons, Pelled, & Smith, 1999).

Today, these key processes to leverage diversity continue to relate to the exchange, discussion, and integration of task-relevant information (Van Knippenberg & Mell, 2016), which are closely related to team learning (see Edmondson et al., 2007 for a review). In short, knowledge diversity in and of itself does not necessarily produce performance benefits, unless enabling conditions such as deliberate process choices are also present. Only then can knowledge be most productively used.

The processes through which teams can benefit from members' diverse knowledge can be thought of as team-learning behaviors; concretely, these include "asking questions, seeking feedback, experimenting, reflecting on results, and discussing errors or unexpected outcomes of actions" (Edmondson, 1999, p. 353). Subsequent research has studied one or more of these behaviors, including going over problems and mistakes (Carmeli & Gittell, 2009), and discussing team goals, processes, or outcomes (Schippers, Edmondson, & West, 2014). In general, these studies show a relationship between these types of team-learning behaviors and team performance.

More specifically in the context of cross-boundary teaming, several studies have shown that effective team-member interaction across boundaries can drive team performance, helping teams solve a complex problem or innovate with a successful new product or service. For example, Fay, Borrill, Amir, Haward, and West (2006) surveyed 66 multidisciplinary, breast care teams and 95 primary healthcare teams and found that team processes, notably interaction frequency, had a positive effect on innovation. In another study of 224 corporate R&D teams, Reagans and Zuckerman (2001) found that exchanges between individuals with diverse knowledge attributes was crucial to maximizing teams' performance, while Harvey, Cohendet, Simon, and Borzillo (2015) found that videogame development teams (scriptwriters, designers, artists, and programmers) would produce blockbuster games by bringing their members' "extra-curricular" expertise into the project (salsa dance, graffiti or textile art, extreme sports, snowboarding and skateboarding, etc.).

Not surprisingly, we observe that achieving the benefits of knowledge diversity is more difficult in extreme teaming than in traditional teams. Most team-diversity studies typically examine the effects of knowledge diversity in bounded and reasonably stable teams, rather than focusing on individuals collaborating in transient teams or in fluid collaboration arrangements. We are just starting to see studies of people working in fleeting team-like arrangements (e.g., Mortensen, 2014; Valentine & Edmondson, 2014), hence team-diversity research has not examined the process through which a group of diverse individuals become a team that is fully prepared to solve a new and complex problem and achieve radical innovation. There are some challenges that may very well make it harder to tap into the conditional benefits of knowledge diversity during extreme teaming efforts.

EXTREME TEAMING CHALLENGES

When organizations convene groups of individuals with diverse backgrounds to develop a new product or service or solve a complex problem, the challenges of teamwork are particularly intense (Edmondson & Nembhard, 2009). There are challenges that pertain to the feelings and emotions at play in the newly formed relationships between cross-boundary project participants, and another set of challenges concerns the relatively difficult integration of diverse knowledge and skills.

Interpersonal Challenges: Emotions and Relationships

Extreme teaming means that individuals who would not normally associate with each other have to develop into a cohesive group; that is, birds of a different feather must suddenly flock together.

Several theories have introduced interpersonal reasons explaining why this is challenging. For instance, social identity theory (Tajfel, 1978) explains intergroup behaviors in relation to the discrimination of in-group members ("us") against out-group members ("them"), while social categorization theory (Turner, 1982) emphasizes individuals' tendency to categorize themselves into certain groups that share a series of values and interpretations. Similarity-attraction theory (Byrne, 1971) is another explanation that posits that people like and are attracted to others who are similar, rather than dissimilar, to themselves.

Drawing from these theories, numerous empirical studies have shown that unclear social identity or multiple social identities are detrimental to team cohesion (Webber & Donahue, 2001). This is important for innovation because individuals are more willing to take risks when they feel they can anticipate others' reactions, which allows them to develop trustworthy bonds with them (Williams, 2001). For instance, with higher perceptions of trust, cross-boundary team members will be more willing to (1) offer useful knowledge to another team member (Andrews & Delahaye, 2000), and (2) listen to and absorb another team member's knowledge – for instance, by experimenting with new knowledge (Mayer, Davis, & Schoorman, 1995). In contrast, when individuals do not know what to expect from other team members, it is more difficult to develop a psychologically safe environment where they are not concerned about interpersonal risks such as being ridiculed, punished, or rejected by others when engaging in learning behaviors, including asking for help, admitting errors, and seeking feedback (Edmondson, 1999). The more confident team members are that the others on the team will not embarrass, reject, or punish someone for speaking up, the more willing they are to take interpersonal risks such as sharing knowledge that can benefit the team by preventing problems or contributing to its common goal – knowledge that might otherwise be withheld (Edmondson, 2004; Kasl, Marsick, & Dechant, 1997).

Concerns about interpersonal risk affect a team's ability to produce creative ideas (Harvey, 2014; Van de Ven, 1986). Research indicates that when individuals first join a team, their propensity to influence others, and share their opinions with them, influences the team's ability to create divergent ideas (De Dreu & West, 2001; Gruenfeld, Martorana, & Fan, 2000). Specifically, minority dissent, which arises when group members who have a more unusual perspective than others choose to express their view, brings more perspectives to a team, and forces those with a dominant opinion to consider alternative solutions to problems (Nemeth, 1986; Nemeth & Kwan, 1987). This often helps teams generate more divergent, original, and complex ideas (Farh, Lee, & Farh, 2010; Gruenfeld, Mannix, Williams, & Neale, 1996). As a result, while extreme teaming efforts may have more perspectives than more stable, rigidly bounded teams, this broader range of knowledge is of little use if it cannot be leveraged via the adoption of learning behaviors by team members.

In a similar fashion, collaboration between the new members of a team has similar effects on the innovation process as does collaboration between diverse members. While new members are more likely to create divergent or novel idea combinations, which should promote more innovative outcomes, much of this success is contingent upon how far teams utilize knowledge from new members (Rink, Kane, Ellemers, & Van der Vegt, 2013).

Fear of interpersonal risk not only affects a team's ability to generate ideas, but also makes it more difficult to adequately coordinate work and implement new ideas (Mumford & Gustafson, 1988; West, 1990). Previous research makes it clear that a team's understanding of the world and its ability to sense changes and adapt its activity according to the knowledge acquired depend on the relationships between its members (Edmondson et al., 2007). Without regular, high-quality communication between team members, those engaged in extreme teaming endeavors will struggle to develop the emergent states necessary for coordination and adaptation, such as shared mental models and efficient transactive memory (Burke et al., 2006).

Decreased amount of exchanges between team members reduces each team member's respective capability to locate and solicit the knowledge available from all members (Faraj & Sproull, 2000), and such teams will struggle to promote shared meaning and adjust its course through reflective practices (Edmondson et al., 2001). Besides, these teams will lack in relational coordination – defined as "a mutually reinforcing process of interaction between communication and relationships carried out for the purpose of task integration" (Gittell, 2002, p. 301). Essential for complex, ongoing work operations like hospitals and airlines to better anticipate task interdependence (Kellogg, Orlikowski, & Yates, 2006; Wageman, 1995), such coordination comprises informal/on-the-spot/unplanned communication. Without it, decision-making is slowed down, impeding execution and reactivity (McDonough, 2000; Wheelwright & Clark, 1992). Task conflict also tends to become too difficult to reconcile and extremely frustrating for those involved, leading to additional relationship conflict that ultimately hinders team cohesion (Simons & Peterson, 2000), and performance (De Dreu, 2006; Hülsheger, Anderson, & Salgado, 2009). In such cases, team members tend to lose interest in the team's mission, feel less part of the overall endeavor, and make more errors (Hoegl, Weinkauf, & Gemuenden, 2004).

Managing emotions and relationships for collaboration is thus crucial to the success of extreme teaming. Doing so effectively, we argue, sets the stage for overcoming the technical challenges that lie ahead.

Technical Challenges: Knowledge and Skills

The technical challenges of extreme teaming relate to integrating diverse knowledge and skills. Knowledge is socially embedded and context-dependent (Brown & Duguid, 2001), such that it can be difficult to transfer knowledge developed in a previous context and simply apply it to a new one (Carlile, 2002, 2004). The more people engage in a particular context, solving problems, interacting with peers, and producing artifacts, the more robust the system of interpretation they develop. As a result, crossing knowledge boundaries involves dealing with viewpoints and interests that are often taken-for-granted.

Most people take the norms and values within their own professions, organizations, or industries for granted, sharing a set of largely unquestioned assumptions (Cronin & Weingart, 2007). When tackling new problems or operating in novel settings, those involved in extreme teaming endeavors must inevitably negotiate uncertainty regarding other project participants' assumptions and how these relate to their own assumptions (Skilton & Dooley, 2010). Extreme teaming may only succeed if these "technical challenges" are managed, allowing project participants to communicate efficiently with one another and to build on each other's knowledge and skills to generate new insights.

Individuals with completely different perspectives face serious communication problems when they interact together. When there is too much distance between team members' knowledge, it is particularly difficult for them to learn from each other because they have trouble recognizing the value of each member's input (Gardner, Gino, & Staats, 2012). As a result, they tend to become reluctant to invest time and energy to share their point of view. For instance, in an inductive study of six product development teams in three different industries, Seidel and O'Mahony (2014) showed that technical challenges lead to disunity within teams if left unmanaged.

Scholars have found that knowledge and action are interconnected through people's work practices (Gherardi, 2000) and situated, or localized, in the work context (Sole & Edmondson, 2002), which makes it difficult to move knowledge from one context to another. As Orlikowski (2002) points out in her study of high-tech product development, skillful practice is not based on experts' application of a priori domain knowledge, but instead emerges from practitioners' ongoing and situated actions as they engage with their environment. This emergent and collaborative process through which the participants ought to engage in transforming their current knowledge into new knowledge that fuels the transformation of others' knowledge can be extremely challenging.

Technical challenges can run deeper than language and interpretation differences, relating to people's interests (Carlile, 2004). Practitioners from significantly different domains follow distinctive rationalities, which have been described as "regimes of worth" (Boltanski & Thévenot, 2006) and "principles of evaluation" (Stark, 2011). People may strive to attempt different goals that their profession, organization, or industry encourages. For instance, NGOs and corporations, for instance, tend to bind individuals to diverging interests (Battilana & Dorado, 2010). In terms of professions, Ritti (1968)'s study at the development laboratories of a large electronics company is a good illustration of deep technical challenges. The nearly 5,000 engineers he surveyed showed their desire for technical excellence to be generally in harmony with business goals. However, scientific output was the least of their concerns, which clearly clashed with their scientist counterparts. Other research also points to conflicting interests between science and business innovation (e.g., Gittelman & Kogut, 2003). An implication of these findings is that scientists and engineers entering the same project will perceive their role and evaluate their efficacy differently, but also see the appropriate goals and routines of the team in accordance with their respective interests.

Hence, team-member interactions often must include representations for cross-boundary team members to broaden understanding of the problem at hand, and to find and adapt approaches to solving it (Edmondson & Harvey, 2017). This sort of adaptation has been described in several other field studies, including a yearlong ethnography of technicians, engineers, and assemblers on the production floor of a Silicon Valley semiconductor manufacturer. In this study, Bechky (2003a) noticed that intergroup differences were rooted in products' conceptualization, as well as in their production process. Members of different groups had to co-create common ground, so that each could understand how other groups' knowledge fit into their own context, thus developing a collective mental model. As an illustration, Bechky noticed an assembler showing a product's physical parts to an engineer to demonstrate a problem. The two experts could then link the physical production process (assemblers' practice) with the conceptual one (engineers' practice), and identify a solution that considered, and accommodated, the multiple perspectives involved. In such a situation, simply trying to transfer knowledge from one group to another, without such a process of deep engagement, would not have worked. As a result, those participating in extreme teaming efforts must do more than engage in mere information sharing or task conflict; they must engage in perspective taking (Boland & Tenkasi, 1995; Hoever, Van Knippenberg, Van Ginkel, & Barkema, 2012). In other words, technical challenges will have to be addressed for extreme teaming endeavors to succeed.

CAN LEADERSHIP ADDRESS THESE CHALLENGES?

Research suggests that leadership has an important role to play in helping people engaged in extreme teaming overcome the interpersonal and technical challenges they face. Those who lead innovation endeavors often have a high-level view of team processes, task environments, and team objectives (Morgeson et al., 2010), and are in a unique position to provide assistance. Exploring those leadership practices is thus of great importance (Lord, Day, Zaccaro, Avolio, & Eagly, 2017).

Emergent states and team-member interactions form part of a reciprocal pattern. In previous work, we suggested that team members' cross-boundary exchanges allow emergent states to be adjusted and reframed (Edmondson & Harvey, 2017). We believe this process can be managed with specific leadership functions. As team members work across boundaries, they have the chance to reevaluate their own perceptions and reflect on the project or their own approach to it. This also influences the emergence and development of cognitive states, such as transactive memory and shared mental models, or emotional states, such as psychological safety and team cohesion. As Marks et al. (2001) have proposed, and Waller, Okhuysen, and Saghafian (2016) further emphasized, emergent states are inherently dynamic, and vary as a function of the behaviors that team members adopt – and, we would add, the representation or objects that they use.

Previous findings hint at the leadership functions that matter for extreme teaming, because they have been shown to address the interpersonal or technical challenges of teamwork. For instance, by reinforcing the kind of behavior they expect from team members, providing feedback on whether members have met these expectations, and rewarding those who do, leaders may convey certain messages with regard to emergent states, such as goal priorities (Dragoni & Kuenzi, 2012). Coordination of information is another leadership function that gains importance when task demands exceed the expertise of any single member of a team (Mumford, Zaccaro, Connelly, & Marks, 2000), such as in situations of extreme teaming.

Leaders also influence team-member interaction and other emergent states. Team members tend to notice the behavior of the leader (Tyler & Lind, 1992), such that his or her responses to team members speaking up either help creating an atmosphere of psychological safety or damage it. Studies have shown that people are more likely to take interpersonal risks within their team if they see the leader as someone who is available and approachable (Edmondson, 1996), who invites input and feedback, and models openness and fallibility (Nembhard & Edmondson, 2006). In their study of 43 cross-functional new product teams, Lovelace, Shapiro, and Weingart (2001) found that leader effectiveness mitigated the challenges of task disagreement and helped team members to express task-related doubts, in addition to directly affecting innovativeness. Leadership effectiveness was measured by behaviors such as encouraging individual initiative, clarifying individual responsibilities, maintaining a strong task orientation, emphasizing group relationships, and demonstrating trust in organizational members.

Recent work moves beyond diversity climate and focus on diversity mindsets instead (Van Knippenberg, van Ginkel, & Homan, 2013). Diversity mindsets clarify diversity-related goals and procedures how to achieve these goals, the main goals being the elimination of intergroup bias and facilitation of team learning. These mindsets are thought to positively moderate the effects of knowledge diversity when they are accurate, shared, and when there is awareness of sharedness. As such, diversity mindsets might also be an effective means to reduce social uncertainty in diverse work groups by clarifying interpersonal conduct. The literature would suggest that leadership functions that support extreme teaming address these questions.

Moreover, depending on the thickness of the technical challenges they face, previous research has shown that those involved in extreme teaming must engage in relatively deep conversations to create common ground and shared meaning but also make use of knowledge representations, such as objects, stories, and metaphors that help practitioners traverse knowledge boundaries (Okhuysen & Bechky, 2009). For instance, "boundary objects" such as diagrams, drawings, and blueprints, have been shown to facilitate cross-boundary teaming (Henderson, 1991, 1999; Star & Griesemer, 1989), as have prototypes and models (Bechky, 2003a; Leonard-Barton, 1995; Wheelwright & Clark, 1992). Those representations that are sufficiently stable to circulate and be recognized across boundaries, yet flexible enough to adapt to the various interpretations or interests, have been shown to reduce technical challenges by promoting comradeship (Bechky, 2003b; Kellogg et al., 2006) and facilitating the sharing of diverse knowledge to mobilize action (Bechky, 2003a; Carlile, 2002). Yet, if left unmanaged, such representations can also have the unintended negative effect of making differences between groups more salient without providing the desired and necessary common ground to bridge them (Ewenstein & Whyte, 2009). Some research suggests that these challenges can be managed with the help of a leader who oversees team activities (Ancona & Caldwell, 1992; Brown & Eisenhardt, 1995; Levina & Vaast, 2005), but not much is offered with regard to such a function.

Leaders can influence team-member behaviors and introduce certain objects at key times during the teaming process. At the point when they enter a new setting, team members may have only the most cursory understanding of collective states. To change this, leaders may act in ways that facilitate conversation with other project participants, which they can then all use to glean further insights, further cultivate certain states, and so on. Interacting with other team members and reflecting on progress are examples of activities that can help to clarify roles, instill a sense of belonging, and boost self-efficacy. They can also help clarify the team's goals, norms, and routines, which in turn shape collective states. While this all sounds very challenging, we believe that there are leadership functions that can contribute toward the effectiveness and success of extreme teaming.

A NOTE ON METHODS

Research Approach

Our review of prior research suggests that theory on how team leaders support extreme teaming is limited. We thus use a qualitative multiple-case study approach (Eisenhardt, 1989) to induce contextualization, vivid description, and an appreciation of subjective views (Lee, 1999; Locke, 2001). We base our methodological decision on the fact that the replication implicit in systematically comparing and contrasting features of multiple cases increases confidence in emergent theory (Eisenhardt, 1989; Eisenhardt & Graebner, 2007; Yin, 2009). This approach, which involves eliciting novel elements of theory from contextualized patterns of behavior, is amenable to grounded theory methods (Glaser & Strauss, 1967; Strauss & Corbin, 1998). It is particularly well suited to answering the "how" element of our research question (Edmondson & McManus, 2007), that is, "How do leaders support extreme teaming?"

We were fortunate to encounter, through our academic work, a broad range of empirical settings relevant to our focus – that is, groups in which individuals with diverse backgrounds (in terms of their occupations, organizations, and industries) had developed into high-performing, temporary teams tackling multimillion-dollar projects. We report findings from five of these cases (key elements of the projects are reported in Table 2).

We chose these five case studies because they offer a clear view of extreme teaming leadership practices and highlight the interpersonal and technical challenges involved. These challenges included a lack of shared experience or prior interpersonal interaction. Since teams lacked the common context of previous work together, which might have informed their organizing principles (Kogut & Zander, 1992), they had to set up routines and behaviors from scratch. Further, the absence of shared history made it harder for teams to cultivate strong relationships, which could have made it easier to combine diverse knowledge and skills.

The projects were also chosen because of their considerable complexity. Not only were team members uncertain about the ends they were working toward, but goals and solutions were sometimes in conflict. When project participants did converge on a particular solution, defining and completing the requisite tasks

Table 2. Overview of Focal Projects.

Project	Industries	Budget	Number of Informants Interviewed	Number of Interviews	Number of Meetings Observed	Documents Collected
Project Anna	Architecture; Construction; Design; IT	> $10 million	10	20	6	Meeting minutes; PowerPoint presentations; Project reports
Project Bianca	Real estate; Energy; IT; Software; Government	< $10 million	9	14	2	Meeting minutes; PowerPoint presentations; Project reports
Project Fiona	IT; Craft	< $10 million	13	23	6	Meeting minutes; PowerPoint presentations; Project reports
Project Sofia	Healthcare; Software	< $10 million	11	14	2	Meeting minutes; PowerPoint presentations; Project Reports
Project Willa	Engineering; Architecture; Construction; Government	> $10 million	11	13	0	Meeting minutes; Project Reports; Contracts

required navigating a high degree of interdependence. The presence of numerous, interdependent, conflicting, and shifting goals and approaches generated project complexity (Campbell, 1988; Wood, 1986). Each project fully manifested the conditions articulated by Carlile (2004), which include differences in accumulated knowledge (novice vs. expert in the same domain); task and goal dependence (i.e., the need for interaction between project participants); and novelty in the setting (new customer needs or product requirements, first-time collaborations). Carlile posits that, as these attributes scale up, communication will become more complex, bringing correspondingly greater difficulty in managing knowledge across boundaries.

Finally, the five projects we selected were regarded as resounding successes by the participating organizations, and all received national or international recognitions. We were thus especially well positioned to identify leadership patterns across these cases of extreme teaming that could contribute to leadership theory. Since some individuals are less comfortable with publicizing their leadership, we opted for pseudonyms throughout the book. We also wished to keep the emphasis on leadership functions and activities, not on individuals.

Data Collection and Analysis

We carried out multiple individual and group interviews with participants and leaders from the various organizations involved in the projects. These were supplemented with documents and other artifacts including meeting minutes, progress reports, and PowerPoint presentations. Nonparticipant observation of meetings was also used for projects already underway, in parallel with our interviews. Interviewees varied widely in terms of their level of expertise, experience, position within their organization, and position within the project. Some were recent hires, while others were recognized experts in their field with over 25 years' experience at their organization. They were asked to participate by email, following snowball sampling. Participation was voluntary and interviewees were ensured confidentiality. No one declined to participate.

Our data collection focused on project leadership activities and behaviors. Leaders were variously referred to as "the primary leadership group," "core decision-makers," "the management team," and "the business group." At least one leader from the organizations involved in each project was interviewed. Interviews varied in length from 45 to 180 minutes. After the first round of interviews, follow-up interviews were conducted with project leaders to clarify and/or explore emerging themes more deeply. This cycle was repeated until conceptual saturation was reached.

Data analysis followed established procedures for theory-building from inductive research (Denzin & Lincoln, 2011; Glaser, 1978). A fine-grained, in-depth, inductive reading of the case evidence on the leadership activities in a single project yielded a list of first-order codes. These were consolidated, via axial coding, across all five cases, and the data structured into second-order concepts and along more general aggregate dimensions (Corbin & Strauss, 1990). Tentative theoretical explanations were suggested by replication, in successive cases, of patterns identified in the data (Eisenhardt, 1989; Glaser & Strauss, 1967; Yin, 2009). By using these to challenge and extend existing theory, we gradually resolved them into the concepts that informed our core ideas about leadership functions.

To address the fact that our account was but one interpretation of the data (Van Maanen, 1998), and concerns that the data were shaped to theory rather than vice versa (Wodak, 2004), we triangulated first between data types and then across analysts. Given that different people have different perspectives on leadership (Fleenor, Smither, Atwater, Braddy, & Sturm, 2010), we only considered leadership practices if they were both perceived by the project participants and intended by the leaders. This gave us confidence that we had indeed found the "actual" leadership practices that enable extreme teaming. Moreover, since only one of the two authors was involved in data collection for each case, each of us was always in a position to challenge and interrogate the other.

CHAPTER TAKEAWAYS

- Research on team development acknowledges that it is a complex and messy process, but consistently identifies the importance of certain emergent states that influence team development and team effectiveness. Emergent states can be affective, cognitive, or motivational, and may influence interactions that take place among team members, which in turn shape team performance.
- The classic input-process-output model of team effectiveness implies that diverse teams should outperform homogeneous teams because they can tap into a wider range of perspectives (inputs). However, the benefits of knowledge diversity are conditional on team processes that enable teams to put their diverse inputs to good use.
- Putting diverse knowledge to good use is challenging for two reasons. First, people tend to categorize each other into group categories and we are usually attracted to similar others. Thus, individuals from different groups must work hard to develop trusting relationships so as to engage in behaviors that support integrating diverse knowledge. Second, boundaries (language, interpretation, and interests) are needed for knowledge to develop in any domain, and knowledge developed in one domain cannot be readily transferred and applied in another. As a result, integrating diverse knowledge often requires concentrated efforts in the transfer, translation, or negotiation of the knowledge to be used during extreme teaming.

- Several studies suggest that leadership can affect team-member behaviors and introduce certain objects at key times during the process of extreme teaming but have not specifically identified the leadership practices that do this.
- Our case studies documented some unusual successes, from the perspective of involved participants and their organizations. Our data were collected through interviews, observation, and review of documents. They were analyzed and triangulated to find patterns across the five case studies of extreme teaming.

PART II

FOUR LEADERSHIP FUNCTIONS FOR EXTREME TEAMING

4

BUILD AN ENGAGING VISION

Every project leader articulated an engaging vision in a similar fashion. Innovating across organizational boundaries is a complex endeavor, which involves a high degree of cohesion among different people with diverse complementary skills. But seeing and appreciating the complementarity of diverse expertise can be elusive. Differences are more likely to produce conflict or misunderstanding than synergy. An engaging vision that became shared by the individuals involved helped projects overcome the risks of confusion and frustration to work hard to align their actions to focus on achieving a common goal. This leadership function was comprised of two practices: (a) making values explicit and (b) articulating a challenging target. Before delving into each of these two leadership practices, which are synthesized in Table 3, the story of how the vision for Project Sofia was developed is offered in Showcase 1.

MAKING VALUES EXPLICIT

We observed leaders, typically at the beginning of a project in team meetings – and in one-on-one conversations – emphasizing one particular set of ideals, principles, and beliefs related to why the newly formed team should pursue the aims that brought them

Table 3. Leadership Function 1: Build an Engaging Vision.

Leaders present a vision that appeals to project participants, and help align their actions to focus on achieving a common goal

Practices	*Definitions*	*Examples from Dataset*
Making Values Explicit	To emphasize a set of ideals, principles, and beliefs that resonate with the larger view of why the newly formed team should aim to accomplish the work that brought them together in the first place	"The project could act as a stepping stone from traditional architecture to a future more about conserving resources and building more delicately and sustainably" [Manager, Project Willa]
Articulating a Challenging Target	To highlight the "unique" nature of the project through a challenging target that serve to clarify the project's constraints and the process to reach the project's goal	The budget was set at $10 million, project size was 60,000 square feet, and delivery deadline was [specific date]. The new building was to have offices, workstations, and a 5,000-square-feet [brainstorming space] that included conference rooms and a gallery of customer's works demonstrating how [new modeling technology] supported innovative design outcomes [Document, Project Anna]

together in the first place. Leaders portrayed projects in ways that imparted their significance and conveyed to project participants that they were engaged in something important. For instance, leaders in both Project Willa (discussed in more detail in Chapter 6) and Project Sophia advocated the value of benevolence. For Willa, leaders characterized the goal as an inspiring precedent for local and foreign designers aiming for more sustainable solutions, the second as in service to the greater good of society with healthy children. When describing the project, Project Willa's leaders would use very strong terms that made unequivocal what were the

SHOWCASE 1: Project Sofia

Serendipity played a crucial role in the genesis of Project Sofia. Finding themselves seated at the same table during an event organized at the city hall, the CEOs of Healthcare Company (HC) and Software Company (SC) suddenly perceived a fit between what they were each trying to accomplish. A conversation ensued about their organizations' respective strategies. Having become one of the leading medical centers in the world, HC strove – through its internationally renowned research centers – to constantly develop new methods of diagnosis and treatment at the forefront of science for the benefit of patients. Meanwhile, artistic, research, and training goals were core to the mission of SC, an internationally acknowledged leader in the development of immersive technologies and augmented reality, along with creative use of high-speed networks. By the end of the event, the CEOs each had agreed to designate an individual to explore whether cooperation might generate synergies to advance their respective missions.

Immersive technologies were a hot topic in the healthcare sector, and many children's healthcare institutions were eager to evaluate their potential for helping young patients. But the precise nature of immersive technologies, and their utility in practice, remained unclear to most healthcare professionals. SC's interest in developing products that employed all manner of technologies to deliver engaging entertainment experiences, typically targeted at young adult customers, made it seem an unlikely partner for HC. But this business strategy did not immediately reveal SC's desire to contribute to the betterment of society through high-end research, which was genuine and lurking beneath the surface.

HC assigned Denis and SC assigned Jacques to work together to explore the potential for collaboration between the two organizations. Denis knew that clinicians in many departments at HC were interested in technology. Psychiatric clinicians, in particular, witnessed how contact with technology could alter the behavior of traumatized children plagued by sociability challenges. Denis sought out groups that exhibited the greatest readiness to engage in collaboration with SC.

Jacques, meanwhile, was pondering how to open a channel of communication and focus conversation between interested parties in the two organizations. He needed a problem that could be framed in a way that would require contributions of resources and expertise from both organizations, and to articulate a vision of collaboration toward its solution. Early meetings, he determined, must include all interested parties lest an element that underscored the organizations' collaboration potential be missed.

Denis and Jacques agreed on the necessity of continuously reminding interested parties of the importance of making a meaningful contribution to the lives of patients and their families. Including patients and family members in meetings with HC physicians, nurses, and researchers and SC designers, programmers, and artists served to focus exchanges on personal experiences in the healthcare system. The meetings fostered a singular engagement with the vision of improving the well-being of children afflicted by disease. This compelling objective resonated with all involved, rapidly gained visibility, and gradually crystalized into a project.

Subsequent meetings aimed at refining the vision into a set of clearly defined objectives were held on-site at SC to afford clinicians and patients the opportunity to observe first-hand work being done around immersive technologies. Physical and emotional reactions to the technologies were recorded and shared. Jacques prepared – in collaboration with Denis and using animation techniques derived from design thinking and gamestorming – diverse real-life scenarios intended to spur idea generation. Animation was employed to facilitate productive dialogue among clinicians and patients assembled into larger groups to share the ideas generated in the role-playing scenarios. The exercise was intended to impart an understanding not only of the potential, but also of the limitations, of available immersive technologies and, for parties affiliated with SC, of the clinical world in which healthcare practitioners operated and technical constraints associated with it.

SC designers, programmers, and artists came away from the meetings with an appreciation for clinical work and patient experiences that enabled them to perceive and discuss relevant applications of technology to a

> variety of real-life scenarios over the spectrum of patient experience, from private, to semiprivate, to public. HC physicians, nurses, and researchers became acquainted with SC's mission and portfolio of immersive technologies. Collaborative investigations were undertaken of potential solutions to real-life issues; taking how things were at the moment as a benchmark, the collaborators imagined how they might be if their organizations worked together, and transformed what they envisioned into a framework that established clear targets for all involved.
>
> Avenues of exploration articulated in the course of the meetings were slated for later detailed investigation and prioritization. Documentation of available resources was disseminated throughout the two organizations, and selected projects were assigned names. Follow-on meetings were promptly scheduled to maintain the momentum built up during the idea generating sessions. Denis and Jacques both viewed continued patient involvement as essential as a source of ideas and feedback and as a clear reminder of the purpose of the collaboration.

values of the group. For instance, a manager from the design company communicated that "the project could act as a stepping stone from traditional architecture to a future more about conserving resources and building more delicately and sustainably," while a Senior Associate stated that "the project will contribute to the 'greening' of the city as a catalyst that encourage people to consume more efficiently and aspire to improve the environment conditions." These two examples were not uncommon ways for project leaders and participants to describe what they were trying to accomplish.

Distinguishing their project's ideals, principles, and beliefs from those used by outsiders was a way for leaders to make values explicit. For instance, Project Bianca's (discussed in more detail in Chapter 7) emphasis was put on the potential discovery that could result from the exploration in which the group engaged. Similarly, Project Anna's leaders' efforts to explicitly associate with it the values of innovation, openness, and transparency were evidenced in the following remarks by the CEO of the architecture company involved in the project.

> *Our industry has gone backwards in productivity; 1975 to 2000 was a dark period because we became controlled by non-value added activities, especially increased litigation and the dramatic increase in the prominence of attorneys. This intrusion caused us to focus more on documenting behavior rather than improving the product we were creating.*

Statistics were cited to substantiate the referenced "dark period," and team leaders spoke openly against legal professionals, deemed responsible for the increase in nonvalue added activities that, the leaders insisted, promoted "silo-ing" and "parochial self-interest," and impeded product innovation. Clashes between professional groups in the industry were exploited to coalesce participating companies' employees around a shared vision. A senior director from the design company emphasized how self-serving behavior of, and values embraced by, attorneys and insurance companies were counter to the interests of project participants. He repeated that the group needed to distance themselves from conservative values of conformity and tradition, presenting the project as an opportunity to "do something really innovative" and "move away from traditional methods."

Values were also made explicit through decisions leaders made throughout their respective project. Some decisions, and the reasons behind them, were specifically discussed during meetings with project participants. For example, Project Anna's leadership sent a strong message to project participants when they fired the millwork vendor they had just hired for its capabilities and experience working on one of the benchmark projects. Project leaders gathered everyone together and explained that they decided to compensate the vendor for its efforts and end its contract when they realized it would not share its pricing method with the group. They emphasized the lack of behavioral alignment to explain why they had broken off that key relationship, making this decision very value-consistent within the context of the project. Responding in this way to a vendor's refusal to embrace core project values – specifically, openness and transparency – even at the risk of jeopardizing the entire project exemplified and reinforced those values.

The decisions made by other parties could also serve to make values explicit. For instance, in Project Anna, when responding to resistance to sign a deal involving various organizations, a senior director from the architecture company said: "Lawyers are not ready to embrace innovative product development, and you can't blame them. […] And the insurance companies don't believe in trust for a reason… it's not in their best interest." This empathy – along with specific reinforcing of a new way of working across industry boundaries and a new set of values – had a powerful impact in bringing the group of project participants together.

Finally, values cannot be created from scratch, and the leaders engaged in the extreme teaming efforts that we studied also tapped into the history of the organizations involved to make the project values explicit. For instance, Project Fiona (discussed in more detail in Chapter 5) leaders espoused the art and science of making embedded in craftsmanship and referred to these as being part of both organizations' DNA. Furthermore, the organizations involved in Project Anna had begun to "philosophically align" as early as a few years preceding their project. According to a Technician from the architecture company involved in Project Anna, "They [employees from Builder Company and Architecture Company] met at various conferences and were allies in advocating for industry-wide process change." The technology company had also started to embrace the same values. According to their VP:

> *This project is a perfect opportunity to demonstrate what [Technology Company] has been promoting for years. We realized long ago that the building industry was highly fragmented, highly inefficient, not very sustainable and not very profitable. We have been advocating the need for a process revolution in the building industry, with technology as a catalyst, for several years now. The project is a chance to walk the talk.*

Across all five projects, leaders engaged in behaviors that helped build an engaging shared vision. More often than not, the details of the vision evolved as the projects unfolded. As members' expertise was integrated, new possibilities came in to focus. This evolution of vision can be celebrated, not discouraged, because it allows diverse experts to influence it early on and then for the team to keep influencing it as the work evolves. This helps create engagement. Finally as projects get under way, insight about end users' needs may deepen and shift, calling for changes in the vision, or perhaps in how it's articulated. The set of values remaining stable for project participants, it served as the leading light throughout their uncertain endeavor.

ARTICULATING A CHALLENGING TARGET

Throughout the extreme teaming cases we studied, leaders spent a considerable amount of time articulating the challenge that the group had to tackle. They aimed to gradually clarify the overarching objective and the constraints the group had to manage to succeed, but also encourage project participants to view their work as "unique," "unusual," and "fun." Observed a principal at the architecture company involved in Project Willa: "This is a chance to do something really innovative. This movement to [innovative product development] from traditional methods requires a leap of faith." Other people involved in the project were well aware of this, knowing that the group was going to try to use new material, which was originally developed for the space industry and had never been used for major buildings. As a technician involved in the project mentioned: "Demonstrating that the design provided an accepted level of safety was an especially challenging task because of both the unconventional structure and the innovative material." A clear, precise, and measurable target served to guide task division and execution and encouraged commitment to a compelling vision, as well as provided an objective format by which progress could be monitored and assessed.

Leaders expressed a challenging target early on, and kept communicating it using various means. Remarked the director of the software company participating in Project Sofia: "We were not keen on the white page. We framed the project with a strong intention: decrease the number of household incidents and encourage the integration of best practices for at-risk parents." Identifying stakeholders was one of the ways for project leaders to clarify the target for the group and the constraints associated with reaching that target. Doing so made it more "real" or "tangible" according to the project participants we interviewed. To define their challenging target, leaders would also point out what the project's goal was not. For instance, during a brainstorming meeting, the director from the healthcare company involved in Project Sofia claimed: "We do not want to look into therapeutic art, but explore deeper pediatric implications of immersive technology." We noted that using such contrary examples elicited meaning for project participants and increased team cohesion as others would then refer to that very contrary example in following conversations.

Providing (and stressing) a particular timeline also helped imprint the challenging target that the newly formed group was hoping to achieve, as one of the professionals from craft company working on Project Fiona mentioned: "By the end of 2015, we need to have a product that will allow kids to experiment with craftsmanship in new ways." During the Chilean mine rescue, leaders continuously reminded project participants of their challenging target to focus efforts toward their common goal – there was no time for fighting, competing, and blaming because "we were risking lives by doing that." As the drillers, geologists, and drill profilers worked to enhance drilling accuracy, the project leadership continued to remind project participants not to allow speed to be compromised for accuracy. The Minister himself came to the work site frequently to remind everyone of the scenario in which they would reach the refuge but not fast enough to matter. Pointing to the consequences of not reaching their goal was thus another way to imprint the challenging target the project participants were trying to achieve.

Similarly, another professional from the architecture company involved in Project Willa indicated: "[Project managers] made sure everyone understood the grand goal of converting a highly ambitious concept into a developed design supported by full calculations to be submitted to the local authorities within four months." Adding accurate details to the timeline was clearly one of the approaches for leaders to imprint the challenging target further. Project Anna provided another illustration as the VP from the technology company involved in the project tapped into the characteristics specified by LEED Platinum Certification for Commercial Interiors to emphasize the challenging aspect of the target that the group was trying to achieve. A field document from Project Anna also shows some of these details:

> *The budget is set at $10 million, project size is 60,000 square feet, and delivery deadline are [specific date]. The new building should have offices, workstations, and a 5,000-square-feet [brainstorming space] that includes conference rooms and a gallery of customer's works*

> *demonstrating how [new modeling technology] can support innovative design outcomes.*

We observed that the project leaders believed in mentioning the various constraints the project had to deal with, often specified in considerable detail, generally communicated in documents, such as contracts and PowerPoint presentations, and occasionally articulated in meetings. "Everyone needs to be in the same boat facing the same direction," maintained the CEO of the builder company involved in Project Anna. "You can't have people rowing in different directions. If you don't stay focused on the holistic final result there will be no profit for you, or for anyone else."

Projects were not meant to appear in any way easy or "regular," project leadership wanting to ensure that each such endeavor was understood to be difficult and project teams to have no room for "passengers." This message invariably hit home in the projects we studied, and although stressful, helped project participants commit to the vision.

CHAPTER TAKEAWAYS

- We found that leaders built an engaging vision that became shared by the individuals involved in our five cases of extreme teaming.
- This leadership function was comprised of two practices: (a) making values explicit, and (b) articulating a challenging target.
- Leaders, typically at the beginning of the project in team meetings and in one-on-one conversations, made values explicit. They emphasized a particular set of ideals, principles, and beliefs that resonated with the larger view of why the newly formed team should aim to accomplish the work that brought them together in the first place.

- Leaders articulated a challenging target by spending a considerable amount of time expressing and defining the challenge that the group had to tackle. They aimed to gradually clarify the overarching objective and the constraints the group had to manage to succeed, but also encourage project participants to view their work as "unique," "unusual," and "fun."
- Building an engaging vision helped individual project participants overcome the risks of confusion and frustration to work hard to align their actions to focus on achieving a common goal.

5

CULTIVATE PSYCHOLOGICAL SAFETY

The unusually high level of expertise diversity in the teams we studied was vital to undertaking their innovation goals in these challenging projects. Diverse expertise, however, would have been to no use without a learning environment in which people felt able to share knowledge, experiment with new ideas, seek feedback, and talk about mistakes. These learning behaviors were vital to developing original insights and adjusting the course of a project as more was learned. Research has shown that learning behaviors are particularly challenging in front of "strangers" – especially in the absence of psychological safety (Edmondson, 1999). Each project was fraught with moments during which people could have been concerned about their reputation or image – and could unproductively hold back their ideas or concerns, due to an unwillingness to take undue interpersonal risks.

We expected to observe several meetings with awkward silences that we have all experienced at the beginning of new encounters or exchanges during which project participants would speak half of their mind, being more mindful of the words they used than the ideas they expressed. For this not to occur, psychological safety had to be established. Project leaders (from each organization) played a crucial role in making this happen. We observed that project leaders enacting two practices that helped cultivate an environment of psychological safety – a learning environment: (a) displaying

Table 4. Leadership Function 2: Cultivate Psychological Safety.

Leaders foster rapports that give rise to the shared belief that the project is safe for interpersonal risk taking

Practices	***Definitions***	***Examples from Dataset***
Displaying Authentic Care	To pay close attention to the project participants' viewpoints and difficulties, and provide feedback and support to deal with them	"When you lead such a project, you need to be visible. People need to know where to reach you, that you will pick up if they call and be attentive to what they have to say" [Director, Project Sofia]
Framing Cross-Boundary Work as a Resource	To acknowledge that project participants' different perspectives are a boon to be leveraged, not a difficulty to be managed	"At the advisory meeting, we got reminded of the value of end-users' input about their needs and design ideas" [Designer, Project Anna]

authentic caring and (b) framing cross-boundary work as a resource. This leadership function, along with the two practices, is synthesized in Table 4. Before delving into each practice, we present the story of how a learning environment was cultivated in Project Fiona (Showcase 2).

DISPLAYING AUTHENTIC CARING

All project leaders conveyed a sense of desire to achieve collective success rather than personal glory. They expressed a belief in the potential of teamwork, and showed genuine care for the project participants. For instance, leaders regularly checked how people were emotionally coping with project demands. While the work was important, they also paid close attention to the well-being of project participants and would not hesitate to ask if they were "okay" or if they needed a break. These "small things" appeared

SHOWCASE 2: Project Fiona

In a response to people's rapidly accelerating interest in making things themselves, Project Fiona was launched with the support of top management at Craft Company (CC) and IT Company (ITC). CC and ITC were interested in exploring the proliferation of makerspaces, fab labs, and a DIY culture in the United States, particularly with respect to their implications for children and education. The novelty and freshness of the cross-industry collaboration elicited such enthusiasm that discussion of the project was becoming pervasive throughout both organizations, occurring as frequently during business meetings and in conversations around the water cooler as in published accounts and formal forums of communication.

One might expect the leadership of a project so highly visible and generating such immense excitement to project the air of confidence and competence managers are wont to cultivate in the presence of their peers, but project leaders Mathieu at CC and Marine at ITC were surprisingly and refreshingly candid about the exploratory nature and uncertain trajectory of their joint endeavor. Asked, for example, how their newly formed team proposed to accomplish its goals, Mathieu and Marine, not without the self-assurance that accrued to their status, were likely to reply, "we don't know." To them, such a response acknowledged simply that the path to success was not yet defined, that they were embarked on a new interdisciplinary journey "with open minds," an expression they often used during team meetings.

Believing it important for the team to achieve a level of comfortable interaction before engaging in project work, Marine and Mathieu organized two outside events to which all project participants were invited. These "get acquainted" sessions provided a casual forum in which to exchange anecdotes about life and work, informally re-hash the week's news, and crack a joke or two, or laugh at someone else's.

Marine and Mathieu reserved some time for preliminary pleasantries in their planning of meetings for exploring paths along which to pursue the team's objectives. Any and all, even "terrible," ideas were entertained, Mathieu relating examples of successful projects born of futile ideas as a way to encourage consideration of anything, likely or not, that might lead to the ultimate eureka. This philosophy and format fostered and facilitated exchanges between project participants at various hierarchical levels in both of the participating organizations. A white board and marker were further catalysts, available to any who wished to use them, the marker serving additionally as an invitation, to any to whom it was presented, to share their thoughts and ideas. Lacking explicit time pressure and deadlines, project participants were encouraged to think aloud. Although the first few meetings generated no ideas deemed worthy of pursuit, the team had begun to gel and the project leaders were optimistic. They showed appreciation for everyone's participation, frequently ending meetings by thanking everyone for their input.

As Marine was preparing for a team meeting, her assistant, Charles, proposed to her and a couple of her colleagues an idea that had occurred to him while watching a TV show, which he believed might be applicable to the project. College students in the show had created a mobile tailgate, a 20-foot shipping container set up with a bar, a TV, and everything one needed to throw a party. It could be dropped just about anywhere; one could rent or buy it. Recognizing that, with appropriate tweaks, the idea could have significant potential Marine invited Charles to present it at the meeting. Having initially acknowledged his idea to be "crazy," Charles was nervous, but Marine assured him that crazy ideas were more than welcome. The worst-case scenario, she reminded, was not very bad – the group might not use it and might simply move on to the next idea.

Marine opened the meeting with the customary invitation to anyone who wanted to starts things off, and Charles took the cue. Marine reframed the idea presented by her assistant to ensure that everyone understood it. To Charles' considerable surprise and pleasure, the idea was not only well received, but became the foundation of the product the team ultimately designed. The group developed the first ever mobile makerspace equipped with laser cutting machines, mini 3D printers, ICT laptops kitted out with 3D modeling and printing software, and an arsenal of traditional tools such as saw, screwdrivers, and wire cutters. Equipped with everything needed for kids to develop prototypes and build finished products, and in the process acquire an understanding of manufacturing, it could go to elementary schools and show kids that creativity and innovation was not reserved to large corporations, and could be fun and self-fulfilling.

In subsequent meetings and during presentations to internal stakeholders, Marine and Mathieu often characterized Charles' idea as a crucial input to the project. They emphasized that their tack from the outset had been to encourage project participants to share their interests as a way to arrive at perhaps unexpected but ultimately workable ideas.

Marine and Mathieu maintained close communication with the subgroups involved in the various facets of product development, often visiting them in their offices or participating in their meetings, and even assuming administrative duties for project participants facing crucial delivery deadlines. The project leaders' frequent presence was perceived by project participants to have the purpose not of checking on their work, but of seeking to lend a hand in any capacity in which they might be helpful. Marine and Mathieu conducted themselves and behaved as peers of the project participants whose labors they directed and oversaw. Marine was able to accelerate aspects of the project by making connections that facilitated decision making, and she and Mathieu encouraged the groups to share difficulties with the entire project team to tap resources and networks both within and outside both organizations.

to go a long way for the newly formed groups as project participants felt that "everyone was there for each other." This was apparent in the way individuals expressed their experience on the project but also in the way they interacted with each other. There were conflicts, of course, but conviction of each other's good intentions prevailed and kept the environment psychologically safe, preventing personal attacks and hard feelings.

In projects we studied, leaders approached project participants differently than what is often observed in high-profile project work. They were distinguished by interaction perceived by project participants to be cooperative and the project leaders' initiation of contact being, not for the purpose of "checking on things," but rather to offer useful information or guidance. A project director from the architecture company involved in Project Anna, according to a superintendent from the builder company, "was always the first to say, 'What can I do to help?', and really meant it." Leaders were open to discussing failures, and took the time to explore potential avenues for dealing with obstacles that project participants faced along the way.

This leadership practice was also illustrated by a director of the healthcare company that participated in Project Sofia; "When you lead such a project," he observed, "you need to be visible. People need to know where to reach you, that you will pick up if they call and be attentive to what they have to say." Project leaders were generally not overcommitted by their companies, and thus could be present for their project participants. A director from the software company involved in Project Sofia deemed "authentic accessibility" fundamental to the project's success. "It's key," he observed, "to understand the language of both organizations to facilitate communication. Especially at the beginning, I would always be around to answer questions, facilitate interaction across groups, and slowly enable the two cultures to merge together." In short, the need to be accessible was important but leaders went further into making sure project participants understood or felt that their accessibility was authentic.

Leaders' displays of authentic care were apparent during group meetings. To discourage tensions that might impede candid discussions of proposed solutions, project leaders regularly invited project participants to voice their opinions, and enthusiastically probed and explored new ideas and concerns that were presented. That project participants were not afraid to speak their minds and question each other's beliefs is evidenced by the following snippet from our field notes of a meeting between participants in Project Fiona.

> *The director at the information technology company participated in an operational meeting with [first-level employees from both organizations]. Some people at the meeting appeared pleasantly surprised to see him there. He engaged quickly during the meeting by stating: "We [his colleagues and himself] are not here to simply sell our solutions. [...] We are not here to serve a customer, but to build a partnership that can potentially create a whole new market for us. [...] On our end, all options are open." [...] The meeting turned into a brainstorming session, and a free-flowing enthusiasm that could be felt from both parties was evidenced with participants who felt comfortable enough to stand up and begin writing on boards and walking about as they enthusiastically expounded their ideas. The director offered guidance on several propositions, and engages with the group in finding ways through which he could help them make progress.*

As the preceding vignette from Project Fiona illustrates, leaders took part in operational meetings to hear and build on project participants' viewpoints. They would candidly present a flexible position, leaving room for others to pitch in with their own ideas and opinions. Leaders would follow up on their own exchanges with project participants, which would set the stage for further valuable exchanges between project participants themselves, nurturing an environment conducive to learning. Project participants reported that leadership on these projects was particularly attentive and responsive to their ideas, as illustrated by several technicians from the craft company: "[Director] would not only listen to our viewpoints, but would also come back to us during the following week to update us on what he had found out in relation to what we had said." Thus, authentic care went beyond listening to project participants; it also meant acting on their ideas.

Similarly, a principal from the technology company involved in Project Anna underlined the importance of these first impressions: “The early communications we had with [Builder Company CEO] were authentic, and showed us that we were not dealing with a contractor that was only interested in his side of the project.” A director from the software company made a similar remark regarding his behavior as a leader of Project Sofia: “We were very honest right from the beginning. We told [Healthcare Company employees]: ‘We have this pool of technologies, which we think might have some potential to be adapted to your context, but we do not know exactly what you guys need. We’ll learn that together.’” This sentiment of “togetherness” was an important part of the message that leaders communicated through authentic care.

FRAMING CROSS-BOUNDARY WORK AS A RESOURCE

Beyond displaying authentic care to help everyone feel comfortable expressing their thoughts and ideas, leaders also cultivated a psychologically safe environment for learning by framing cross-boundary work as an essential resource to the project’s success. The distinct perspectives of project participants were viewed as, and acknowledged to be a boon to be leveraged, not a difficulty to be managed. Leaders made sure to acknowledge that conflicting views were going to arise and that, while demanding, this process was deliberate and thoughtful. They maintained that project participants’ input was essential to project success using terms like “learning opportunity” and “potential for discovery” when referring to conflict emerging from cross-boundary work. The director of the architecture company participating in Project Anna recalled how a manager from the builder company had framed cross-boundary work as a resource by emphasizing the value and relevance to the project of both organizations' internal knowledge, explaining that,

> *In a normal project, without the communication with the contractor, we may not have proceeded. [The builder company's] manager instilled confidence in our respective expertise, and made us feel we could pull it off if we integrated what we all knew.*

A designer involved in Project Anna also noted: "At the advisory meeting, we got reminded of the value of end-users' input about their needs and design ideas." During that meeting, we observed again the director of the architecture company making a parallel between that input and the input of other specialists on the team. Clearly, project leadership was adamant that solutions and products represented the integration of many ideas from manifold project participants, and made no attempt during meetings to downplay, and in fact were keen to highlight, differences of opinion.

The director of the software company involved in Project Sofia had a similar stance: "Without involving users, we might have missed on various key dimensions. Developing something that you think is useful is one thing, but you need users to tell you what is actually desirable, usable, or just accessible." This was further exemplified with the following remarks from a pediatrician and a researcher, involved in Project Sofia:

> *Of course, [Software Company Professionals] bring another perspective, different ideas. They bring another dimension: while our approach is very much a clinical one, they take more of a technological and artistic perspective. [Healthcare Company Pediatrician 1, Project Sofia]*

> *We definitely benefited from our differences. Us, we are always with our same literature, our similar ideas... them,*

they are creatives [...] they brought some extraordinary ideas. [Healthcare Company Researcher, Project Sofia]

Leaders were keen in making project participants aware of others' perspectives but also instill confidence in their own. For instance, all project leaders we observed insisted to welcome input from everyone without resentment or blame. The overall view of the projects we studied was that it was impossible to attribute any single idea or part of an idea to any single person. From what was communicated and understood, bringing various professionals together was not a burden but the only way to co-create their projects. The solution was not a design with a single author but more the work of an empowered and well managed collective.

In the projects we studied, conflicts and disagreements were not matter of winning or losing a battle. Instead, they were perceived as part of problem solving. A professional from the energy company involved in Project Bianca mentioned: "We took the time to listen to each other because we knew we would also have the time to make our point, and at the end of the day we were to put our egos to the side and decide what made the most sense for the project." As a result, each project experienced moments when concerns voiced by a project participant engaged in productive dialogue yielded a new or more sophisticated solution to a problem. For example, the managing director of the architecture company involved in Project Willa remembered: "In an emergency meeting held four weeks before the deadline, project participants were invited to voice their concerns a few weeks in advance. [...] Architects were persuaded to abandon initial design concepts, and go in a new direction." In Project Sofia, "every time someone presented something new," recalled a designer at the software company, "we [project participants] were invited to challenge it if we thought it needed to be challenged or to support it if we felt like supporting it. We knew others could see things differently, and that this was what gave us the potential to innovate." If left unresolved, most of the feedback from each project participant was incorporated into the existing work and presented at the beginning of the next meeting as key points that had to be addressed, hence contributing to framing cross-boundary work as a resource.

Project leaders insisted on the value of holding meetings with people from various backgrounds, and acknowledged the different contributions that were made by several project participants by emphasizing their unique perspective on the problem. In writing through emails or verbally at the end of a meeting, leaders continued throughout the project to make sure everyone was aware that representatives of all partner organizations needed to attend certain meetings that would lead to important decisions because their viewpoints would shed light on an essential part of a problem. "The distinctive difference between this and other projects," remarked an engineer with the engineering company that participated in Project Willa,

> *was that we brought together engineers of all disciplines and started brainstorming before the architectural design had commenced, and we stuck together throughout the project. At the very beginning, specialists from different disciplines knew that their input was important, and we understood that our implication had to last throughout the project.*

Displaying authentic care and framing cross-boundary work as a resource helped cultivate psychological safety. It allowed for broad and deep learning throughout each project as team leadership remained true to its commitment to care for others and put stock in their contribution, as attested by reported conversations between project participants at different hierarchical levels. Groupthink was largely avoided thanks to an environment conducive to learning, in which individuals felt entitled to express their own thoughts and ideas.

CHAPTER TAKEAWAYS

- Leaders in extreme teaming projects must recognize the need to cultivate a psychologically safe environment for learning.
- In the projects we studied, this leadership function was comprised of two practices: (a) displaying authentic caring and (b) framing cross-boundary work as a resource.
- Leaders were highly visible in their projects and used this visibility to display authentic caring. Their interactions were perceived as cooperative; they regularly checked how project participants were emotionally coping with the situation. They also showed care for what people were thinking during group meetings and would not hesitate to reach out individually when necessary.
- Leaders framed cross-boundary work as an essential resource to the project's success. They insisted that the distinct perspectives of project participants were a resource to be leveraged, not a difficulty to be managed. They referred to conflict emerging from a clash of different perspectives as a "learning opportunity" and "potential for discovery."
- Cultivating psychological safety made it possible for those involved in the projects to feel comfortable sharing knowledge, experimenting with new ideas, seeking feedback, and talking about mistakes to develop original insights and adjust the course to fit with the lessons learned.

6

DEVELOP SHARED MENTAL MODELS

In our research, projects had leaders who helped ensure that project participants figured out each other's expertise and understood it well enough for the team put it to integrate at the right time. A central element of this challenge related to understanding each other's work and perspectives sufficiently to be able to identify areas of necessary interdependence. This in turn allowed project participants to develop shared mental models and take advantage of interfaces between areas of expertise, and corresponding areas of modularity. Project leaders engaged in a set of activities that helped develop shared mental models and put them to good use. These were (a) diagnosing interfaces for knowledge-sharing and (b) leveraging boundary objects. Before examining each of these leadership activities, synthesized in Table 5, we present the story of how Project Willa's leaders dealt with this challenge in Showcase 3.

DIAGNOSING INTERFACES FOR KNOWLEDGE-SHARING

In addition to framing cross-boundary work as a resource to cultivate an environment in which people felt that their input was valued, we observed that project leaders focused on identifying and managing interfaces throughout their projects. Extreme

Table 5. Leadership Function 3: Develop Shared Mental Models.

Leaders help project participants understand others' expertise and how it applies to the modular task sets

Practices	***Definitions***	***Examples from Dataset***
Diagnosing Interfaces for Knowledge-Sharing	To identify interfaces at which a project requires attention to facilitate discussion and coordination across boundaries	"The project as we know it today is the result of architects and engineers working side by side at key times in the project" [Director, Project Willa]
Leveraging Boundary Objects	To manage objects strategically, in ways that enable the transfer, translation, and transformation of knowledge across boundaries	"I first thought the tools we were using during meetings used a lot of our time for their actual benefits. I changed my mind when we were told to draw sketches. We did all together, in real-time […] their two designers added to mine, and vice-versa. With management guys at the meeting, we could integrate their branding desires as we progressed" [Engineer, Project Fiona]

teaming required transferring not only information, but also translating different interpretations and transforming individual interests to build shared meaning and understanding. We found that leaders would plan for specific moments during which intense cross-boundary work was needed, and strategically organized for these encounters. In some ways, they were curating a repertoire of knowledge-sharing needs and opportunities.

Just as in a relay race, exchanges of the baton between project participants were crucial to the outcomes in the projects we studied. The identification and orchestration of interfaces between project participants who did not know each other well and worked in different fields could easily make the difference as to whether development activities were transferred effectively from

SHOWCASE 3: Project Willa

Project Willa was organized to design and construct an aesthetically remarkable, environmentally sustainable building to serve a visible public function, in time for a major international event. The objective – which we can summarize as to build a memorable, iconic structure that reflected specific cultural values, integrated well with the landscape, inspired its diverse visitors and employees, and minimized energy consumption – was subject to a global competition. The winner – that is, the proposal selected for building, would face ambitious financial goals and an inflexible deadline.

Being cross-disciplinary, cross-organizational, cross-industry, and even cross-continental, the Willa project easily satisfied the criteria for extreme teaming. Led by Tom, senior structural engineer at a global engineering firm that had launched the collaborative effort, Willa involved more than 80 individuals from four organizations (engineering, architecture, and construction firms and a government agency), 20 disciplines (ranging from architecture to fire engineering and materials science), and four countries. Immense diversity of technical expertise had to be accommodated by collaborators who shared neither mother tongue nor time zones.

Occasionally, participants came together face-to-face to confront critical tasks or interim deadlines, the overarching task initially being to brainstorm design ideas. Teaming across cultures presented a significant challenge at first. An attempt to build shared understanding by exploring how cultural meanings of design elements differed across nations was facilitated by some members' familiarity with more than one national culture, and the willingness of several individuals to spend time in other firms and countries as a way to help bridge national and professional cultural gaps. The presence of these boundary-spanners engendered in others in local offices curiosity about the traditions, norms, practices, and expectations of other groups.

Willa's commitment to develop an unconventional structure using novel materials made ensuring an acceptable level of safety in the building design an especially challenging task. The project had to work closely with government authorities to assure that the building's performance was consistent with requirements set forth in local regulations.

Beyond safety, the Willa project was committed to being "green." One strategy for achieving environmental sustainability was to minimize reliance on energy sources derived from fossil fuels by incorporating solar energy in the design. A revolutionary design for solar heating and lighting, refined over a period of weeks, was expected to reduce heating energy consumption by more than 30% and the need for artificial lighting by 55% relative to comparably sized buildings. To minimize the building's impact on already strapped local water supply systems, the team designed innovative ways to harvest and recycle water, and advanced design strategies for water use. Beyond its structural accomplishments, the Willa project became a source of inspiration and pride to local community residents as well as to observers worldwide.

In any extreme teaming effort, it is important to establish guidelines for boundary management. Guidelines are needed for specifying points at which separate teaming activities must come together to coordinate resources and decisions. Tom and his team adopted a strategy for "interface management" that divided the project into modules based on physical and time attributes. Each module was owned by a subteam. An interface existed when anything touched or crossed a boundary. Regular interface coordination meetings were held to manage physical, functional, contractual, and operational boundaries. Through disciplined documentation and care, the project avoided costly mistakes that might otherwise occur at boundaries.

The result, after many months of intense teaming, was a spectacular structure – delivered on time and on budget – that fully deserved its instant reputation as a technologically, environmentally and aesthetically revolutionary design.

one group to another, or from one process stage to another. Recalled the director of the real estate company in Project Bianca: "We gave ourselves a methodology on which the various professionals could rely to execute their project. [...] We knew when new input would be useful." Added the director of the software company involved in the project: "Intermediation became more and more prominent for me. My main role during projects was to have good timing to jump in and provide intermediation between [the software company] and [the real estate company] professionals." Project Sofia's leaders similarly assured a smooth transition through the different stages of development by instituting a formal process of knowledge translation.

With more complex products, multiple design streams may develop concurrently, requiring leaders to identify touch points between project participants to orchestrate their respective activities. Among the challenges faced during Project Willa were integrating and coordinating the project's many interfaces, which involved multiple stakeholders and oftentimes conflicting demands. Coordinating requirements for all stakeholders was a complex task that required a delicate balance. Project leaders introduced a strategy for "interface management" that divided the final product into "volumes" based on physical and time boundaries set forth in a project volume register, each volume of which was "owned" by a subproject team (empowering decision clusters, as illustrated with the next activity). An interface occurred when anything touched or crossed a boundary between volumes. High- and low-level interfaces were distinguished and captured in a register, and coordination meetings involving all relevant parties were regularly held. Orchestrating interfaces in such ways led to a more efficient and better-coordinated design process that promoted optimal use of materials and improved execution.

For key meetings, Project Anna's leadership insisted on inviting certain professionals from the several involved organizations, and suggested, and sometimes even requested, that project participants work in locations in which they would directly interact with one another. Similarly, "the project as we know it today," explained a director from the architecture company working on Project Willa, "is the result of architects and engineers working side by side at key times in the project." Leaders realized that not all mediums offered adequate richness to support the type of communication that was needed to manage ambiguities and uncertainties, and facilitate mutual understanding. Rich synchronous interaction (such as that allowed by face-to-face meetings) was encouraged for highly ambiguous information while less rich asynchronous communication was enough for the mere transfer of information. The value of doing so was appreciated by project participants. Observed an architect with the architecture company involved in Project Anna: "I see the value of co-location for our meetings. When I come to the job site, I see how everyone speak for the design and understand why we design what we design." A coordinator from the participating software company explained how insisting to bring a diverse group of professionals together at the same table at key times had benefited Project Sofia:

> *[Professionals from the healthcare company] having conversations with me, and then me going to our programmers and designers, would have been counterproductive. I'm from [the software company], but I know close to nothing about programming. We needed everyone together to reflect on the project and co-create. It was important that we sit them all at the same table for these meetings.*

Leaders of the projects we studied showed a mastery of timing for bringing project participants together. They made sure they were present during these occasions to reframe some of the inputs from the participants so that everyone involved would be able to understand and apply the insights offered. This meant taking the next step from orchestrating interfaces to leveraging boundary objects.

LEVERAGING BOUNDARY OBJECTS

Project leaders tended to organize strategic cross-boundary encounters in singular ways, not over the phone or Skype, but in person where project participants could work on problems together, side-by-side, using objects that helped them communicate with one another. Project Willa's team leaders' insistence on physically gathering individuals from the organizations involved to develop design and documentation drawings at key times during the project facilitated deep and enduring collaboration among project participants. They believed that these encounters were too important for the future of the project to happen offline. Custom-developed scripts and tools were employed to bring various disciplines together to develop the overall model and design drawings. "The design and documentation drawings allowed the client, potential contractors, and mechanical specialists to see different parts of the building in the format they needed," explained a 3D technician with the design company that was participating in the project. Added a senior structural engineer at the design company:

> *The capacity to attach building information to the 3D digital model allowed for a seamless exchange of information between architects, engineers, contractors, and other stakeholders. [...] Creating 3D representations of the various stages of the design process and simulating real-world performance [were among the ways the] technology was used to streamline the process and enhance the quality of the design solution.*

Similarly, a director at the architecture company involved in Project Anna noted the benefits of using such objects when the project reached a point where important decisions ought to be made.

At the meeting, everyone projected the latest model and began a real-time walk-through in and around the atrium. The static drawings shown during the presentation were now kinetic. All of a sudden people started sitting back down as we virtually "walked" around the space and by the end of the meeting we said, "you know what, it's worth it!"

During Project Sofia, boundary objects also served "to speed things up," as a director from the healthcare company remarked: "Clinicians could understand technology's potential faster by seeing products, touching them, and testing their use." Similar evidence was found in Project Fiona, where an engineer from the IT company mentioned:

I first thought the tools we were using during meetings used a lot of our time for their actual benefits. I changed my mind when we were told to draw sketches. We did all together, in real-time [...] their two designers added to mine, and vice-versa. With management guys at the meeting, we could integrate their branding desires as we progressed.

Furthermore, leveraging boundary objects was as beneficial to deal with potentially conflicting issues, helping in transforming certain interests. Project leaders would feel the need to manage interfaces when they sensed that conflict could occur, becoming personal and further hampering execution, if not managed thoughtfully. Elaborated a designer from the software company participating in Project Sofia:

At one point, there were a lot of demands coming from [the healthcare company]. They wanted to add "voice over" but we had agreed on visual information. [The software director] could feel trouble brewing, and quickly suggested we get together to discuss it. We had them over, *and we showed them all the work that this would require. We would have had to pay for recording sessions, review the code structure to enable sound, and implement all of that.*

Project leaders leveraged certain objects during those moments to avoid confusion or misunderstandings that might arise and engender conflict between the different parties. “The meetings were wrapped up with project leaders going through what was discussed with a simple black-and-white PowerPoint deck,” explained a technician with the software company involved in the project. The deck became very useful as the project progressed and necessary adjustments needed to be made. It crystalized exchanges, and further the “sharedness” level of the mental models that were being developed across project participants.

Recall the Chilean mine rescue, which provides clear instances of this third leadership practice. Leaders planned such moments in advance and picked individuals with the best communication skills to manage them, for instance, managing the transition between experts developing the capsule design and those figuring out how to build it. They too used various objects to facilitate communication, understanding, and negotiation across boundaries. By doing so, leaders enabled the project participants to better understand each other’s requirements before some of the work was changing hands but also helped prevent communication breakdowns and brokered hidden knowledge between collaborators operating in different industries or areas of expertise.

CHAPTER TAKEAWAYS

- Extreme teaming leaders promoted the development of shared mental models among participants who were necessarily engaged in tasks with complex interdependencies.

- This leadership function was comprised of two practices: (a) diagnosing interfaces for knowledge-sharing and (b) leveraging boundary objects.
- Leaders worked to identify and manage interfaces throughout the projects. By curating the knowledge-sharing needs of their project, they were able to plan for those moments in which intense cross-boundary work was most needed and to organize for these encounters between project participants.
- Leaders used *boundary objects* to help project participants communicate with one another during cross-boundary encounters. Sometimes, an insistence on physically gathering individuals involved on certain tasks allowed them to leverage models and drawings, PowerPoint presentations, and other "boundary objects" to help translate interpretations across boundaries, so as to avoid the confusion and misunderstanding that might have arisen and possibly led to conflict.
- Developing shared mental models helped ensure that project participants figured out each other's expertise and understood it well enough to put it to use at the right time. This is turn allowed them to take advantage of interfaces between areas of expertise, and corresponding areas of modularity.

7

EMPOWER AGILE EXECUTION

The projects we studied presented high levels of novelty and uncertainty. Making sure the teams were flexible enough to make small and sometimes large adjustments throughout project execution – as more was learned – was a core leadership challenge and one that all of the project leaders we interviewed considered essential to success. At the same time, it was the expertise of the project's diverse participants that informed the need for, and the nature of, those adjustments. We thus observed leaders encouraging a way of operating that empowered experts and integrated learning into the work. Instead of separating action and learning – say into planning and then execution – project participants built learning into action, enacting a form of *execution-as-learning*. This way of working required adopting a learning orientation or mindset – in contrast to a performance or efficiency mindset. To foster this mindset, we found that project leaders tended to focus on (a) enabling expert decision clusters and (b) providing room to maneuver. This leadership function is summarized in Table 6 (and illustrated in Showcase 4).

Table 6. Leadership Function 4: Empower Agile Execution.

Leaders encourage project participants to take charge of non-modular tasks that fall within their realm of expertise

Practices	***Definitions***	***Examples from dataset***
Providing Room to Maneuver	To afford project participants the "space," both physical and intellectual, to explore and make progress on the project	"Specialists on the project were dealing with materials and structures that were quite new in some ways. […] This meant that we had to give our people the freedom to progress and be able to fail and stand up again. […] The people who were collaborating in the design on a day-to-day basis did not have to get involved in the contractual and commercial conversations." [Manager, Project Willa]
Enabling Expert Decision Clusters	To delegate decision-making authority over certain non-modular tasks to the project participants with adequate expertise	"[E]ach professional needed to be able to make ultimate decisions, and from Day 1 each needed to be supported with sufficient resources to do so." [Director, Project Anna]

ENABLING EXPERT DECISION CLUSTERS

Decision rights were often distributed in the projects we studied. While some aspects of the work were modular, many of them did not require cross-boundary efforts to be realized. Explained a director from the builder company involved in Project Anna regarding these non-modular activities, "[E]ach professional needed to be able to make ultimate decisions, and from Day 1 each needed to be supported with sufficient resources to do so." Project Anna's project management team, for example, had authority to determine the optimal course for pursuing project goals including the *latitude* to move money, without executive

SHOWCASE 4: Project Bianca

Project Bianca's goal was to exploit data in an innovative urban retrofit project in Europe to optimize the energy needs of a set of connected buildings and their inhabitants. The project involved 10 Global 500 companies from several sectors, government agencies and a handful of start-ups. A key challenge was to configure and manage a distribution and consumption grid to manage energy use. The project was ambitious and full of uncertainty. Thomas, a relatively young project manager at Property Company (PC), was asked to lead it, with a focus on ensuring that relationships among the many partners remained positive throughout. PC was transitioning from a traditional real estate to a more open business model to better suit what it saw as a new generation of buildings, communities, and cities. Believing that the future of the industry would require innovative, technology-enabled solutions to the construction and management of real estate projects, PC had been eager to invest in the experimental and ambitious Bianca Project. Building on PC's recent achievements with green office spaces and residential buildings designed to produce more energy (from renewable sources) than they consumed in carbon-based energy sources, the Bianca project sought to find a way to achieve synergy by connecting differentially purposed buildings in a smart, interactive network that allowed a real-time exchange of energy consumption data and sharing of energy produced.

The project's multi-organizational composition made decision-making challenging. This required the design of mechanisms to ensure that everyone's input and interests were taken into consideration, without becoming overly bureaucratic or bogged down by excess meetings or conflicts. A thoughtful multi-layered structure emerged that was highly effective in ensuring coordination and creativity. Some decisions were made at monthly meetings coordinated by Thomas' team – meetings that included all 10 industrial partners in the consortium. Thomas believed, however, that not all decisions should be made in such a manner.

To help to assure agile execution, at the project's initiation, Thomas invited the partners to carefully review the division of labor, responsibility, and budget. To emphasize the variety and complexity of the issues that would need to be addressed, which far exceeded his technical expertise and for which *ad hoc* decisions would have to be taken swiftly if the project was to adhere to its timeline, Thomas requested that participants propose what they would contribute and how they would work together. As it became clear to him that not all participants needed to be steeped in all aspects of the project, Thomas began to associate specific aspects with the individuals best suited to manage them.

Thomas organized aspects of the project into subgroups in which domain experts, who had planned to interact relatively informally, could more fully exercise their expertise. Mostly focused on technical issues crucial to the project, such as energy production, information architecture, and aggregator optimization, the subgroups also fostered a sense of community. One partner was chosen to lead each subgroup. All partners could contribute to and assist with the work of any subgroup, but the designated partners were responsible for the outcomes of their subgroups. Thomas identified, with the input of all partners, milestones for each of the subgroups, which were free to pursue them as they thought best.

In the case of disagreement, subgroup leaders had the authority to make a final decision on an issue. Subgroups were the project's fundamental operational units, and it was required that subgroup leaders be respected. Decisions concerning the overall project were made together at the monthly meetings with all partners.

This governance structure helped the project cultivate community-oriented behaviors. Participants in the project – each one both a project team member and at the same time a representative of his or her employer's interests – viewed themselves not only as employees of their respective companies but also as members of their subgroups; this sense of dual belonging tended to discourage value-claiming behavior. By delegating project management to one of the subgroups, moreover, Thomas reduced the administrative hassles that subgroups owning the more technical aspects of the project would otherwise face.

By the end of the project, the group had helped put in place over a thousand connected homes, a handful of commercial offices, state-of-the-art efficient urban lighting, collective energy storage systems, and photovoltaic energy production facilities. Local residents and enterprises were enjoying numerous benefits, from reduced energy consumption to the increased attractiveness of the area. Each of the organizations involved had gained capabilities that were valuable to future research and deployment going forward from the venture. Finally, new divisions and companies had been generated by the project to integrate disparate expertise and thus better respond to new market demands.

level approval, from one budget to another to assure that resources were available where and as needed. Leaders across the projects we studied were keen at recognizing project participants' knowledge and skills, and quickly leveraging it for their project by giving them the autonomy needed within their area of expertise.

The essential balance that project leaders demonstrated was to give a considerable degree of autonomy in carrying out their work to individuals and subteams while also making participants understand their responsibilities to the entire project in consistent ways. Participants whose expertise was recognized and acknowledged by leaders, and also by other project participants, reported being comfortable making decisions. Recalled a designer from the software company in Project Sofia:

> *[Professionals from the healthcare company] would come up with lots of suggestions, so we had to decide on which would add the most value for the effort. [...] At one point, we explained [to healthcare company participants] that, of course, there are good suggestions that can be made, but we couldn't execute all of them. We had to decide and prioritize.*

Mutual respect flourishes in successful extreme teaming projects. Project Sofia, for example, the healthcare professionals saw a presentation of some of the software company's past endeavors and were impressed by what they learned about the partner's expertise. Expert decision clusters facilitated changes when needed. Importantly, developing shared criteria with which to prioritize proposals "helped move the project forward a lot faster," according to the Director at the healthcare company who had witnessed personal conflict result from groups questioning decisions they considered dubious because of a lack of mutual understanding of each other's expertise and how it might come together.

Understanding boundaries of authority was key, especially in enabling expert decision-making, across every one of the extreme teaming efforts we studied. The projects all manifested a clear authority structure, albeit not always defined graphically (only Project Anna and Project Bianca had formal organizational charts). Decision-making layers were nonetheless clear, however, because project participants knew whom to contact in the event of a problem or conflict and who had authority over, and exercised decision rights in, specific domains. Project leaders did not want every new situation to be debated between all project participants, slowing down execution. During the Chilean mine rescue, leaders did not rely on seniority to find ideas but rather on novelty and credibility. For instance, they listened to the innovative suggestion of a 24-year-old and empowered him to work with a new supplier to implement the idea. The leaders trusted the young expert to broker the technology and to communicate effectively with those directly involved as the experiment unfolded. In another instance, project leaders were quick to call for an accuracy test on the measurements of drill holes to figure out who should oversee the measurement of drilling accuracy going forward. This recognition of expertise – based in directly observable data – smoothed future decision-making in that domain. Furthermore, once one person was appointed to a decision-making position, that person was responsible to appoint everyone under him or her. Doing so established a clear line of command with clear accountability.

PROVIDING ROOM TO MANEUVER

In addition to inviting project participants to take ownership of certain areas of their project leaders also found ways to give them room, both physical and intellectual, in which to explore the alternative solutions. One project participant called this, "room to maneuver." "Specialists on the project" explained a manager at the design company in Project Wilma:

> *are dealing with materials and structures that are quite new in some ways. [...] This means that we have to give them the freedom to progress [...] to fail and stand up again. [...] The people who are collaborating in the design on a day-to-day basis do not have to get involved in the contractual and commercial conversations.*

Project leaders employed various means to shield participants from "administrative noise" that might divert precious cognitive resources from the crucial problems at hand. To provide participants an opportunity to work "in a bubble," Project Wilma's leaders established four design streams, in part, to provide technical experts the freedom to act within their domain and to help avoid task overload on key staff such as senior engineers and associates, who might be at risk of creating decision-making bottlenecks. Leaders in other porjects created formal positions to manage such tasks as internal reporting, commercial issues, or coordination of technical interfaces.

Various technologies were "put on the table" in Project Sofia, according to a director of the software company. Recalled a researcher in the healthcare company: "We were interested in evaluating the potential of as many technologies as possible."

The group sought outside input both to develop and to assess competing design options. Patients, for example, gave the team feedback, exploring analyses of the different technologies and providing insights into design options in a series of workshops jointly organized by the two organizations. Observed a manager in the healthcare company: "Both organizations were open to exploration at the beginning of the project, and this was helpful for all participants, I think … to push out any thoughts they had, and everyone felt we were open, so we would not dismiss anything without a reason." Similarly, participants in Project Fiona quickly started experimenting with various options, and left them open for users to guide the project through its final design. As mentioned by a development specialist from the craft company involved in the project, "nothing on the product was set. […] Everything could be easily modified […] We knew the users were going to tell us what they liked, what tools should be there and how it should be organized." This practice of generating diverse ideas from which the best could be culled increased project confidence in execution in all five projects we studied.

Physical space was also provided to support intellectual stimulation. Although all development work was done at the software company, which was quite different from the hectic environment of the healthcare company (a tertiary-care hospital), at certain points in the project, some technologies were tried out onsite. Explained a hospital-based clinician: "We installed a small dome in my unit to test some of our techniques with immersive technologies. After a few experiments, we realized some of our assumptions were off, and discovered some interesting additional options." Here again, we observe the room to maneuver that leadership gave project participants to enable agile execution. Similarly, team members from the craft company in Project Fiona were relieved from regular duties to complete project tasks, and were provided with a large space in which to work together to focus on developing the product.

All leaders we studied were keen in providing project participants with room to maneuver – that is, with the space they needed to explore alternative avenues in their project. Importantly, leaders ensured that discarded options were not judged as "failures" but rather as valued lessons that informed next steps. This required instilling a learning mindset aimed at exploring various options, learning fast from ongoing actions, and integrating what worked into final options that were designed before execution – an approach that is reminiscent of prior models on teaming for innovation (e.g., Edmondson, 2013). For instance, none of the three initial design options developed for Project Anna was ultimately selected, project participants learned from them and applied that learning to the development of a fourth option that incorporated attributes of the first three. "Since we were unsure about the options we had developed," a VP of the technology company participating in the project mentioned, "we asked our partners to sit down to develop a fourth design option: one within budget that incorporated certain attributes of the first three options." For Project Wilma, rapid prototyping was part of the strategy for engineers, IT specialists, and architects to produce a coordinated system of modeling. "Unlike the traditional approach, in which architectural design was restricted by the boundaries of structure, we used the structure as an element of the design," explained a project manager at the engineering company involved in the project.

CHAPTER TAKEAWAYS

- Project leaders promoted a way of operating that empowered agile execution, or "execution-as-learning."
- This leadership function was enacted through two practices: (a) enabling expert decision clusters and (b) providing room to maneuver.

- Leaders created decision clusters by allocating decision rights over tasks that did not require cross-boundary efforts, empowering experts to proceed independently. This involved recognizing project participants' expertise, so as to leverage it for the project, giving them autonomy within their area of expertise – and ensuring coordination between areas of expertise.
- By providing the "space," both physical and intellectual, leaders provided room to maneuver to the expert decision clusters. Project participants could thus quickly explore several options and push the envelope within their realm of expertise.
- Empowering agile execution allowed for the exploration of several options and rapid knowledge accumulation within one area of the project. Doing so supported quick decision-making and the fast realization of independent tasks.

PART III

LOOKING BACK AND MOVING FORWARD

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8

A MODEL OF LEADERSHIP FOR EXTREME TEAMING

Though teaming across boundaries is increasingly regarded as a useful strategy for innovation, prior research has done little to specify the activities that allow organizations and individuals to be leading this kind of extreme teaming effectively. Our inductive research provides a foundation with which to theorize these activities, as well as to combine them into a tentative model for leading extreme teaming. In this chapter, we combine the diverse and dynamic practices introduced in the four chapters in Part II of this book into a single framework. Our hope is that this framework will help people guide the necessarily unpredictable journeys that extreme teaming will generate.

Around the world, numerous attempts to integrate new technologies, methods, or tools to innovate within and across sectors are underway. We suggest that, due to the inherent challenges of extreme teaming, such endeavors are unlikely to succeed without deliberate, thoughtful leadership. As we have emphasized, complex technical and social problems nearly always require experts from different organizations and sectors to collaborate in search of novel solutions – what we have called extreme teaming. Yet, as we saw in Chapter 3, extreme teaming brings new challenges, stemming from the interpersonal (emotions and relationships) and technical (knowledge and skills) difficulties of assembling a diverse group of strangers to take on a demanding task. We now look at how the four leadership functions described above specifically address these two categories of challenge.

ASSEMBLING THE FOUR FUNCTIONS INTO A COHERENT WHOLE

The four leadership functions – realized through eight leadership practices –support extreme teaming by helping to overcome interpersonal and technical hurdles to success. They also help motivate effort and remove barriers to progress.

As depicted in Fig. 1, two of the functions (and thus four practices) target emotions and relationships. First, building an engaging vision (by making values explicit and articulating a challenging target) engages participants' emotions and helps them see each other as sharing a sense of purpose and aspiration. Second, cultivating a psychologically safe environment for learning (by displaying authentic care and framing cross-boundary

Figure 1. Model of Leadership for Extreme Teaming.

	INTERPERSONAL Focused on Emotions and Relationships	**TECHNICAL** Focused on Knowledge and Skills
MOTIVATE Funnel Energy	Build an Engaging Vision	Empower Agile Execution
FACILITATE Remove Barriers	Cultivate Psychological Safety	Develop Shared Mental Models

work as a resource) also relates to emotions and relationships. In psychologically safe work environments, people are less reluctant to share ideas, admit mistakes, and engage in other behaviors that could make them vulnerable in the eyes of others (Edmondson, 1999). Psychological safety promotes respectful, collegial candid conversations, and is vital to making progress in uncertain, interdependent environments (Edmondson, 2012). In sum, the left-hand side of the 2×2 model shown in Fig. 1 captures the leadership functions that facilitate interpersonal interactions in the service of learning and innovation.

The other two leadership functions can be seen to target the technical challenges of integrating knowledge and skills across expertise domains. First, by promoting the development of shared mental models (through orchestrating interfaces and using boundary objects), leaders help experts bridge the very real gaps they face – in terminology, values, timeframes, tasks, and more (e.g., see Edmondson & Reynolds, 2016). Second, when they work to enable agile execution (providing room to maneuver and creating expert decision clusters), leaders are addressing some of the thorny, task-based technical dimensions of extreme teaming. Because it is difficult for different areas of expertise to come together and work seamlessly, experiments are both essential and invaluable. But this experiment-and-learn approach must be led – it cannot simply be left to unfold in an *ad hoc* manner. In sum, the right-hand column in Fig. 1 contains the leadership functions that are more technically oriented.

Next, we consider the rows in Fig. 1: The four leadership functions serve two additional managerial purposes: they motivate and they facilitate. Two of the leadership functions are largely about generating and funneling energy; two are largely about removing obstacles. In this remainder of this chapter, we flesh out the model, mobilizing past research on teams and combining it with our field research, to take a deeper look at each leadership function and practice. We conclude the chapter with a review of the whole model.

Build an Engaging Vision: Motivation/Interpersonal

The rationale for engaging in the interpersonal challenges of teamwork is frequently an engaging vision – the opportunity to accomplish something worthwhile that none of us could do alone. Members must agree on, and feel proud of, their team's shared goal, as well as feel good about the process it will use to achieve it (Mullen & Copper, 1994). In our context, participants must care enough to expend the energy required for extreme teaming, and a shared vision plays a powerful role in motivating that effort. When people share a compelling vision of the future, they are more willing to work hard to bring it about (Hackman, 1990).

Note that we use the term "build" – when talking about the emergence of shared vision in extreme teaming – deliberately. An engaging vision may be relatively straightforward in the case of a stable, well-bounded team in a single organization, because its objective supports a specific organizational mission and its *modus operandi* follows norms and routines largely defined by a shared and familiar organizational context. In contrast, the novelty and uncertainty inherent in the settings of our extreme teaming case studies give rise to an equally powerful need to nurture team cohesion around an engaging vision – but a far less realistic chance of crystalizing a clear, engaging vision in advance. Instead, the vision must be able to develop as more is learned, and it must be open to being partially reshaped by input from diverse camps. Vision, for extreme teaming, is thus both leader led and collectively built. Although project participants may be focused on contributing to a common problem, they are likely to be motivated by different goals. For instance, NGOs and corporations tend to bind individuals to diverging interests (e.g., Battilana & Dorado, 2010). In contrast, to more traditional homogeneous teams, where individuals typically come together around a common purpose that is relatively singular and stable (e.g., Salas, Dickinson, Converse, & Tannenbaum, 1992), establishing a goal for those engaged in extreme teaming is more complex and fluid. Without common goals to help direct and organize the behavior of diverse individuals, team performance can suffer (Martocchio & Frink, 1994). But without participation in helping build that engaging vision, motivation and performance can also suffer.

Leaders in our case studies acted publicly to nurture team cohesion around an engaging vision through two practices: making values explicit, and articulating a challenging target. The targets were generally ones to which all could agree – bold, audacious performance indices that everyone hoped the joint work could help realize. Values, similarly, were shared, abstract, and enduring. But the details of a vision sometimes come into focus slowly and are, by design, allowed to morph. We argue that the leadership practices that give rise to an evolving shared vision work by motivating people, affecting emotions and giving rise to trusting relationships across boundaries.

Leaders in our research worked consistently to motivate the assembled group, and emphasized the difficulty of their endeavor. To do so, they used various ways to make the values of their project explicit. Because values strike such a deep chord in people, they can be a source of motivation. Moreover, they are relatively stable, and can thus serve as a rallying point around which strangers can unite, despite their many differences. By evoking a consistent set of values, leaders motivated the integration of individual and group goals. Ambiguity and uncertainty were acknowledged, but at the same time subordinated to the value-identity that leaders had imprinted on the project. Furthermore, leaders turned the challenge to be achieved into a target, with a timeline, clear goals, and specific details of individual project participants' tasks. Recent work by Carton (2017) on how JFK helped NASA employees connect their daily tasks to the organization's highest aspirations also emphasizes similar practices.

The leaders' actual activities can be linked to theory in multiple ways. First, goals have been presented as fundamental to contemporary treatments of work motivation (Klein, Wesson, Hollenbeck, & Alge, 1999). Cognitive evaluation theory, for instance, posits that individuals can be motivated by intrinsic factors such as deep personal interest in an activity or topic (Deci & Ryan, 1985), and goal-setting theory points to the individual performance benefits of goals that are specific yet stretching (Locke & Latham, 1990). In short, to feel motivated to put effort into a task, people must be interested and challenged. Similarly, in work groups, team members must feel motivated by their collective goal to persevere towards its achievement. Team members must see "the bigger picture," and feel attached to it, in the sense that they internalize the collective goal as their own (Crown & Rosse, 1995). Other theories, such as expectancy-value theory (Klein et al., 1999; Vroom, 1964), posit that team goals must be perceived as important and achievable for team members to become engaged in a project. Making the team goal public has also been found to spark team commitment towards it (Hollenbeck, Williams, & Klein, 1989). This commitment has been shown to be closely related to team cohesion (Mulvey & Klein, 1998), and to positively influence team performance (Aubé & Rousseau, 2005; Hecht, Allen, Klammer, & Kelly, 2002).

Research also shows that goals must be tangible enough to define member tasks, coordinate their actions, and develop efficient work procedures (Klein & Mulvey, 1995). Conversely, Kozlowski and Ilgen (2006) suggest that when teams commit to a particular goal, it allows them to manage ambiguity more easily, because roles fall into place as the path to the goal becomes more tangible. Indeed, leaders' orientation towards reaching a particular target, and performing well against it, has been shown to help create cohesion around it (Misumi & Peterson, 1985).

Cultivate Psychological Safety: Facilitation/Interpersonal

Scholars who investigate why some teams perform better than others have paid increasing attention to team learning (see Edmondson et al., 2007 for a review). Team learning represents a continuous process of reflection and action, through which teams acquire, share, combine, and apply knowledge (Argote, Gruenfeld, & Naquin, 2001). It captures the extent to which team members reflect upon the team's objectives and activities, communicate with each other about both, and make changes accordingly. In terms of behaviors, team learning indicates that team members are "asking questions, seeking feedback, experimenting, reflecting on results, and discussing errors or unexpected outcomes of actions" (Edmondson, 1999, p. 353). Most empirical studies have shown performance benefits of learning behaviors in work groups (e.g., Bresman & Zellmer-Bruhn, 2013; Mitchell & Boyle, 2015; Somech & Drach-Zahavy, 2013). These benefits are especially significant for teams tackling creative or innovative tasks who can tap into a broad range of members' expertise (Bell et al., 2011; Sanner & Bunderson, 2015).

Learning benefits are relatively straightforward on simple tasks for which knowledge is codified and outcomes are well known in advance. However, the benefits are less clear when a task is complex and outcomes are uncertain (Edmondson, Winslow, Bohmer, & Pisano, 2003; Singer & Edmondson, 2008). This is the nature of the settings for extreme teaming, as well as precisely the kinds of settings in which learning is crucial to success (e.g., Ancona & Bresman, 2007). In uncertain situations, especially in those that require innovation, an exploratory response to errors or unexpected outcomes of actions is more valuable than seeking to confirm existing views. However, such an exploratory response does not come naturally, since it means challenging existing assumptions and experimenting with new behaviors and possibilities, with the aim of learning – and quickly (Edmondson, 2012).

In such situations, the costs of learning are often more evident than the performance benefits, and thus leadership must explicitly consider the role of emotions and relationships when they seek to facilitate team learning.

Team psychological safety – or members' shared sense that it is safe to take interpersonal risks within the team (Edmondson, 1999; see Edmondson & Lei, 2014 for a review) – has been found to be significantly associated with team learning in a wide variety of settings, including in inter-organizational product development teams (Bstieler & Hemmert, 2010) and other temporary project teams (Choo, Linderman, & Schroeder, 2007), as well as R&D (Post, 2012), manufacturing (Edmondson, 1999; Lee, Swink, & Pandejpong, 2011), and intensive care units (Tucker, Nembhard, & Edmondson, 2007). Psychological safety promotes inquiry in teams: it facilitates the emergence of trusting relationships among group members, which in turn makes it easier for team members to put themselves in vulnerable positions in the group context. Feeling psychologically safe within work groups is critical, because concern over image costs such as losing credibility generally makes it difficult to adopt learning behaviors (Argyris, 1982; Edmondson, 1996). As Schein recently observed, "We live in a pragmatic, problem solving culture in which knowing things and telling others what we know is valued. […] Having to ask is a sign of weakness or ignorance, so we avoid it as much as possible" (2013, p. 10). For instance, adopting an experimentation behavior under high evaluative pressure has proved to be problematic (Lee, Edmondson, Thomke, & Worline, 2004). Some failures are inevitable during experimentation, but not all environments are conducive to learning from failures. People thus need psychologically safe environments to openly discuss errors or unexpected turns of events (Carmeli & Gittell, 2009), or to experiment and voice their ideas and concerns to other team members (Edmondson, 2012). This, in turn, gives rise to productive dialogue (Tsoukas, 2009), perspective taking (Boland & Tenkasi, 1995), and learning and performance for the team.

Leaders in our case studies of extreme teaming addressed project participants' emotions and relationships in ways that cultivated psychological safety. They enacted two leadership practices that helped the newly formed group engage in learning behaviors, and gain and develop knowledge about the complex, interconnected problems it was facing: displaying authentic caring and framing cross-boundary work as a resource. By doing so, leaders nurtured positive emotions and relationships, hence facilitating productive dialogue among parties with different, sometimes conflicting, perspectives.

Displaying authentic caring is not about trying to gain the respect of others in the project, which is how leadership is all too often perceived in project settings (e.g., Pinto & Kharbanda, 1995; Pinto & Slevin, 1988). Instead, authentic caring must be just that – authentic concern for others' well-being. The leaders we studied tended to project a considerate demeanor and to act in ways that showed that they valued others and cared about their state of mind; they did this by engaging in helpful behaviors, by listening carefully to what others said, and by inquiring into others' experiences on the project. This helped participants feel that their problems were genuinely important to project leaders, not just "one more thing on their plate." Instead of attributing responsibility for problems (i.e., blaming), leaders focused their energies on how to resolve them. Their approach was neither passive nor reactionary, but rather active and forward-looking. In such an environment, participants could feel confident that public embarrassment or private derision was unlikely. That helped them bring up "weird" ideas, mistakes, and other project issues without fear.

In addition to building a psychologically safe environment and positive relationships, leaders underscored the value of each participant's input by framing cross-boundary work as a resource.

How people feel about what they know has been shown to moderate the relationship between psychological safety and knowledge sharing among coworkers (Siemsen, Roth, & Balasubramanian, 2009). When people gave feedback, leaders worked hard not to respond defensively, and instead took an active interest – such as by asking clarifying questions, probing for more insight, and involving others in the conversation. In these ways, project leaders embodied a belief that an appropriate understanding of the complexity the group faced necessitated support from all its parties in contemplating alternative hypotheses. Rather than seeking to prove what one set of individuals already believed, leaders' behaviors and attitudes influenced the group to seek discovery through iterative exchanges across knowledge and organizational boundaries. They invited dissent without disrespecting individuals who wanted to express their opinions. This leadership practice paved the way for exploratory responses to errors, rather than confirmatory ones: Leaders did not reinforce accepted assumptions, naturally promoting an advocacy orientation on the part of the most powerful or most expert individuals involved. Approaches to analysis and problem-solving led the teams to embrace ambiguity and openly acknowledge gaps in knowledge by inviting action and inquiry. Overall, this leadership practice helped nurture psychological safety by making project participants feel that their time and expertise was valued, and that their personal involvement could have an impact.

These findings extend several past studies that have emphasized the relational aspects of leadership in facilitating the emergence of psychological safety (Edmondson, 2004; Nembhard & Edmondson, 2006). They suggest that leaders can model relational behaviors by encouraging collaboration and open communication, and through sincere behaviors. This also chimes with organizational scholars' explorations of positive working relationships (Dutton, 2003; Dutton & Heaphy, 2003; Dutton & Ragins, 2007). When relationships are of sufficiently high quality, people feel valued and safe enough to engage in learning behaviors and in challenging tasks together. As active participants in their organization, not interlopers, they are free to say what they really think and feel (Kahn, 2007). In other words, high-quality relationships instill a feeling of psychological safety that "operates at a level of sorts, which, once triggered, enables people to move more closely toward one another in positive ways" (Kahn, 2007, p. 280).

Collaboration helps people get to know each other more intimately. When members find collaboration a positive experience and develop positive expectations of others' intentions and behaviors, they are more willing to open up and leave themselves exposed. In addition, when relationship-focused leaders encourage open communication, project participants feel they are safe to speak up and express their views freely (Edmondson, 1999, 2003). Thus, the more members sense a genuine openness in their mutual interactions, the more likely they are to accept becoming vulnerable. Dutton (2003) noted that by paying attention to relationships, leaders also learn how to cultivate connections, and that their willingness to embody openness and emotional accessibility builds a foundation for high-quality relationships. Several recent survey-based studies have demonstrated the positive relationship between relational or inclusive leadership, team learning, and performance (e.g., Carmeli, Brueller, & Dutton, 2009; Carmeli, Tishler, & Edmondson, 2012; Choo et al., 2007; Hirak, Peng, Carmeli, & Schaubroeck, 2012). Drawing from our case studies of extreme teaming, we offer a fine-grained view of how leaders can encourage sincere behaviors, cultivate a safe environment where members can count on each other, and encourage vulnerability.

Develop Shared Mental Models: Facilitation/Technical

Mental models are defined as cognitive representations that allow individuals to depict, explain, comprehend, and predict events and phenomena in the settings in which they find themselves (Johnson-Laird, 1983). Although essentially individual, these cognitive structures can be shared in a group, thus becoming group-level phenomena (Kozlowski & Klein, 2000; McComb, 2007). In such cases, group members are tapping into a shared understanding of the situations they encounter, and are thus better able to determine which actions to take, and how and when to take them (Mohammed, Hamilton, Tesler, Mancuso, & McNeese, 2015). Put differently, shared mental models are "knowledge structures" or "road maps" that team members share, which help them recognize each other's needs as they make progress towards their goal (Cannon-Bowers, Salas, & Converse, 1990; Stout, Cannon-Bowers, Salas, & Milanovich, 1999).

Shared mental models become especially pivotal in the face of uncertainty, because they encourage team adjustment and adaptation – team members will interpret changes in the task environment in an analogous way, allowing them to respond as one (Entin & Serfaty, 1999; Klimoski & Mohammed, 1994). Shared mental models are thus an important mediator of the diversity–performance relationship; numerous prior studies have demonstrated the performance benefits of shared mental models in diverse teams (e.g., Edwards, Day, Arthur, & Bell, 2006; see DeChurch & Mesmer-Magnus, 2010 for a meta-analysis of 23 empirical studies on the subject).

Shared mental models are also significant when the team's task is complex. For complex tasks, performance depends not only on whether group members actually share knowledge but also on their ability to develop and implement effective strategies (Hackman, 2002; Salas, Sims, & Burke, 2005). A classic example is product designers creating designs that can't be manufactured within cost and quality constraints, despite the ability of manufacturing experts to see the problem immediately, upon reviewing the designs (Clark & Fujimoto, 1991). In some cases, this is merely a matter of passing information from one expert to another. In others, however, the gap between the experts is caused by "sticky" information (Von Hippel, 1994), which cannot be readily transmitted and comprehended due to a lack of underlying specialized knowledge that provides context. As discussed in Chapter 3, people from different backgrounds tend to interpret and solve problems differently, making it difficult for them to communicate and share ideas with each other (Cronin & Weingart, 2007; Dougherty, 1992). As a result, those engaged in extreme teaming face technical challenges characterized by information asymmetry (Stasser & Stewart, 1992; Stasser & Titus, 1985) and other knowledge boundaries (Carlile, 2004; Edmondson & Harvey, 2017), where team members hold disparate but nonetheless crucial information about the problems they are trying to solve.

Shared mental models facilitate the necessary learning processes among team members, enabling efficient communication and fruitful cross-boundary collaboration where people draw on diverse knowledge to solve complex problems or make difficult decisions (Kozlowski & Chao, 2012). Indeed, previous research shows that a shared understanding of the learning requirements of the task is related to the quality of knowledge sharing among the team members (van Ginkel, Tindale, & van Knippenberg, 2009). In such situations, shared mental models ensure that team members can quickly make sense of suggestions made by their teammates; that they discuss strategies that are aligned with the group objectives; and that they provide each other with relevant feedback at the right time (Salas, Cooke, & Rosen, 2008). Yet, it is difficult to develop shared mental models in the case of extreme teaming because unless people look physically different from each other, they tend to assume that they all see the situation the same way, and won't make the effort required to truly understand one another (Phillips, Northcraft, & Neale, 2006). For that reason, outside interventions are often needed for newly formed teams to develop shared mental models that are accurate and extensive (Mohammed, Ferzandi, & Hamilton, 2010).

Cannon-Bowers (2007) has argued that team members rarely make a conscious and deliberate effort to develop shared mental models. Cross-training – that is, "an instructional strategy in which each team member is trained in the duties of his or her teammates" (Volpe, Cannon-Bowers, Salas, & Spector, 1996, p. 87) – has remained the main intervention to develop shared mental models (see Cannon-Bowers, 2007, for training strategies designed to increase the convergence and "sharedness" of team members' mental models), which in turn positively influences performance (Marks, Sabella, Burke, & Zaccaro, 2002). Surprisingly, the literature has rarely examined the role leadership can play in facilitating the development of shared mental models.

The two leadership practices that helped project teams develop shared mental models (diagnosing interfaces for knowledge sharing and leveraging boundary objects) were vital to effective coordination. Coordination was a daunting job in the projects we studied; designing work was more than a matter of planning, directing, and controlling a bundle of tasks. Rather, leaders were obliged to make judicious use of diverse expertise to solve a series of interconnected problems in a timely fashion. This meant helping to ensure that project participants shared information with each other at the right time, but it also meant ensuring that those involved on the same task, or an interdependent series of tasks, had a common knowledge repertoire to draw from, and knowledge objects to leverage, so they could comprehend each other's activities and requirements despite knowledge gaps. We contend that these practices were about breaking down barriers caused by the boundaries or gaps between individuals' knowledge and skills.

Diagnosing interfaces becomes meaningful when an issue or decision affects more than one area of expertise, such that discussion and coordination are needed to resolve the issue or make the decision. As Baldwin and Clark (2000) note, the process of modularization – or splitting up the elements of a project according to a formal plan – make complexity manageable and enable parallel work. Moreover, the introduction of modular work (task sets that can be accomplished independently of others) into a complex project helps to mitigate future uncertainty, by reducing the number of interconnections affected by future changes in a single aspect (or module) of the project's output. By adopting what Baldwin and Clark call a modular architecture, a project also makes it easier to substitute new modules for old, without having to change every aspect of a project output (whether product, service, or system).

Diagnosing interfaces and opportunities for modularity is an important leadership practice for extreme teaming. At the outset, there is considerable uncertainty about what ideas and actions might help to solve the problems faced by complex projects' participants. However, this uncertainty is exacerbated by participants' limited understanding of others' expertise. Therefore, as we found throughout our cases, there is a need for leaders to facilitate the sharing and learning process that gives rise to shared mental models. These models, in turn, help the team determine the nature and structure of the modules – or subsystems – required, and to identify the interfaces at which thoughtful conversation and coordination are needed. Although knowledge sharing is an obvious challenge, the work needed to make it happen is frequently neglected. This makes leadership essential; it plays a gap-closing role, through a set of practices that help to surface and translate relevant expertise among the subgroups in a project. Leveraging participants' capabilities is essential to any innovation project, but participants in extreme teaming typically need help to bridge the knowledge gaps between areas of expertise. This observation is consistent with recent work by Lingo and O'Mahony (2010) emphasizing the importance of a sequential integration process where certain contributors are strategically brought together at certain phases of the project, and kept apart at others.

Our findings also suggest that bringing project participants together must be managed tactically, because teams with divergent mental models have been shown to fail to discuss and integrate members' uniquely held perspectives or ideas (Harvey, 2013). Objects such as design and documentation drawings, or custom-developed scripts and tools, served to spark dialogue about important issues at times when shared mental models were *under construction*. They helped translate expert knowledge across domains to support important decisions. Our findings also showed how leaders mobilized objects to reach agreement on how certain tasks needed to be accomplished, what deadlines project participants needed to accomplish the task, and at what pace activities should take place. For knowledge integration to succeed, it is likely that these practices should continue throughout an extreme teaming initiative.

Empower Agile Execution: Motivation/Technical

Extensive research finds that leaders play an important role in enabling teams facing new tasks to perform well, by empowering team members and developing the team (Conger & Kanungo, 1988; Kozlowski & Ilgen, 2006). Empowerment stimulates learning behaviors in self-managed teams (Cohen & Ledford, 1994); teams with the latitude and ability to experiment, and develop and implement potential solutions as they see fit, are likely to learn as they go about accomplishing their work. In contrast, when groups are denied substantial freedom, they may fall into a fire-fighting mode of merely reacting to problems rather than pushing on towards a goal (Kirkman & Rosen, 1999). This is especially important in the case of extreme teaming, because no single project participant or leader would be able to tackle the challenge alone. Significant effort is required early in such projects to cultivate *agile execution* – essentially, the ability to identify, study, and experiment with potential opportunities, and to decide which ones to exploit further.

Empowerment enables the expression of intrinsic motivation (Thomas & Velthouse, 1990). Motivation is increased when individuals are given the authority to make decisions (Seibert, Silver, & Randolph, 2004) – that is, to be delegated authority over certain tasks. Yet, individuals only feel empowered when they are competent to make the decisions delegated to them, and feel comfortable being held accountable for the outcomes of those decisions (Spreitzer, 1995). Tuckey, Bakker, and Dollard (2012) found that leaders empower work group members by giving them the opportunity to self-determine and accomplish tasks in areas they had mastered, and Burpitt and Bigoness (1997) found an impact of empowerment on risk-taking, innovativeness, and adaptability. Team researchers (e.g., Srivastava, Bartol, & Locke, 2006) have operationalized empowerment by leadership as the granting of autonomy that makes it more meaningful to complete skilled tasks, and thereby boosts individual motivation – provided team members believe that they have an impact on the tasks over which they are given autonomy (Chen, Kirkman, Kanfer, Allen, & Rosen, 2007).

We can identify several reasons that empowerment supports agile execution. First, empowerment helps people approach their work proactively (Crant, 2000). Individuals who are empowered take more initiative and show greater perseverance in the face of challenges. Deci and Ryan (1985) found that the more people perceive that they are autonomous, the more initiative they take in work-related situations. In team research, team proactivity is expressed through commitment to continuous improvement and the capability to address potential issues that arise in innovative ways (Kirkman & Rosen, 1999; Power & Waddell, 2004).

In the cases we studied, leaders established expert decision clusters and gave them room to maneuver – to determine their own actions, plans, and schedules within their scope of expertise.

Because extreme teaming endeavors bring together individuals with diverse task-related resources, giving everyone authority over the same tasks would cause difficulty. Therefore, leaders gave project participants autonomy over tasks that corresponded to their profiles and expertise. In this way, they instilled motivation and elicited effort by allowing participants to freely explore the areas over which they were granted autonomy, with the aim of developing solutions that drew from their own areas of expertise. In short, project leaders enhanced experts' capabilities and influence on the team by forming subgroups and giving them decision rights over modular parts of the project, thus leveraging their ability to self-regulate.

While team researchers have tended to see subgroups as negative – or as potential sources of conflict and performance losses (e.g., Lau & Murnighan, 1998), subgroups can be beneficial for extreme teaming, as was true in our case studies. Our findings depict leadership practices focused on creating and enabling subgroups among individuals who display great heterogeneity in terms of disciplines and organization affiliation. In this context, it was not easy for leaders to bring people with similar expertise or viewpoints on a problem together, within a single subgroup, because project participants could still identify with those outside their subgroup – those who had different expertise might, for instance, have the same organizational affiliation. Thus, the subgroups were not, in these projects, polarizing entities representing different factions with distorted perceptions of reality and biased opinions, risking potential deadlock (Earley & Mosakowski, 2000). Rather, subgroups were typically able to consider another subgroup's ideas – and the extreme teaming context made them eager to do so, because of the explicit goal of teaming up to innovate. In this way, subgroups were able to converge on plans and ideas, because of what members of different subgroups shared (Brewer, 1991; Roccas & Schwartz, 1993).

In a related vein, prior research has shown that lone contrarian experts have trouble influencing group decisions unless they can find someone else who shares their viewpoint (e.g., Stasser et al., 1989). Even when most of the team disagrees with an idea, members will be receptive to the idea as long as it has more than one backer. Thus, a team member will have a better chance of getting a new idea taken seriously or implemented if he or she is part of a subgroup with members who are generally sympathetic to his or her views. Gibson and Vermeulen (2003) showed that the strength of heterogeneous teams' subgroups positively influences the learning that takes place in the team overall. Throughout the extreme teaming projects we studied, it was especially useful for people with similar perspectives to tackle certain tasks together, away from the opposition of individuals with a different viewpoint. Within such subgroups, project participants will find it easier to enjoy a richer exchange of information and rapid decision-making to study their preferred option or to tackle a set of independent tasks. For instance, it would have been difficult to engage in rapid learning around the different options that were explored in each project we studied if those involved had lingering concerns over the selection of the option to pursue. Forming subgroups helps experts find other project participants who share similar viewpoints to dig into and develop ideas more quickly.

Ambidextrous Leadership for Extreme Teaming

As emphasized throughout this book, realizing the performance potential of extreme teaming takes more than simply getting a diverse group of people into the same room. The reality of extreme teaming brings significant interpersonal and technical challenges. Most fundamentally, individuals who would not normally mix with each other must coalesce into a group with enough psychological safety for experimenting and learning to take place. Further, crossing knowledge boundaries involves dealing with viewpoints and interests that bring taken-for-granted assumptions. Several well-established theories have explored the complexity of both challenges, but theory on the leadership functions that help organizations succeed at extreme teaming is less well developed. Our model takes a small step towards a deeper understanding of this important aspect of management. By spanning different streams of research outside the core leadership literature, we believe our model offers the foundation for a more sophisticated and integrative leadership theory.

Our findings build on and extend the task- and person-focused leadership functions identified in the literature on team functional leadership. Converging on their purpose (motivation and facilitation) and orientation (interpersonal and technical challenges), the functions we analyzed may help resolve inconsistencies in the team leadership literature – for instance, Behrendt, Matz, and Göritz (2017) recently presenting "fostering coordination" as non-task related. Considering leadership functions in the context of extreme teaming challenges gives our model the concreteness necessary to support building new theory, while providing rich detail that connects the four functions we identified to a wealth of prior psychological theory and empirical research. Categorizing leadership functions based on motivation and facilitation makes sense in the context of two prior streams of psychological research: first, motivation and action theories that explain how people determine and achieve their goals, and, second, group and engagement research that examines the conditions under which people invest effort in a shared endeavor. There is little overlap between these streams historically, and we hope that our work will stimulate new, testable, integrated hypotheses and empirical research.

The other major body of organizational literature from which we draw studies knowledge in organizations and sheds light on the importance of ensuring a fit between the nature of the project and the approach taken by the team. Managing the technical challenges inherent in extreme teaming requires understanding the nature of the knowledge boundaries present within a project (Edmondson & Harvey, 2017). It also means identifying tasks for which knowledge may be simply transferred across boundaries, rather than translated or negotiated (Carlile, 2004). Several recent meta-analyses found that the performance benefits of knowledge diversity were contingent on the types of task undertaken (e.g., Bell et al., 2011; Van Dijk et al., 2012). In short, not every task requires major issues to be resolved, or new agreements created. Project participants, in some cases, can develop integrative solutions without deeply sharing each other's knowledge – thereby "transcending knowledge differences rather than traversing knowledge boundaries" (Majchrzak, More, & Faraj, 2012). Consistent with this idea, our model shows that one of the key leadership skills for extreme teaming is the ability to assess the types of tasks to be tackled by project participants and respond accordingly with motivation- or facilitation-oriented leadership practices.

Some tasks necessitate reciprocal interaction, due to inherent interdependencies, throughout their completion. Through intense collaboration and the development of shared mental models between the various parties, new, integrative solutions to vexing problems can be reached. Yet this requires considerable facilitation from leaders. Diverse mental models are neither revealed and reconciled simply by leadership fiat, nor can a leader simply install identical mental models in every participant's head. Allowing time for self-reflection and discussion that often relies on boundary objects for clarity is the essence of perspective making and perspective taking (Boland & Tenkasi, 1995), but even this process must be managed. In their study of project teams within an architecture firm, Ewenstein and Whyte (2009) found that boundary objects had the unwanted drawback of accentuating differences between groups, without forging the desired and necessary consensus to overcome them. To engage in deep conversations with people from other groups is demanding, and incurs a risk of fomenting interpersonal conflict that can undermine relationships and call future teamwork into question (Edmondson & Nembhard, 2009; Edmondson & Smith, 2006). Therefore, leaders must work hard to facilitate mutual understanding through some of the practices discussed in this book.

In contrast, some tasks do not necessitate interaction, and it is possible to empower a subteam to explore and make decisions within a realm of expertise. If a task can be broken down into simple, relatively independent components (modules), team members can work on the modules separately, with only minimal interaction and updating of other subteams, working on other modules. In such settings, leaders may opt for a motivation-oriented strategy where work is accomplished without deep interpersonal and knowledge exchange – particularly when there are protocols to support the teaming efforts. When time is tight, as is so often the case, saving time by minimizing non-essential interactions is valuable. For example, Faraj and Xiao (2006) showed, in a study of trauma care, that teams found treatment solutions not by sharing deep knowledge but by using relatively simple protocols to distinguish among the roles and responsibilities of separate disciplines (anesthesiology, nursing, and surgery). In the same way, leaders in our case studies developed decision-making protocols and made the necessary space for experimentation within them.

The effective project leaders we studied thus handled their considerable technical challenges in a manner that is reminiscent of the ambidextrous leadership advocated by Tushman and O'Reilly (1996; O'Reilly & Tushman, 2016). Extreme teaming endeavors present good examples of groups facing pressure to both explore and exploit opportunities simultaneously; that is, the ambidextrous leader must foster both efficiency and learning processes in parallel. In line with previous views on ambidextrous leadership (e.g., Vera & Crossan, 2004), the leadership functions we identify show a transition from one type of activity to another. Our model differs, however, in that the transition is not from top-down control to bottom-up ownership, but rather from leadership practices focused on motivation to those concerned with facilitation to tackle interpersonal and technical challenges. This new set of categories offers promising opportunities to further our understanding of leadership in the context of extreme teaming, which is becoming increasingly common today.

CHAPTER TAKEAWAYS

- The four leadership functions fall into two sets of categories. First, they address either an interpersonal or a technical challenge, and, second, they serve primarily to motivate effort or to facilitate progress. Each category finds support in – and adds to – prior research on team effectiveness and on knowledge in organizations.
- Building an engaging vision is about motivating project members, primarily through influencing emotions and relationships. Prior research has shown that goals that are challenging but realistic, and public, are most motivating.
- Cultivating psychological safety helps overcome interpersonal challenges. Research has shown that team learning is heightened when team members feel able to take interpersonal risks, fostering productive dialogue, perspective taking, and learning and performance in teams. Leaders play a vital role in building psychological safety. Our findings add evidence of leadership practices that may facilitate extreme teaming.
- Developing shared mental models facilitates work on technical challenges. Our findings show that leaders of extreme teaming facilitate efficient communication and fruitful cross-boundary collaboration, in part by helping teams identify those tasks that

are independent and those that are interdependent. The two practices we identified are consistent with prior work on team effectiveness and with studies of knowledge in organizations that emphasize the objects used during cross-boundary innovation processes.

- Empowering agile execution motivates effort and addresses technical challenges. Groups need the latitude and ability to experiment, to develop and implement potential solutions as they see unfold, and to learn. This is in line with past research on team leadership, and elaborates the behavioral practices that leaders can enact for this function.
- Our model exemplifies the notion of ambidextrous leadership. Leaders of extreme teaming must transition between facilitation- and motivation-oriented practices and must continuously tackle both interpersonal and technical challenges. Moving nimbly from one type of activity to another fulfills four leadership functions necessary to support extreme teaming.

9

DIRECTIONS FOR FUTURE RESEARCH AND PRACTICE

Recall from Chapter 3 that the team leadership literature can be organized into four streams: leader–member exchange, transformational leadership, shared leadership, and functional leadership. We found the functional leadership perspective to be most useful for understanding and enabling extreme teaming. Its focus on leader–team interactions and its attention to creating the conditions for effective performance is particularly useful for thinking about how leaders (along with other project participants) can observe and anticipate shifting situations, and intervene in thoughtful ways to steer team activities accordingly. In contrast, measuring leaders' characteristics, traits, or styles is unlikely to be a fruitful approach, theoretically or practically, in explaining or facilitating extreme teaming.

We chose to focus on the activities leaders can enact to tackle the shifting challenges and complex perspectives involved in teaming up across organizations to innovate. In so doing, we hope to provoke further research and practical experiments in the field, through which knowledge can be expanded in this important emerging domain. Our work was necessarily exploratory and future researchers and practitioners can help to advance knowledge of how to promote teamwork and effective innovation in cross-sector projects. Some of this work has already begun (e.g., Davis, 2016; Furr, O'Keeffe, & Dyer, 2016), and strategy, management, and organizational behavior researchers will no doubt continue this important line of work.

IMPLICATIONS FOR RESEARCH

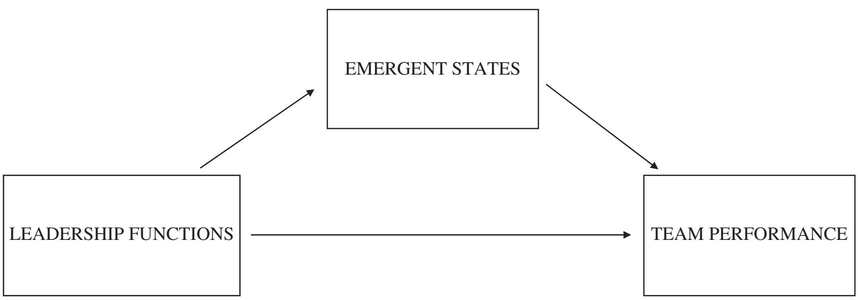
As we have argued, prior leadership taxonomies do not precisely reflect the challenges of extreme teaming. As outlined in Chapter 3, both interpersonal and technical challenges make it difficult for individuals with different expertise and organizational and industry affiliation to come together within a project designed to tackle an unfamiliar, complex problem. Our in-depth look at the activities of leaders involved in extreme teaming projects led us to identify four leadership functions, each formed of two specific practices and related to one emergent state. We believe our findings offer important avenues for future research.

First, most leadership taxonomies have been developed without considering contextual features (Johns, 2006; Maloney et al., 2016), despite long-standing recognition that the effectiveness of leadership approaches is situationally contingent (e.g., Fiedler, 1964, 1967, 1971, 1972, 1978). Our research examines leadership functions in team-like arrangements in which people collaborated in time-limited projects that spanned boundaries including expertise, organization, and industry. In each case, the goal was to develop and implement an innovative product to address a novel challenge or solve a complex problem. Future research is needed to more precisely and accurately develop the leadership functions that help teams facing intense ambiguity and uncertainty in innovation project spanning organizational and industry boundaries. Situational opportunities and constraints affect the occurrence and meaning of leaders' activities, and additional case studies of extreme teaming are needed to identify the leadership functions that matter most. Polar cases, in particular, would now represent a good approach to build on our findings. Given limited prior theory and empirical evidence, scholars should continue to rely on inductive theory building using embedded, multiple cases (Edmondson & McManus, 2007; Eisenhardt, 1989) alongside a synthetic strategy to generate convincing relationships (Langley, 1999). This would enable them to better understand how leaders support extreme teaming, drawing on failures as well as successes. Such data would support the development of short cases describing each extreme teaming endeavor and the functions enacted by leaders, which could then be used to suggest further patterns related to team leadership and the performance of extreme teaming endeavors.

After building a better understanding of extreme teaming leadership functions, scholars might then further explore the mechanism behind the leadership–performance relationship in such settings. Drawing a connection between leadership functions and emergent states, for instance, is a key contribution of our work, because it shines a light on an area in which further research is greatly needed. We know that team member interaction enables or inhibits emergent states that can be affective, cognitive, or motivational (Edmondson & Harvey, 2017). These states include team cohesion, shared mental models, psychological safety, and empowerment; as well as transactive memory, potency, and more (Marks et al., 2001). Scholars have suggested that leadership plays a crucial role in shaping team member interaction so as to influence the emergence of these team-level states (Mathieu et al., 2008). However, we know very little about which leadership function is related to the emergence of which specific state; team leadership scholars mostly look at team effectiveness as an outcome of good leadership. In our findings, we take a small step in that direction by identifying four leadership functions and associating each with a specific emergent state evident in our data. For instance, we argued that by showing concern for work group members' well-being, taking time to discuss their concerns patiently, or giving them honest and fair answers, leaders primarily fostered psychological safety. To further our understanding of extreme teaming, and of team leadership and team effectiveness in general, scholars need to specify more precisely which leadership function enables the development of which emergent state. Our analysis is based on qualitative data, hence the lack of quantitative measurement for the emergent states noted. Yet, this opens the door for much more research in that direction.

In the long run, drawing on our leadership model, scholars may experiment with time-lagged research designs that consider leadership functions and emergent states, over time, during extreme teaming efforts. It is often assumed that team leadership directly influences team performance (Morgeson et al., 2010). However, as presented in Fig. 2, we can make the case that the leadership–performance relationship may be partially mediated by emergent states. Those leading extreme teaming efforts influence the emergence of certain states, which in turn affect team performance. Few studies have looked at leadership functions and emergent states *together* to understand the mechanism that underpins team performance, and what makes some extreme teaming efforts more successful than others. Understanding mediation relationships is vital to theory development and refinement (Mathieu,

Figure 2. Mediated Leadership–Performance Relationship.



DeShon, & Bergh, 2008), and testing this mediating pathway can help organizations make practical interventions that are more effective and efficient (Rucker, Preacher, Tormala, & Petty, 2011).

Clearly, there are different types of extreme teaming. Further contextual features should be included in the model shown in Fig. 2 in future research, so as to build a contingency framework that may serve as a tool for managers and practitioners. It will be useful to identify the team leadership functions needed for particular types of extreme teaming. To illustrate, in a meta-analysis, Joshi and Roh (2009) compiled data from nearly 9,000 teams in 39 studies in varied organizational settings and discovered that effect sizes associated with different occupation- and industry-level moderators varied significantly across studies. As we also have argued (c.f., Edmonson & Harvey, 2017), teaming across boundaries is not uniform across occupations or industries. There is a difference between engineers from BP teaming up with designers in a Tel Aviv start-up and those same engineers teaming up with nurses and doctors at a public healthcare institution in Canada. The nature of the boundaries that must be spanned for such extreme teaming to succeed may have a significant influence on the leadership functions that should be deployed.

Even beyond the boundaries to be spanned, future inquiries will need to consider the impact of other potential constraining or enabling contexts for extreme teaming. Political, economic, and legal factors, for instance, may influence the nature and applications of the leadership functions identified in this book. The study of how their interplay influences outcomes and emergent states is a promising avenue for future work, given that context affects how people think, feel, and interact (Kozlowski & Klein, 2000).

Our account of extreme teaming leadership is limited to studying the activities that took place within a small set of cross-sector innovation teams. Further, we did not observe activities taking place in other parts of the participating organizations, such as senior management support, access to experts or resources, or other external factors that may have affected project outcomes. Despite extremely diverse task-relevant resources, these innovation projects had in common that they lacked the full set of resources needed within their boundaries. Moreover, they did not work free from external scrutiny. The success of these projects most certainly depended, in part, on their ability to reach into the team environment to obtain important resources and support (c.f., Katz & Allen, 1985; Ancona & Caldwell, 1988). Leaders probably had to bargain and negotiate with other managers within their organization and outside, and build coalitions that facilitated the development of their project. The activities of executive championing a project, for instance, have been shown to positively influence the performance of software development project teams (Guinan, Cooprider, & Faraj, 1998), self-managed teams (Druskat & Wheeler, 2003), and R&D teams (Hirst & Mann, 2004). Other scholars have shown how important it is for leaders to protect the team from outside pressure or interference, especially when the team is dealing with a lot of uncertainty (Faraj & Yan, 2009). Future research looking at extreme teaming endeavors could focus on the extent to which leaders establish linkages and manage interactions with parties outside the team for the purposes of representation, information-seeking, or coordination, and thus complement our model with a set of externally focused leadership functions and practices.

IMPLICATIONS FOR PRACTICE

The trend that motivated this book is one with profound practical implications in its own right. Specifically, as the problems organizations and societies face become more and more challenging, the frequency of extreme teaming is likely to continue and grow. Cross-sector innovation in business ecosystems is on the rise (Davis, 2016) because of the need for technical depth and diverse skills at the same time; both are vital to the production of innovation (Wuchty et al., 2007). Diverse teams are more likely than individuals to develop innovative solutions, yet, as we have argued in this book, bringing diverse experts together in ways that they overcome differences in understanding and interests to create value requires determined, courageous, enabling leadership. Our exploratory qualitative research is a first step toward identifying factors that increase the effectiveness of extreme teaming. The most important practical implication of our research is thus to begin to test our ideas for leadership practices that facilitate extreme teaming in more projects and more contexts than we have been able to do thus far.

Culture Clash

An emphasis on leadership emerged from our qualitative data, in large part, due to the centrality of *culture clash* in cross-sector work. The clash of cultures that occurs when diverse professions and organizational members convene can quickly stymie collaboration (Edmondson & Reynolds, 2016). The boundaries between professions comprise points of vulnerability that can give rise to delays and failures in projects.

Culture comprises taken-for-granted aspects of how members of a group behave – whether that group is a country, a profession, or a company (Schein, 1985). Because culture is taken for granted, it feels natural and right to members of a group. When people from different cultures come together, they quickly encounter thwarted expectations, which give rise to misunderstandings, and in turn to conflicts. These dynamics lead people to make unflattering conclusions about the intentions or abilities of those in the other groups. And once we have made them, our negative attributions seem to us accurate and true. For this reason, the barriers created by cultural differences are difficult to overcome. They are subtle – all but invisible. Crossing knowledge boundaries is thus more difficult than it first appears, due to the subtle complexities of knowledge domains (Van de Ven & Zahra, 2016).

Some of the most important taken-for-granted assumptions relate to timeframes, work, priorities, and values. Only by exposing and exploring these assumptions can people from different cultures overcome the barriers they impose to work together effectively. This, of course, creates a crucial role for leadership.

The basic ways in which fields of expertise differ most critically can be sorted into the technical and interpersonal. Technical dimensions of cultural differences include timeframes and tasks. Interpersonal dimensions emphasize norms and values. Technical and interpersonal assumptions are equal partners in creating confusion and inhibiting progress. Helping extreme teamers become boundary-crossing tourists, eager to learn from other professional cultures, is a necessary first step in innovating together. Our recognition of the two broad elements of culture clash gave rise to our leadership framework, where these two elements came together with the fundamental need for both motivation and barrier removal in ensuring project progress.

Leadership that Overcomes Culture Clash

The leadership that works to facilitate extreme teaming is at first glance paradoxical. It embraces contradictory elements – like clarity and flexibility, ambition and inclusiveness, confidence and curiosity. Just as an organization's culture that enables innovation contains superficially contradictory elements like high standards and failure tolerance (Edmondson, 2012), leading extreme teaming projects focused on innovation has a similar need to navigate competing elements.

The most significant practical implication of our work thus lies in helping to guide future leaders of extreme teaming efforts. To navigating the difficult job that lies ahead, our research suggests that a learning mindset goes a long way toward helping traverse the inherently uncertain territory of extreme teaming. Enacting this mindset takes effort and a willingness to embrace a set of four leadership functions that are remarkably different from the top-down model of implementation of known plans that often exist as a kind of taken-for-granted model in the minds of new leaders. A crucial leadership role is that of creating psychologically safe interpersonal spaces to team up despite technical and interpersonal barriers.

Practitioners may ask themselves which leadership function is the most important, or which one comes first, second, third, and fourth. We believe that all four functions are necessary to make progress – and that the timing and sequence of each will vary in a dynamic way over time, depending on how projects unfold. Clearly, participants need to agree on a guiding vision to get started on their work. They also need to feel safe and empowered as they engage in the necessary risk-taking and testing that come with the territory. They must embrace the chance to learn as they go – to execute as a learning process.

Consider the importance of the following emergent states: If people agree on the vision, they will be open to considering various ways to get there. If people feel psychologically safe, they will be willing to share and learn from each other. If people feel empowered, they will engage more and take initiatives that can spur learning. If people share mental models of each other's expertise, they will be more able to efficiently pivot the project when needed. Each of these emergent states matters, to greater and lesser degrees, at different times in a project. There is no best way forward. No precise playbook to follow. Figuring out which leadership practice to exercise at which time will take judgment and intuition. However, it may be helpful to start with a focus on emotions and relationships, rather than with a focus on tasks and technology. We suspect that without building a social context in which people feel able to take interpersonal risks and feel motivated to appreciate others' unique contributions, it is difficult to make progress on technical dimensions of the collaborative work.

Our research underscores the technical rationale for extreme teaming (that is, the kinds of novel, and often "wicked," problems these groups confront necessarily require multiple areas of expertise to proceed), but also that a team's ability to achieve the potential for synergy is limited by the potential for misunderstanding and/or lack of goodwill across expertise groups. In short, the technical expertise that people bring to a project cannot be leveraged without interpersonal skill and goodwill. Moreover, once projects get off on the wrong foot – where heated conflicts or disdainful interactions have taken place – it is difficult to reverse the affective tide and help people reengage with each other, with enthusiasm. It is even more difficult to turn an unsafe environment for learning into a psychologically safe one, than it is to build a safe environment in the first place. These challenges fall into the domain of leadership responsibilities. Our model provides a way of thinking about these challenges, by categorizing leadership functions based on whether they motivate or remove barriers, and whether they address technical or interpersonal domains. We also identified specific practices that we believe are worthy candidates for practical experiments by project leaders who face some of these challenges. We hope and expect these experiments to further inform the development of more robust and actionable models for leading extreme teaming in the future.

CONCLUSION

This book presents ideas and practices that emerged from our qualitative research into a small set of cases of extreme teaming. In each case, a fluid and porous group of individuals from different occupations, organizations, and industries sought to develop into a high-performing innovation team. In each case, the rationale for undertaking a novel and risky project was a belief in the potential upside of the innovation for their organizations. Participants passionately believed that members of their partner organizations brought something to the collaboration that they, and their own organizations, lacked. These precious resources made the struggles of extreme teaming worthwhile.

Our case studies documented innovation journeys, in which the teams developed and changed, stumbled and got back on track, only to adapt again. All of the projects produced innovative outputs; whether they reached the full potential synergy their collective talents would enable can never be answered definitively. But we can say, with confidence, that the results were consistently regarded by the participating organizations as outstanding successes. They garnered prestigious international awards, results that bolstered our confidence in the value of the leadership activities we observed to support the extreme teaming.

The leadership practices we observed addressed both interpersonal and technical challenges – in alternating rhythms that supplemented and reinforced each other. Our analysis of the empirical data we collected led us to aggregate these practices into four leadership functions that we believe foster the success of extreme teaming. Hopefully, they can help you on your journey to studying or leading extreme teaming.

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